Projection Obstruction for Factorized States

Nonlinear Dynamics, Conditional ℓ^1 Diagnostics, and Exact Short-Time Defect Bounds

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Abstract

For finite one- and two-local quantum lattice dynamics starting from a product state, we establish the exact short-time two-site correlation-defect lower bound $D_e(t) = \|\rho_e(t) - \rho_a(t) \otimes \rho_b(t)\|_1 \geq \nu_e t - 24g^2t^2$, where ν_e is the initial normal product-retraction rate on edge e and g is the full incident interaction norm. All-order commutator bounds yield explicit Taylor coefficients, finite-horizon remainders, and a volume-uniform Lindbladian open-system extension. Lifting expectations to polynomials over cyclic operator subspaces gives a finite onset-or-invariance criterion; for k-body connected cumulants, this combines with a Steiner-tree onset bound $\tau_G(S)$, an algebraic cancellation stratification, an incidence-qubit tree construction attaining $\tau_T(S)$, and a boundary-checked local persistent-zero certificate.

Geometrically, we compare the canonical marginal-product retraction against nearest-product trace-norm approximation, derive a dual tangent-space distance formula and a pure-state Pythagorean rate identity $\nu^2_{\text{global}} = \sum_e \nu_e^2$ (together with its cut-edge cluster extension and an intra-cluster entanglement counterexample), and construct explicit state families showing that global and edge-wise trace-norm defects are not uniformly equivalent. Along time-dependent variational principle

(TDVP) product trajectories, we prove an edge-decomposed Duhamel error bound weighted by Lieb–Robinson light-cone factors.

At the observer and cochain level, we quantify the physical visibility gap between projection-limited local observers and global section defects. On an n-qubit Ising ring ($n \geq 5$) and on higher-degree graphs of girth at least 4 and minimum degree at least 2, we construct a three-body cycle-cochain adapter on the ring and degree-dependent stabilizer observers on regular graphs. At the exact graph-state time $t_* = \pi/(4|J|)$, every two-body edge defect vanishes identically ($D_e(t_*) = 0$, Local Green), yet the ring cycle class is nonzero and the product-state minimax defects remain strictly positive ($r_2 \approx 0.6823278038$ on the ring, and $r_d^{\mathrm{Q}} > 0$ on d-regular graphs, Global Red). These results isolate the physical and geometric core of projection obstruction on quantum lattices, while companion investigations across this series examine higher presheaves, gauge classifications, adaptive numerical prototypes, and operational foundations (§1.2).

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1. Introduction

Projection-based approximations are central to many-body quantum physics and numerical simulation. Methods such as mean-field theory, Hartree–Fock, and product-state variational ansätze replace a correlated many-body state with a factorized product of local states, reducing an exponentially large state space to a tractable product family. Mathematically, this replacement is implemented by a nonlinear retraction Π. For density matrices, its smooth product extension lives in the affine space of trace-one Hermitian operators; the positive subset has rank boundaries, where a linear tangent space and a feasible tangent cone must be distinguished (§§5.10.2 and 5.11.1). While the computational utility of such retractions is well established, a quantitative mathematical theory of the error they introduce requires answering three foundational questions: how the nonlinear geometry of Π shapes the projected vector field and its normal leakage; how local edge residuals aggregate into global diagnostics under explicit observer axioms; and how fast two-body and k-body correlation defects grow at short and intermediate times.

In this work, we focus on the core quantum lattice dynamics and the physical gap between local and global observation:

1. **Exact Short-Time Local Defect Bounds, TDVP Duhamel Enclosures, and Connected Cumulant Onset (§5 and §6):** For any finite one- and two-local Hamiltonian with incident interaction norm g evolving from a product state, Theorem 5.1 establishes the exact two-site correlation-defect lower bound $D_e(t) \geq \nu_e t - 24g^2t^2$ (extended to open Lindblad dynamics in §5.4.1), where $\nu_e = \|Q_e\dot{\rho}_e(0)\|_1$ isolates the normal component of the initial two-site velocity after projecting out one-local tangent motion (Lemma 5.3). Sections 5.8–5.10 extend this estimate to all Taylor orders, establish all-time local dynamical limits under a uniform incident bound and spatial density limits under the additional translation-invariance hypotheses, without invoking Lieb–Robinson bounds, and derive an edge-decomposed Duhamel observable error bound along TDVP trajectories weighted by Lieb–Robinson light-cone factors (Theorem 5.10.A) together with a cut-edge Pythagorean cluster rate law (Theorem 5.10.B). Sections 5.11–5.17 compare the marginal retraction Π with nearest-product trace distance, establish pure-state Pythagorean rate identities, establish the Steiner-tree onset bound $m_{\mathrm{onset}} \geq \tau_G(S)$ for k-body connected cumulants $\kappa_S(t)$, and classify algebraic cancellation strata and local persistent-zero certificates.
2. **Local Marginal Geometry and 0-Cochain Representations (§2–§4):** In §2 and §3, we

establish the smooth product retraction geometry on the affine space of trace-one Hermitian operators, isolating tangent and normal projectors P_e and $Q_e = I - P_e$. In §4, Theorem 4.3.A represents the two-site correlation defect $D_e(t) = \|\rho_{uv}(t) - \rho_u(t) \otimes \rho_v(t)\|_1$ as the fiber norm of the 0-cochain difference $s_{\text{true}} - s_{\text{prod}}$ of two closed marginal sections, whose unweighted ℓ^1 summation $C(t) = \sum_e D_e(t)$ is characterized by replica additivity (RE3) and singleton calibration (RE5).

3. **Physical Visibility Gaps and Graph-State Minimax Separation (§7):** Section 7 separates raw projection residuals, induced 1-cochains, and quotient cohomology classes $\Phi([\omega]) = \inf_\alpha \|\omega - d_0\alpha\|_1$. In §7.4, we formalize projection-limited local observer families and exhibit a three-patch carrier with full edge coverage where every local observer reports zero defect ("Local Green") while the global compatibility defect norm is $I(s) = 6 > 0$ ("Global Red"). In §7.8, we construct an explicit three-site stabilizer cochain adapter on an n-qubit Ising ring ($n \geq 5$) and establish the Higher-Degree Graph-State Minimax Theorem (Theorem 7.12) on d-regular graphs ($d \geq 2$, girth ≥ 4): at the exact graph-state formation time $t_* = \pi/(4|J|)$, every two-body edge defect $D_e(t_*)$ vanishes identically because $\rho_{uv}(t_*) = I_4/4$, yet the ring cycle quotient class $\Phi(\omega(t_*)) = 1$ and the product-state minimax defect ($r_2 \approx 0.6823278038$ on the ring, and $r_d^{\mathrm{Q}} > 0$ on d-regular graphs) remain strictly positive.

Conventions. Norms are denoted $\|\cdot\|$ with appropriate subscripts ($\|\cdot\|_1$ for trace norm on trace-class operators or ℓ^1 norm on cochains, $\|\cdot\|_\infty$ for operator/supremum norm, and $\|\cdot\|_2$ or $\|\cdot\|_{\mathrm{HS}}$ for Euclidean/Hilbert–Schmidt norm); scalar absolute values are denoted $|\cdot|$.

Notation

Symbol	Meaning
X	ambient trace-class or Banach state space of Hermitian operators
$\mathcal{M}$	declared smooth product family; for density states, its positive subset has rank boundaries
Π	C^2 product-state retraction
T_t	ambient unitary or Lindblad dynamics semigroup
S_t	projected dynamics family $\Pi T_t\|_{\mathcal{M}}$
$\bar{B}(m)$	projected tangent vector field $D\Pi(m)Am$
$D_e(t)$	two-site trace-norm correlation defect $\|\rho_e(t) - \rho_a(t) \otimes \rho_b(t)\|_1$
$\widehat{D}_e(t)$	dimension-normalized defect $D_e(t)/d^2$ (for equal endpoint dimensions d)
$C(t)$	edgewise correlation defect sum $\sum_{e \in E} D_e(t)$
P_e, Q_e	tangent and normal projectors at a product state ($Q_e = I - P_e$)
n_e	initial normal velocity residual $Q_e \dot{\rho}_e(0)$
ν_e	local normal leakage rate $\|n_e\|_1$
γ_e	dimension-normalized rate ν_e/d^2 (for equal endpoint dimensions d)
g	full local incident interaction norm $\max_{v \in V} \sum_{X \ni v} \|h_X\|_\infty$

1.1 Related Work and Positioning

This paper connects to six main lines of research:

1. **Mean-field, Hartree–Fock, and product-state variational methods.** Product-state ansätze make many-body quantum evolution computationally tractable by restricting states to $\mathcal{M}$. Rather than proposing a new variational ansatz, we quantify the normal leakage rate ν_e, the short-time trace-norm defect $D_e(t)$, and cohomological diagnostics obtained through separately declared observation adapters and cycle repairs.
2. **Time-dependent variational principle (TDVP) and projected dynamics.** TDVP projects the Schrödinger or Lindblad vector field onto the tangent bundle $T\mathcal{M}$ of a variational manifold (Haegeman et al. [14]). While global and local error bounds for variational quantum dynamics via standard Duhamel variation of constants are studied in Martinazzo and Burghardt [10] and Zoufal, Sutter, and Woerner [11], our focus in §5.10.2 is the exact local decomposition of the generator residual into partial-trace-annihilating edge normal operators ($\mathrm{Tr}_u\, n_e = \mathrm{Tr}_w\, n_e = 0$) and their coupling with incident Lieb–Robinson spatial suppression weights.
3. **Projection-operator (Nakajima–Zwanzig) methods.** Classical projection-operator formalisms eliminate unresolved degrees of freedom at the cost of a non-Markovian memory kernel. Our short-time bounds in §5 analyze the instantaneous normal velocity $n_e = (I - P_e)\dot{\rho}_e(0)$ and the moving retraction residual $F_e(t) = \rho_e(t) - \rho_a(t) \otimes \rho_b(t)$, while §2.2 explains why the projected family $S_t = \Pi T_t|_{\mathcal{M}}$ need not be a semigroup. An exact memory-kernel closure is not derived here.
4. **Lieb–Robinson locality bounds.** Whereas Lieb–Robinson bounds [1, 3, 4, 9, 25] control spatial light-cone propagation across lattice distance, our short-time lower bounds (Theorems 5.1 and 5.5) and infinite-volume dynamical limits (Theorem 5.7) rely on direct finite-support commutator combinatorics without causal-cone truncation. When bounding errors of spatially localized observables along a TDVP trajectory, Theorem 5.10.A incorporates Lieb–Robinson light-cone weights $w_{O,e}(\tau)$ to exponentially suppress distant edge leakage.
5. **Presheaf and cohomological obstructions in quantum information.** Sheaf-theoretic and cohomological methods have been widely used to study contextuality and local-to-global consistency. Here we work with observer visibility kernels (§7.4), and an explicit 1D cycle cochain adapter (§7.8) that detects graph-state multipartite correlations invisible to all two-site marginals.
6. **Entanglement timescales and correlation distance.** The pure-state normal rate identity $\nu_e = 2\|\chi\|$ in §5.3.1 connects directly to the correlated uncertainty governing Yang's entanglement timescale [15] and Cresswell's Rényi-entropy rate [16], while $D_e = \|\rho_e - \rho_a \otimes \rho_b\|_1$ is Hall's trace-norm correlation distance [13], whose relationship to minimum product distance [17] is quantified in §5.11.

1.2 Research Series and Sibling Packages

To keep the present paper focused on quantum lattice dynamics, local correlation bounds, and physical visibility gaps, specialized and background developments are coordinated with companion packages in the `002_projection_obstruction` series and `coda_atlas`:

Sibling Paper / Package	Formal Title & Version / Commit	Source	Key Anchors	Results used & Scope
PO-001	*From Covering-Edge Descent Complexes to Functorial Presheaves and Coherence Defects* (v2.0.0)	`002_projection_obstruction/001_from_covering_edge_descent...`	Theorem 2.1, §3	Saturated-chain presheaves, degree-2 composition curvature cochains $K_{c,b,a}$, and operator cocycle coboundaries.
PO-006 (GS-003)	*Vector and Quantum Observer Geometries: Gauge Relativity and Trace Metric Bounds* (v0.4.0)	`002_projection_obstruction/006_vector_and_quantum_observer_geometries`	§§2–4	Non-scalar observer geometries, Schatten-norm operator bounds, dual witness certificates, and phase metrics.
PO-007 (GS-001)	*M-Compatible Gauge Classification for Banach Presheaf Coboundaries: The ℓ^1 Maximal-Symmetry Endpoint* (v4.7.0)	`002_projection_obstruction/007_m_compatible_gauge_classification`	Theorem 2.2, §3	M-compatible gauge classification for Banach presheaves; maximal linear symmetry selecting weighted ℓ^1.
PO-008 (GS-002)	*SO(k)-Invariant Gauge Rigidity for Declared Euclidean Stalks: Local ℓ^2 Selection and Outer ℓ^1 Aggregation* (v2.0.0)	`002_projection_obstruction/008_so_k_invariant_gauge_classification`	Theorem 1.1, §§2–3	$SO(k)$-invariant gauge rigidity on Euclidean stalks; local ℓ^2 selection with outer ℓ^1 aggregation.
PO-009 (GS-000)	*Coordinate-Wise Additivity and Local ℓ^1 Gauge Selection* (v6.1.1)	`002_projection_obstruction/009_coordinate_wise_additivity`	Theorems 1–3, §2	Coordinate-wise additivity and local ℓ^1 gauge selection on finite graph cochains.

Sibling Paper / Package	Formal Title & Version / Commit	Source	Key Anchors	Results used & Scope
PO-033	*Certified Adaptive Quantum Dynamics from Projection Residuals* (v1.0.1)	`002_projection_obstruction/033_certified_adaptive_quantum_dynamics`	§3, §4, Claims PO033-C1–C6	Conditional analytic bounds and uncertified adaptive cluster prototype (`HEURISTIC_UNCERTIFIED`); Lean scope is conditional finite-sum inequalities over an abstract carrier (`Nat`/`Int`). Validated continuous-time ODE integration and conservative quadrature remain open work.
PO-034	*Higher-Cell Product Retractions, Simplicial Cochains, and Finite Retraction Dynamics* (v0.1.0)	`002_projection_obstruction/034_higher_cell_cochain_projection...`	§2, §3, Theorems 4.1–4.3	Higher-cell product retractions, oblique projectors, simplicial cochain dynamics, and finite-dilation Taylor estimates.
PO-035	*Zero-Container Relational Spacetime, Emergent Lorentz Causality, and Witness Holonomy* (v0.1.0)	`002_projection_obstruction/035_zero_container_relational_spacetime...`	§§2–7, Claims RC-01–RC-10	Defect 4-current conservation ($\partial_\mu J^\mu = 0$), discrete relational spacetime, and witness holonomy.
Finite QM	*Finite Quantum Mechanics: Operational Jordan–Lie Foundations, Canonical Traces, and Central Purification* (v0.55.0)	`F004_papers/006_finite_qm`	§§13.5–13.8, Claim FQM-13.8	Operational Jordan–Lie foundations, character cancellation on the sign cube, positive Bell cone extensions, and central-sector purification obstructions (former Appendix E).

Sibling Paper / Package	Formal Title & Version / Commit	Source	Key Anchors	Results used & Scope
Archived Monograph Material	*Projection Obstruction for Factorized States* (pre-streamlining monograph)	Commits `90e6231546` / `d44f760555`; code: `code/retraction_geometry.py`	Former §§3.4–3.14, Appendix D	Banach graph charts, shear retractions, parameter-chart transitions, bounded-left-inverse failure lemmas, and conditional extension reductions (omitted from the 69-page edition).

Every theorem in §§5–7 of the present paper is self-contained under its declared Hilbert-space and interaction-graph premises.

1.3 Conceptual Pipeline

The progression from full many-body dynamics to local and cohomological diagnostics follows six explicit steps:

1. **Ambient state space** X: the affine space $\mathcal{A}_1$ of trace-one Hermitian operators, with parallel subspace of Hermitian trace-zero variations $\mathcal{A}_0$.
2. **Smooth product extension** $\mathcal{M} \subset X$: selected via a C^2 retraction Π on a stated neighborhood. Positivity and boundary feasibility are additional constraints, not consequences of smoothness.
3. **Projected dynamics and normal residual:** tangent projection $\bar{B}(m) = D\Pi(m)Am$, projected family $S_t = \Pi \circ T_t|_{\mathcal{M}}$, and normal velocity $(I - D\Pi(m))Am$.
4. **Induced 1-cochains:**
   - *Rate diagnostic:* instantaneous local leakage $\omega_e^{\text{rate}} = q_e(n_e) \in V_e$ induced by linear observation maps q_e on normal velocity n_e. For identity observation maps, this yields the normal leakage rate $\nu_e = \|n_e\|_1$.
   - *State diagnostic:* finite-time correlation defect $\omega_e^{\text{state}}(t) = \rho_e(t) - \rho_a(t) \otimes \rho_b(t)$, whose trace-norm magnitude is $D_e(t) = \|\omega_e^{\text{state}}(t)\|_1$.
5. **Quotient obstruction vs. coboundary:** given a coboundary differential $d_0 : C^0 \to C^1$, the repair-minimized quotient seminorm $\Phi([\omega]) = \inf_{\alpha \in C^0} \|\omega - d_0\alpha\|_1$ vanishes if and only if $\omega = d_0\alpha$ is exact (in finite dimensions).
6. **Global edge aggregation and short-time connection:** under replica additivity (RE3) and singleton calibration (RE5), local edge diagnostics aggregate via unweighted ℓ^1 summation:
   - the total instantaneous rate tally is $\sum_{e \in E} \nu_e$;
   - the total finite-time state correlation defect is $C(t) = \sum_{e \in E} D_e(t)$; these two diagnostics are physically distinct and connected by the short-time expansion of Theorem 5.5: $|D_e(t) - t\nu_e| \leq 24g^2t^2$.

2. Structural Framework

We consider quantum spin lattices and general Banach state spaces under nonlinear factorization retractions.

2.1 State Space and Product Retraction

Let X denote the real Banach space of Hermitian trace-class operators on a composite Hilbert space $\mathcal{H}_V = \bigotimes_{v\in V} \mathcal{H}_v$ equipped with the trace norm $\|\cdot\|_1$. Let $\mathcal{A}_1 = \{\rho \in X : \operatorname{Tr}\rho = 1\}$ denote the affine hyperplane of trace-one Hermitian operators, with parallel subspace of trace-zero Hermitian variations $\mathcal{A}_0 = \{\Delta \in X : \operatorname{Tr}\Delta = 0\}$. The convex physical state space is $\mathcal{S}(\mathcal{H}_V) = \{\rho \in \mathcal{A}_1 : \rho \geq 0\}$.

The **physical product state manifold** $\mathcal{M} = \{\bigotimes_{v\in V} \rho_v : \rho_v \in \mathcal{S}(\mathcal{H}_v)\}$ forms a smooth submanifold on the interior of the state space (where states have full local rank), with rank boundaries on singular strata (§§5.10.2 and 5.11.1), embedded in the smooth affine product variety $\mathcal{M}_{\text{aff}} = \{\bigotimes_{v\in V} \sigma_v : \sigma_v \in \mathcal{A}_1(\mathcal{H}_v)\}$. On physical density states, the canonical **product retraction**

$$\Pi : \mathcal{S}(\mathcal{H}_V) \longrightarrow \mathcal{M}, \qquad \Pi(\rho) = \bigotimes_{v\in V} \operatorname{Tr}_{V\setminus\{v\}}(\rho),$$

restricts to a well-defined projection onto $\mathcal{M}$. More generally, Π admits a smooth C^∞ extension $\Pi : \mathcal{A}_1 \to \mathcal{M}_{\text{aff}}$ on the trace-one Hermitian affine slice, with tangent derivatives $D\Pi(\rho) : \mathcal{A}_0 \to T_{\Pi(\rho)}\mathcal{M}_{\text{aff}}$ acting on trace-zero variations. On each two-site subsystem $e = \{u, w\}$, the local marginal product retraction $\Pi_e(\rho_{uw}) = \rho_u \otimes \rho_w$ induces an instantaneous tangent projector $P_e = D\Pi_e(m)$ at any product state $m = \rho_u \otimes \rho_w$, and a complementary **normal projector**

$$Q_e = I - P_e.$$

For general covering-edge Banach diagrams and saturated-chain descent complexes over abstract posets (Mac Lane [2]), we refer to companion package **PO-001** (*From Covering-Edge Descent Complexes to Functorial Presheaves and Coherence Defects*).

2.2 Linear Semigroups and Projected Dynamics Families

Let $\{T_t\}_{t\geq 0}$ be a C_0 contraction semigroup on X with generator $(A, \mathcal{D}(A))$ (for closed quantum systems, $A\rho = -i[H, \rho]$; for open Markovian systems, A is a Lindbladian generator). Whenever $\{T_t\}$ does not leave $\mathcal{M}$ invariant, the **projected dynamics family**

$$S_t = \Pi \circ T_t|_{\mathcal{M}}$$

maps $\mathcal{M} \to \mathcal{M}$ locally for small $t \geq 0$, but need not satisfy the semigroup law:

$$S_{t+s} = \Pi T_{t+s}|_{\mathcal{M}} \quad \text{need not equal} \quad S_t S_s = \Pi T_t \Pi T_s|_{\mathcal{M}}.$$

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3. Affine-Chart Criterion for Projected Vector Fields

3.1 Statement and Decomposition

Theorem 3.1 (Affine Chart Criterion). Let X be a Banach space, $\{T_t\}$ a C_0 contraction semigroup with generator $(A, \mathcal{D}(A))$, and $\Pi : W \to M \subset X$ a C^2 retraction onto a C^2 submanifold

M. Assume either that A is bounded on X, or that M is a C^2 submanifold of the graph-norm domain of A such that $m \mapsto Am$ is C^1 along M. Let $\phi : U \subset M \to Y$ be a chart in a flat affine atlas on M, and define the coordinate vector field

$$b(y) = D\phi_{\phi^{-1}(y)} \left[D\Pi_{\phi^{-1}(y)} A\phi^{-1}(y) \right].$$

If $y \mapsto Db(y)$ is nonconstant on a connected chart domain $\phi(U)$, then b is not affine-linear in that chart. Consequently, the projected dynamics family $S_t = \Pi \circ T_t|_M$ cannot have b as an exact affine-linear semigroup generator in the chart ϕ. (This is a coordinate statement in the chosen affine chart and does not rule out nonlinear C^2 coordinate conjugacies.)

Theorem 3.1 rests on two elementary observations:

Lemma 3.A (Affine Criterion). In any flat affine coordinate chart, an affine-linear vector field $b(y) = Ly + c$ has a constant Fréchet derivative $y \mapsto Db(y) = L$. Therefore, if $y \mapsto Db(y)$ is nonconstant, b is not affine-linear in those coordinates.

Proposition 3.B (Non-Invariance versus Projected Motion). For an orbit point $T_t m \in W \setminus M$, the retraction Π changes that point because its image lies in M. This does not imply that the projected map $S_t = \Pi \circ T_t|_M$, wherever defined, moves its initial point: nontrivial projected motion requires an explicit pair (m, t) with $T_t m \in W$ and $\Pi(T_t m) \neq m$.

3.2 Parallel Tensor Fields on Flat Affine Manifolds

Lemma 3.2 (Parallel fields on flat affine connections). Let (M, ∇) be a connected Banach manifold equipped with a torsion-free flat affine connection. If a $(1,1)$-tensor field S on M satisfies $\nabla S = 0$, then S has constant operator components in every connected affine coordinate chart.

Proof. In an affine coordinate chart where Christoffel symbols vanish, $\nabla S = 0$ reduces to $\partial_v S = 0$ in all coordinate directions, so S is locally constant and hence constant on any connected chart domain. Affine transition maps preserve constancy across the affine atlas. □

3.3 Proof of Theorem 3.1 and the Ambient-to-Chart Bridge

Proof of Theorem 3.1. In an affine chart, any affine-linear vector field has the form $b(y) = Ly + c$ with constant derivative $Db(y) = L$; nonconstancy of $y \mapsto Db(y)$ immediately rules out this form. □

Lemma 3.3 (tangent-restricted affine-subspace transfer). Let $L \subset X$ be a closed affine subspace with direction space L_0, and let $M \subset L$ be relatively open, so $T_m M = L_0$ for every $m \in M$. Let $\Pi : W \to M$ be a C^2 retraction and let A satisfy the bounded-generator or graph-norm hypothesis of Theorem 3.1, so that $\bar{B}(m) = D\Pi(m)Am$ is C^1 on M. Let $\tilde{\phi} : L \to Y$ be a continuous affine isomorphism, set $\phi = \tilde{\phi}|_M$, and write $S = (D\tilde{\phi})^{-1} : Y \to L_0$. Then, under the identification $T_m M = L_0$,

$$Db(y) = D\tilde{\phi} \circ D_M \bar{B}(\phi^{-1}(y)) \circ S.$$

Consequently, on this chart, $y \mapsto Db(y)$ is nonconstant if and only if $m \mapsto D_M \bar{B}(m)$ is nonconstant.

Proof. Because $\tilde{\phi}$ is affine, $D\phi_m = D\tilde{\phi} : L_0 \to Y$ is independent of $m \in M$, so $b(y) = D\tilde{\phi}(\bar{B}(\phi^{-1}(y)))$ and the chain rule gives $Db(y)[\xi] = D\tilde{\phi}(D_M \bar{B}(m)[S\xi])$ for $m = \phi^{-1}(y)$. □

Example 3.3 (Projection curvature inducing chart nonlinearity on a flat subspace). On $X = \mathbb{R}^2$, $M = \mathbb{R} \times \{0\}$, $\Pi(x, y) = (x(1 + y), 0)$, and $A(x, y) = (-x, x - y)$ (a contraction generator

on Euclidean $\mathbb{R}^2$), direct calculation gives $\bar{B}(x,0) = (x^2 - x, 0)$, so $b(x) = x^2 - x$ and $Db(x) = 2x - 1$ ($Db(0) = -1 \neq Db(1) = 1$), where nonlinearity arises directly from the projection-curvature term $D^2\Pi(m)(Am, (1,0)) = (x, 0)$.

Proposition 3.C (Ambient nonconstancy without flat hypotheses). Ambient nonconstancy of $D_X\bar{B}(m)$ and a nonzero projection-curvature term $D^2\Pi(m)(Am, \cdot)$ do not by themselves force $y \mapsto Db(y)$ to be nonconstant on a curved submanifold M.

Proof. On Euclidean $X = \mathbb{R}^2$, take the parabola $M = \{(s, s^2) : s \in \mathbb{R}\}$, vertical retraction $\Pi(x,y) = (x, x^2)$, contraction generator $A(x,y) = (-x, 0)$, and global chart $\phi(s, s^2) = s$. Then $\bar{B}(s, s^2) = (-s)(1, 2s)$, so $b(s) = -s$ and $Db(s) = -1$ (constant), even though $D_X\bar{B}$ and the contracted projection-curvature term $D^2\Pi_m[Am, h] = (0, -2sh_x)$ for $m = (s, s^2)$ and $h = (h_x, 2sh_x)$ are nonconstant along M. (Note that while the uncontracted second Fréchet derivative $D^2\Pi_{(x,y)}[h,k] = (0, 2h_xk_x)$ is constant on $\mathbb{R}^2$, its contraction with the generator field $Am = (-s, 0)$ depends nontrivially on the base point m.) □

Omitted chart hierarchy and archived results. The detailed hierarchy of chart-derivative formulas for vertical, sheared, and fiber-nonlinear graph submanifolds over Banach splittings, along with bounded left-inverse failure criteria (former Lemmas 3.4–3.14 in the pre-streamlining monograph), is omitted from this 69-page edition. Readers seeking those full functional-analytic proofs are referred to the archived reviewed monograph revision `90e6231546` (and `d44f760555`), with exact symbolic verification fixtures preserved in `code/retraction_geometry.py`. Sibling package **PO-034** develops higher-cell product retractions and simplicial dynamics, while **PO-001** and **PO-006** develop covering-edge descent complexes and non-scalar observer geometries respectively (see the authoritative sibling reference table in §1.2).

3.4 Residual Taxonomy and Retraction Geometry

Throughout the paper we distinguish three residual objects associated with $(X, \mathcal{M}, \Pi, T_t)$:

1. **Instantaneous normal leakage:**

$$\Delta_\Pi(m) = (I - D\Pi_m)Am,$$

   which is the first-order normal velocity vector leaving the tangent space $T_m\mathcal{M}$.
2. **Finite-time retraction residual:**
   - Under C^1 orbit differentiability:

$$R(t,m) = T_t m - \Pi(T_t m) = t(I - D\Pi_m)Am + o(t).$$

   - For bounded generators A and smooth retractions $\Pi \in C^3(\mathcal{A}_1)$ (as in finite-dimensional quantum spin lattices):

$$R(t,m) = t(I - D\Pi_m)Am + \frac{t^2}{2}\left(A^2m - D\Pi_m A^2 m - D^2\Pi_m(Am, Am)\right) + \mathcal{O}(t^3).$$

3. **Projected-family semigroup defect:**

$$S_{t+s}(m) - S_t(S_s(m)),$$

   which measures the failure of the projected family $S_t = \Pi \circ T_t|_{\mathcal{M}}$ to form an exact semigroup on $\mathcal{M}$ (delegated to companion study **PO-034**).

The short-time quantum correlation bounds of §5 analyze objects 1 and 2 directly on each two-site subsystem.

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4. Conditional Additive ℓ^1 Diagnostic

Theorem 3.1 supplies a chart-level affine-nonlinearity criterion for a declared coordinate vector field. The next question is how to measure projection residuals once a diagnostic has been declared. Theorem 4.2 establishes that, given RE1–RE5, the ℓ^1 norm is the selected diagnostic. The conclusion is conditional on these axioms; relaxing them admits other norms.

4.1 Primitive Axioms for Replica-Extensive Diagnostics

Let E be a finite edge set. For each $e \in E$, let V_e be a finite-dimensional vector space equipped with a faithful local gauge

$$j_e : V_e \to \mathbb{R}_{\geq 0}.$$

A diagnostic is a family

$$J_E : \prod_{e \in E} V_e \to \mathbb{R}_{\geq 0}.$$

We do **not** assume an additive form.

Axiom RE1 — Global gauge Each J_E is a faithful convex gauge:

$$J_E(x) = 0 \iff x = 0,$$

$$J_E(\lambda x) = \lambda J_E(x), \qquad \lambda \geq 0,$$

$$J_E(x + y) \leq J_E(x) + J_E(y).$$

Axiom RE2 — Local magnitude sufficiency There exists a function

$$A_E : \mathbb{R}_{\geq 0}^E \to \mathbb{R}_{\geq 0}$$

such that

$$J_E(x) = A_E\big((j_e(x_e))_{e \in E}\big).$$

Thus the observer sees only independent local magnitudes, not directions or cross-edge phases.

Axiom RE3 — Disjoint replica additivity For disjoint finite edge sets E, F:

$$J_{E \sqcup F}(x \sqcup y) = J_E(x) + J_F(y).$$

Equivalently,

$$A_{E \sqcup F}(u \sqcup v) = A_E(u) + A_F(v).$$

Axiom RE4 — Edge-exchange symmetry If $\sigma : E \to F$ is a bijection preserving local gauge type, and for each $e \in E$ a gauge-preserving fiber identification map $\tau_e : V_e \to V_{\sigma(e)}$ is declared, then for $x = (x_e)_{e \in E}$ and $(\sigma x)_{\sigma(e)} = \tau_e(x_e)$:

$$J_F(\sigma x) = J_E(x).$$

Axiom RE5 — One-edge normalization For every singleton $E = \{e\}$:

$$J_{\{e\}}(x_e) = j_e(x_e).$$

4.2 Replica-Extensivity Hierarchy

Theorem 4.2 (Measurement Hierarchy and Observer Selection). Let E be finite and let the diagnostic family J_E be defined on every finite disjoint union of labelled edge types across the observer class.

1. **RE3 alone gives singleton separability:**
$$J_E(x) = \sum_{e \in E} J_{\{e\}}(x_e).$$
2. **RE1+RE2+RE3 give weighted ℓ^1 relative to the declared local gauges:** if each $V_e \neq 0$, then
$$J_E(x) = \sum_{e \in E} c_e j_e(x_e), \qquad c_e > 0.$$
3. **RE4 controls symmetry:** c_e is constant on every declared edge-type orbit.
4. **RE3+RE5 give the edgewise sum of declared local gauges:**
$$J_E(x) = \sum_{e \in E} j_e(x_e).$$

   RE1, RE2, and RE4 are unnecessary for this last implication.

Observer Selection Framing. RE3 supplies separability across disjoint replicas, while RE5 calibrates singleton terms; together, RE3+RE5 force the edgewise sum of the declared singleton gauges $j_e(x_e)$. The theorem identifies the exact observer axioms under which an ℓ^1 edge tally is selected, rather than deriving a fiber norm from graph topology alone. In Lean 4, item 4 is mechanized as `ReplicaHierarchy.unweighted_sum_of_replicaAdditive_of_singleton_calibration` in `lean4/po000_l001_replica_hierarchy.lean`, and the singleton algebraic cores of items 2 and 3 are mechanized in `lean4/po000_l005_weighted_hierarchy.lean` (`PO000.WeightedHierarchy.singleton_aggregator_linear`, `singleton_weight_pos`, and `orbit_weights_equal`).

Proof. Because
$$E = \bigsqcup_{e \in E} \{e\},$$
finite induction using RE3 gives
$$J_E(x) = \sum_{e \in E} J_{\{e\}}(x_e),$$
establishing item 1.

Assume RE1 and RE2 as well. On a singleton, write
$$J_{\{e\}}(x) = A_{\{e\}}(j_e(x)).$$
The local gauge has full nonnegative range: choose $v \neq 0$; faithfulness gives $j_e(v) > 0$, and positive homogeneity gives
$$j_e\left(\frac{u}{j_e(v)} v\right) = u \qquad (u \geq 0).$$

Positive homogeneity of the singleton diagnostic from RE1 therefore forces

$$A_{\{e\}}(u) = c_e u, \qquad c_e = A_{\{e\}}(1).$$

Faithfulness gives $c_e > 0$. Substitution into the RE3 decomposition establishes item 2. RE4 identifies coefficients on declared type-preserving edge orbits, establishing item 3.

Finally, under RE5,

$$J_{\{e\}}(x_e) = j_e(x_e).$$

Substitution into the RE3 decomposition establishes item 4 directly. No transitivity assumption is needed. □

A direct logical consequence is that RE3+RE5 also imply RE1, RE2, and RE4 whenever the j_e are faithful convex gauges and RE4 ranges over bijections preserving local gauge type.

4.3 Coboundary and Induced-Cochain Specializations

For a covering-edge Banach diagram over a poset (formalized in §7.3.3; see also companion package **PO-001**), the edge data

$$\delta_e(s) = r_{b,a} s(b) - s(a)$$

are a coboundary by construction: $\delta = d_0 s$. With $j_e(v) = \|v\|_{F(a)}$, Theorem 4.2 gives

$$J_P(s) = \sum_{e \in E_{\mathrm{cov}}} \|\delta_e(s)\|_{F(a)}$$

under RE3+RE5, or a weighted sum under RE1+RE2+RE3.

The physical projection residual in §5 is a distinct object. If bounded linear observation maps

$$\mathcal{O}_{e,m} : \ker D\Pi_m \longrightarrow V_e$$

map the normal leakage

$$\mathcal{N}_\Pi(m) = (I - D\Pi_m) A m$$

to

$$\omega_e(m) = \mathcal{O}_{e,m}(\mathcal{N}_\Pi(m)),$$

then $\omega(m) \in \prod_e V_e$ is an induced 1-cochain relative to the chosen observation family. It becomes a coboundary only when a 0-cochain space C^0, a differential $d_0 : C^0 \to C^1$, and a section $s \in C^0$ with $\omega = d_0 s$ are supplied.

4.3.1 Exact Dependencies in the Implication Lattice

The implication lattice of Theorem 4.2 is:

$$\boxed{\text{RE3} \Rightarrow \text{singleton separability}}, \qquad \boxed{\text{RE1} + \text{RE2} + \text{RE3} \Rightarrow \text{weighted } \ell^1},$$

$$\boxed{\text{RE1} + \text{RE2} + \text{RE3} + \text{RE4} \Rightarrow \text{orbit-constant weights}}, \qquad \boxed{\text{RE3} + \text{RE5} \Rightarrow \text{unweighted } \ell^1}.$$

The first and fourth implications are finite disjoint-union identities. The weighted implication uses only the singleton clauses of RE1 and RE2: singleton homogeneity, singleton faithfulness, and dependence on the local gauge magnitude. In `lean4/po000_l005_weighted_hierarchy.lean`, `PO000.WeightedHierarchy.singleton_aggregator_linear` derives $A_{\{e\}}(u) = A_{\{e\}}(1)\,u$ from homogeneity, magnitude sufficiency, and full nonnegative range; `singleton_weight_pos` derives $c_e > 0$ from faithfulness; and `orbit_weights_equal` derives orbit weight equality under gauge-preserving fiber bijections. Companion works [5,8] study broader gauge classifications under transformation-monoid and coordinate-additivity hypotheses.

Note that Theorem 4.2 characterizes the raw edge tally I on a declared 1-cochain, which is distinct from the quotient obstruction seminorm $\Phi([\omega]) = \inf_{\alpha\in C^0}\|\omega - d_0\alpha\|_1$: a positive raw tally $I > 0$ does not imply $\Phi > 0$, and $\Phi = 0$ does not imply $I = 0$.

4.3.2 Quantum Partial-Trace Poset and Two-Site Correlation Cochains

Theorem 4.3.A (Quantum Partial-Trace Poset and Correlation Defect Representation). Let V be a finite set of quantum sites with local Hilbert spaces $\mathcal{H}_v$ ($v \in V$), and let $\mathcal{P} = (\mathcal{P}(V), \subseteq)$ be the finite poset of site subsets ordered by inclusion. For each subset $A \subseteq V$, let $\mathcal{H}_A = \bigotimes_{v\in A}\mathcal{H}_v$ and let $\mathcal{F}(A) = \mathfrak{T}_1(\mathcal{H}_A)$ be the real Banach space of Hermitian trace-class operators on $\mathcal{H}_A$ equipped with the trace norm $\|\cdot\|_1$. For every inclusion $A \subseteq B$, define the restriction map $r_{B,A} : \mathcal{F}(B) \to \mathcal{F}(A)$ by the partial trace:

$$r_{B,A}(\sigma_B) = \mathrm{Tr}_{B\setminus A}(\sigma_B).$$

Then: 1. *Banach Diagram Functoriality:* For any chain $A \subseteq B \subseteq C$, $r_{B,A} \circ r_{C,B} = r_{C,A}$ and $r_{A,A} = \mathrm{id}_{\mathcal{F}(A)}$. Each restriction map $r_{B,A}$ is a positive, trace-preserving linear contraction with $\|r_{B,A}\|_{1\to 1} = 1$. 2. *Covering-Edge Poset Coboundary and Marginal Compatibility:* The covering relations in $\mathcal{P}$ are pairs $e = (A, A\cup\{v\})$ with $v \notin A$. For any 0-cochain $s \in C^0(\mathcal{P};\mathcal{F}) = \prod_{A\subseteq V}\mathcal{F}(A)$, the coboundary operator $\delta = d_0 s \in C^1(E_{\mathrm{cov}};\mathcal{F})$ evaluates along each covering edge $e = (A, B)$ to

$$\delta_e(s) = r_{B,A}(s(B)) - s(A) = \mathrm{Tr}_{B\setminus A}(s(B)) - s(A).$$

Consequently, a 0-cochain s is closed ($d_0 s = 0$) if and only if the family $\{s(A)\}_{A\subseteq V}$ is a partial-trace-compatible marginal hierarchy ($\mathrm{Tr}_{B\setminus A}(s(B)) = s(A)$ for all $A \subseteq B$). In particular, for any global state $\rho_V \in \mathcal{S}(\mathcal{H}_V)$ with single-site marginals $\rho_v = \mathrm{Tr}_{V\setminus\{v\}}(\rho_V)$, both the true marginal section $s_{\mathrm{true}}(A) := \rho_A = \mathrm{Tr}_{V\setminus A}(\rho_V)$ and the factorized marginal section $s_{\mathrm{prod}}(A) := \bigotimes_{v\in A}\rho_v$ are individually closed 0-cochains ($d_0 s_{\mathrm{true}} = 0$ and $d_0 s_{\mathrm{prod}} = 0$). 3. *0-Cochain Difference and Two-Level Induced Defect Cochain:* Because both s_{true} and s_{prod} are closed under linear partial trace—and because on any singleton-to-pair covering edge $(\{u\},\{u,v\})$ one has $\mathrm{Tr}_v(\rho_{uv}) - \rho_u = 0$, while for $|V| \geq 3$ any subset X with $2 \leq |X| < |V|$ serves simultaneously as an upper vertex of $(X\setminus\{x\}, X)$ and a lower vertex of $(X, X\cup\{y\})$—the subset correlation defect

$$D_A(\rho_V) := \left\|\rho_A - \bigotimes_{v\in A}\rho_v\right\|_1 = \left\|(s_{\mathrm{true}} - s_{\mathrm{prod}})(A)\right\|_1$$

is the fiber norm at A of the **0-cochain difference** $s_{\mathrm{true}} - s_{\mathrm{prod}} \in \ker d_0$ rather than the linear partial-trace coboundary of a single section on $\mathcal{P}(V)$. Restricted to the two-site edge family $E \subseteq \binom{V}{2}$, the nonlinear product-retraction residual $\omega_e(\rho_V) := (I - \Pi_e)(\rho_e) = \rho_e - \rho_u \otimes \rho_v = (s_{\mathrm{true}} - s_{\mathrm{prod}})(e)$

defines an induced 1-cochain $\omega(\rho_V) \in \prod_{e\in E} \mathcal{F}(e)$ whose RE3+RE5 ℓ^1 evaluation is the edgewise correlation tally:

$$J_E(\omega(\rho_V)) = \sum_{e=\{u,v\}\in E} \|\omega_e(\rho_V)\|_1 = \sum_{e=\{u,v\}\in E} \|\rho_{uv} - \rho_u \otimes \rho_v\|_1 = \sum_{e\in E} D_e(\rho_V).$$

Proof. For item 1, Fubini's theorem for partial traces on $\mathcal{H}_C = \mathcal{H}_A \otimes \mathcal{H}_{B\setminus A} \otimes \mathcal{H}_{C\setminus B}$ gives $\mathrm{Tr}_{B\setminus A}(\mathrm{Tr}_{C\setminus B}(\sigma_C)) = \mathrm{Tr}_{C\setminus A}(\sigma_C)$, and trace-norm contractivity $\|\mathrm{Tr}_{B\setminus A}(\sigma_B)\|_1 \le \|\sigma_B\|_1$ holds with equality on positive states ($\|r_{B,A}\|_{1\to 1} = 1$). For item 2, $d_0 s = 0$ means $\mathrm{Tr}_{B\setminus A}(s(B)) = s(A)$ on every covering pair $B = A \cup \{v\}$, which by induction along inclusion chains holds if and only if $\mathrm{Tr}_{B\setminus A}(s(B)) = s(A)$ for all $A \subseteq B$; both $s_{\mathrm{true}}(A) = \mathrm{Tr}_{V\setminus A}(\rho_V)$ and $s_{\mathrm{prod}}(A) = \bigotimes_{v\in A} \rho_v$ satisfy this identity since $\mathrm{Tr}_v(\rho_v) = 1$. Item 3 follows immediately by evaluating $(s_{\mathrm{true}} - s_{\mathrm{prod}})(A) = \rho_A - \bigotimes_{v\in A} \rho_v$ and applying Theorem 4.2(4) to $\omega_e(\rho_V) = \rho_e - \rho_u \otimes \rho_v$ with local gauge $j_e = \|\cdot\|_1$. □

Classical control. For

$$\rho_{uv} = \tfrac{1}{2}|00\rangle\langle 00| + \tfrac{1}{2}|11\rangle\langle 11|, \qquad \rho_u = \rho_v = I_2/2,$$

both sections above are closed, while $D_{\{u,v\}} = 1$: the difference from $I_4/4$ is diagonal with entries $(1, -1, -1, 1)/4$. This separable state distinguishes marginal consistency from product independence. Bell extension imposes a further requirement across measurement contexts (§7.10); neither this positive product defect nor a closed marginal section settles it.

4.4 Scope of the Characterization and Related Frameworks

Theorem 4.2 does not assert that every useful diagnostic on a graph must be an ℓ^1 sum, nor that the trace norm is uniquely selected by completely positive trace-preserving (CPTP) monotonicity alone. Rather, it shows that finite replica additivity (RE3) decomposes a global diagnostic into singleton contributions, and singleton calibration (RE5) identifies each singleton contribution with the chosen local gauge.

Related framework connections. The complete transformation-monoid gauge classification, the $SO(k)$-invariance rigidity on Euclidean stalks, and the one-axiom-drop independence counter-models (including `l2_combination_not_additive` and `doubled_sum_not_calibrated` in `lean4/po000_l005_weighted_hierarchy.lean`) are established in companion packages **PO-007** (`007_m_compatible_gauge_classification`), **PO-008** (`008_so_k_invariant_gauge_classification`), and **PO-009** (`009_coordinate_wise_additivity`). Linear measurement dimension counting on repair subspaces is established in the finite obstruction calculus of **PO-001**.

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5. Exact Short-Time Density Production

5.1 Setup and Normalization

Let $\Lambda = (V, E)$ be a finite simple graph, and attach to each vertex $v \in V$ a finite-dimensional Hilbert space $\mathcal{H}_v$. On the tensor-product Hilbert space $\mathcal{H}_V = \bigotimes_{v\in V} \mathcal{H}_v$, let

$$H = \sum_{1\le|X|\le 2} h_X$$

be a Hamiltonian with one- and two-site Hermitian interaction terms. Define the full local interaction norm

$$g := \max_{v\in V} \sum_{X\ni v} \|h_X\|_\infty.$$

This quantity includes one-local fields. If every edge term and the single one-local term at each vertex have norm at most J on a graph of maximum degree Δ, then

$$g \le (\Delta+1)J.$$

For the transverse-field Ising chain,

$$g \le 2|J| + |h|.$$

The first-order normal rate in §6.2 is independent of h, whereas the uniform second-order remainder and certified time window depend on g.

Let the initial state be a product state,

$$\rho(0) = \bigotimes_{v\in V} \rho_v, \qquad \rho(t) = e^{-itH}\rho(0)e^{itH}.$$

For $e = \{a, b\}$, write $\rho_e(t)$ for the exact two-site reduced state and $\rho_a(t), \rho_b(t)$ for its one-site marginals. The raw trace-norm defect and trace distance are defined for arbitrary endpoint dimensions, while the dimension-normalized score $\widehat{D}_e(t)$ is defined when both endpoints have local Hilbert dimension d:

$$D_e(t) := \|\rho_e(t) - \rho_a(t)\otimes\rho_b(t)\|_1, \qquad T_e(t) := \frac{1}{2}D_e(t), \qquad \widehat{D}_e(t) := \frac{1}{d^2}D_e(t).$$

Here D_e is the raw trace-norm correlation defect (cf. Hall [13]), T_e is the trace distance, and $\widehat{D}_e$ is the dimension-normalized score. Throughout the theorems below we work with D_e.

At the product state $\rho_a \otimes \rho_b$, differentiating the product-of-marginals retraction $\Pi_e(\sigma_{ab}) = \mathrm{Tr}_b(\sigma_{ab}) \otimes \mathrm{Tr}_a(\sigma_{ab})$ on the trace-zero tangent space yields

$$P_e Z = (\mathrm{Tr}_b Z)\otimes\rho_b + \rho_a \otimes (\mathrm{Tr}_a Z).$$

Because $L_e = \dot\rho_e(0)$ has trace zero ($\mathrm{Tr}\, L_e = 0$), $P_e L_e$ is the product-tangent component. We write

$$L_e := \dot\rho_e(0), \qquad Q_e := I - P_e, \qquad n_e := Q_e L_e, \qquad \nu_e := \|n_e\|_1, \qquad \gamma_e := \frac{\nu_e}{d^2}.$$

An edge is called *active* when $\nu_e > 0$; this is strictly stronger than having a nonzero commutator $[h_e, \rho_a\otimes\rho_b] \neq 0$, because $Q_e = I - P_e$ projects out all local product-tangent motion.

5.2 Uniform Double-Commutator Bounds

Lemma 5.2 (local derivative bounds). If A_S is supported on a site set S with $\|A_S\|_\infty \le 1$, then

$$\|[H,[H,A_S]]\|_\infty \le \begin{cases} 8g^2, & |S| = 1, \\ 24g^2, & |S| = 2. \end{cases}$$

For $|S| = 1$, one also has $\|[H, A_S]\|_\infty \le 2g$.

Proof. Only terms h_X with $X \cap S \neq \emptyset$ contribute to the first commutator $[H, A_S]$, and their operator norms sum to at most $|S|g$. For each such X, the union $S \cup X$ has size at most 2 when $|S| = 1$

and at most 3 when $|S| = 2$. Only terms h_Y meeting $S \cup X$ contribute to the outer commutator. Applying $\|[A, B]\|_\infty \le 2\|A\|_\infty \|B\|_\infty$ twice yields $4(g)(2g) = 8g^2$ for $|S| = 1$ and $4(2g)(3g) = 24g^2$ for $|S| = 2$. □

By trace-norm/operator-norm duality and unitary invariance, for all $t \in \mathbb{R}$,

$$\|\ddot{\rho}_e(t)\|_1 \le 24g^2, \qquad \|\ddot{\rho}_a(t)\|_1, \|\ddot{\rho}_b(t)\|_1 \le 8g^2, \qquad \|\dot{\rho}_a(t)\|_1, \|\dot{\rho}_b(t)\|_1 \le 2g.$$

5.3 One-Local and Incident Terms Are Tangent

Lemma 5.3 (local normal-rate isolation). For a product initial state $\rho(0) = \bigotimes_{v\in V} \rho_v$ and a one- and two-local Hamiltonian H, every term h_X supported on at most one endpoint of $e = \{a, b\}$ contributes a purely product-tangent operator to $L_e = \dot{\rho}_e(0)$. Consequently, the normal rate $n_e = Q_e L_e$ receives contributions only from interaction terms h_X whose support equals $\{a, b\}$.

Proof. A one-local term on a, or a two-local term on $\{a, c\}$ $(c \ne b)$, contributes after tracing out $V \setminus \{a, b\}$ an operator of the form $M_a \otimes \rho_b$ with $\operatorname{Tr} M_a = 0$, since $\operatorname{Tr}_c[h_{ac}, \rho_a \otimes \rho_c] = [\operatorname{Tr}_c(h_{ac}(I_a \otimes \rho_c)), \rho_a]$ acts only on a and has zero trace. The tangent projection P_e fixes $M_a \otimes \rho_b$:

$$P_e(M_a \otimes \rho_b) = (\operatorname{Tr}_b(M_a \otimes \rho_b)) \otimes \rho_b + \rho_a \otimes (\operatorname{Tr}_a(M_a \otimes \rho_b)) = M_a \otimes \rho_b.$$

By symmetry the same holds for terms meeting e only at b, and terms disjoint from e have zero partial trace over $V \setminus e$. Thus $Q_e = I - P_e$ annihilates all such contributions. □

In particular, one-local field terms (such as transverse or longitudinal fields in Ising models) never contribute to the first-order normal rate n_e, even when the initial single-site states are not field eigenstates; they enter the short-time bound only through the second-order remainder constant g.

5.3.1 Centered Interactions and the Pure-State Rate

Let h_{ab} denote the sum of the terms supported on both endpoints. Lemma 5.3 gives the explicit formula

$$n_e = -i(I - P_e)[h_{ab}, \rho_a \otimes \rho_b].$$

Choose a Hermitian product decomposition $h_{ab} = \sum_\alpha A_\alpha \otimes B_\alpha$, and put $A^\circ_\alpha = A_\alpha - \operatorname{Tr}(\rho_a A_\alpha) I$ and $B^\circ_\alpha = B_\alpha - \operatorname{Tr}(\rho_b B_\alpha) I$. Then

$$n_e = -i \left[\sum_\alpha A^\circ_\alpha \otimes B^\circ_\alpha, \rho_a \otimes \rho_b \right].$$

Indeed, tracing the uncentered commutator over either endpoint gives the corresponding mean-field commutator; subtracting the two product tangents gives exactly the centered expression. Both partial traces of the centered expression vanish. Raw commutators do not generally have that property. For example, $h_{ab} = Z \otimes Z$ and the state $|0+\rangle$ have a nonzero commutator but $n_e = 0$.

For pure endpoint states, write

$$u = \alpha \otimes \beta, \quad \chi = (I - |\alpha\rangle\langle\alpha|) \otimes (I - |\beta\rangle\langle\beta|)\, h_{ab} u.$$

The centered Hamiltonian applied to u is χ, which is orthogonal to u. Consequently

$$n_e = -i(|\chi\rangle\langle u| - |u\rangle\langle\chi|), \qquad \boxed{\nu_e = 2\|\chi\|.}$$

The two possible nonzero eigenvalues are $\pm\|\chi\|$. In the notation of the product decomposition,

$$\frac{\nu_e^2}{4} = \sum_{\alpha,\beta}(\langle A_\alpha A_\beta\rangle - \langle A_\alpha\rangle\langle A_\beta\rangle)(\langle B_\alpha B_\beta\rangle - \langle B_\alpha\rangle\langle B_\beta\rangle).$$

This is the correlated uncertainty in Yang's entanglement timescale [15], so $\nu_e = 2/T_{\text{ent}}$ for the isolated pure bipartite interaction, with $T_{\text{ent}} = \infty$ when the rate vanishes. Cresswell [16] relates the same quantity to Rényi entropy growth. Here the quantity gives a trace-norm rate with an explicit finite-time remainder, and the general theorem also permits mixed product states.

5.4 Exact Short-Time Local Defect Bound

Theorem 5.1 (Exact Finite-System Short-Time Lower Bound). Let $G = (V, E)$ be a finite graph and set $\hbar = 1$. Let $H = \sum_{1\le|X|\le2} h_X$ be a one- and two-local Hamiltonian whose two-site terms are supported on edges of G, with full incident interaction norm

$$g := \max_{v\in V}\sum_{X\ni v}\|h_X\|_\infty < \infty.$$

Let $\Pi_e(\sigma_{ab}) := \mathrm{Tr}_b(\sigma_{ab})\otimes\mathrm{Tr}_a(\sigma_{ab})$ be the product retraction, with $P_e := D\Pi_e|_{\rho_a\otimes\rho_b}$, $Q_e := I - P_e$, and $n_e := Q_e\dot\rho_{ab}(0)$. For every edge $e = \{a, b\}$ and every $t \ge 0$, under a global pure or mixed product initial state $\rho(0) = \bigotimes_{v\in V}\rho_v$, the two-site correlation defect $D_e(t) := \|\rho_{ab}(t) - \rho_a(t)\otimes\rho_b(t)\|_1$ satisfies

$$\boxed{D_e(t) \ge \nu_e t - 24g^2t^2,}$$

where $\nu_e = \|n_e\|_1$. In particular, if $\nu_e \ge \nu_{\min} > 0$, then

$$D_e(t) \ge \frac{\nu_{\min}}{2}t \qquad \text{for } 0 < t \le \frac{\nu_{\min}}{48g^2}.$$

Proof. Define the operator-valued correlation residual

$$F_e(t) := \rho_e(t) - \rho_a(t)\otimes\rho_b(t).$$

The product initial condition gives $F_e(0) = 0$, and differentiation of the product map gives

$$F_e'(0) = Q_eL_e = n_e.$$

Moreover,

$$F_e''(t) = \ddot\rho_e(t) - \ddot\rho_a(t)\otimes\rho_b(t) - 2\dot\rho_a(t)\otimes\dot\rho_b(t) - \rho_a(t)\otimes\ddot\rho_b(t).$$

Trace-norm multiplicativity on tensor products, unit trace of the density matrices, and the bounds of Lemma 5.2 imply

$$\|F_e''(t)\|_1 \le 24g^2 + 8g^2 + 2(2g)(2g) + 8g^2 = 48g^2.$$

Taylor's formula with integral remainder gives

$$F_e(t) = tn_e + \int_0^t (t-u)F_e''(u)\,du.$$

Therefore

$$D_e(t) = \|F_e(t)\|_1 \ge t\|n_e\|_1 - \frac{t^2}{2}\sup_{0\le u\le t}\|F_e''(u)\|_1 \ge \nu_e t - 24g^2t^2.$$

The half-slope interval follows immediately. □

Corollary 5.1.A (System-Size-Independent Family Bound). Consider a sequence of finite quantum lattice systems indexed by size N, with $H_N = \sum_{1\le|X|\le 2} h_{N,X}$, edge sets E_N, active edge subsets $A_N \subseteq E_N$, and full incident norms $g_N := \max_v \sum_{X\ni v} \|h_{N,X}\|_\infty$. Suppose there exist system-size-independent constants $0 < g_* < \infty$, $\eta_* > 0$, $\nu_* > 0$, and $a_* > 0$ such that

$$\sup_N g_N \le g_*, \qquad \inf_N \frac{|A_N|}{|E_N|} \ge \eta_* > 0, \qquad \inf_{N, e\in A_N} \nu_{N,e} \ge \nu_* > 0, \qquad \inf_N \frac{|E_N|}{N} \ge a_* > 0.$$

Then on the uniform near-zero time interval

$$0 < t \le \frac{\nu_*}{48 g_*^2},$$

the total edge-tally defect sum $C(t) = \sum_{e\in E_N} D_e(t)$ satisfies uniform site-density and edge-density bounds independent of N:

$$\frac{C(t)}{N} \ge a_*\eta_*(\nu_* t - 24 g_*^2 t^2) \ge \frac{a_*\eta_*\nu_*}{2} t, \qquad \frac{C(t)}{|E_N|} \ge \eta_*(\nu_* t - 24 g_*^2 t^2).$$

Remark 5.1.1 (Schatten norm conversion). On a two-site Hilbert space of dimension $D = d_a d_b$, converting from the trace norm $\|\cdot\|_1$ to a Schatten p-norm $\|\cdot\|_p$ ($p > 1$) via Hölder's inequality introduces a dimension factor $D^{-(1-1/p)}$ (for $d_a = d_b = d$, a factor of $d^{-2(1-1/p)}$, i.e., d^{-1} for the Hilbert–Schmidt norm $p = 2$ and d^{-2} for the operator norm $p = \infty$).

5.4.1 Open-system Lindbladian generators, parameter uncertainty, and 3-axis separation

Theorem 5.1 extends directly to open quantum systems governed by Markovian Lindblad master equations:

$$\dot\rho = \mathcal{L}(\rho) = -i[H,\rho] + \sum_a \gamma_a \mathcal{D}_{L_a}(\rho), \qquad \mathcal{D}_{L_a}(\rho) := L_a \rho L_a^\dagger - \frac{1}{2}\{L_a^\dagger L_a, \rho\}.$$

In the Heisenberg picture, the adjoint dissipator acts on any operator A_S supported on $S \subseteq V$ via the commutator identity

$$\mathcal{D}_{L_a}^\dagger(A_S) = L_a^\dagger A_S L_a - \frac{1}{2}\{L_a^\dagger L_a, A_S\} = \frac{1}{2} L_a^\dagger [A_S, L_a] + \frac{1}{2}[L_a^\dagger, A_S] L_a.$$

Consequently, $\mathcal{D}_{L_a}^\dagger(A_S) = 0$ whenever $\mathrm{supp}(L_a) \cap S = \emptyset$, and $\|\mathcal{D}_{L_a}^\dagger(A_S)\|_\infty \le 2\|L_a\|_\infty^2 \|A_S\|_\infty$ whenever $\mathrm{supp}(L_a) \cap S \ne \emptyset$. For local interaction supports $X \subset V$ with $|X| \le 2$, define the local Lindblad generator strength

$$\ell_X := 2\|h_X\|_\infty + 2 \sum_{a:\mathrm{supp}(L_a)=X} \gamma_a \|L_a\|_\infty^2, \qquad \Lambda_* := \max_{v\in V} \sum_{X\ni v} \ell_X.$$

When all Hamiltonian terms and Lindblad jump operators are at most 2-local ($|\mathrm{supp}(L_a)| \le 2$), with nonnegative rates γ_a, replacing $2g$ by Λ_* in the support-counting argument of Lemma 5.2 yields, for $\|A_S\|_\infty \le 1$, $\|\mathcal{L}^\dagger(A_S)\|_\infty \le |S|\Lambda_*$ and $\|(\mathcal{L}^\dagger)^2(A_S)\|_\infty \le 2\Lambda_*^2$ for $|S| = 1$ and $6\Lambda_*^2$ for

$|S| = 2$. Thus $\|F''_{ab}(t)\|_1 \le (6+2+2+2)\Lambda_*^2 = 12\Lambda_*^2$, and for each edge $e = \{a, b\}$ with initial normal rate $\nu_{ab} = \|Q_{ab}\mathcal{L}(\rho(0))_{ab}\|_1$, Taylor's theorem gives the volume-uniform open-system lower bound

$$D_{ab}(t) \ge \nu_{ab}t - 6\Lambda_*^2 t^2.$$

Under interval uncertainty $\nu_{ab} \in [\nu_{ab}^-, \nu_{ab}^+]$ and $\Lambda_* \in [\Lambda_*^-, \Lambda_*^+]$, monotonicity on $\mathbb{R}_{\ge 0}$ yields the worst-case enclosure

$$D_{ab}(t) \ge \nu_{ab}^- t - 6(\Lambda_*^+)^2 t^2.$$

This formulation distinguishes three physical mechanisms affecting two-site marginals: 1. *Coherent Hamiltonian dynamics:* reversible unitary evolution under $-i[H, \cdot]$, which generates first-order normal leakage ν_{ab} solely from two-site edge interactions; 2. *Dissipative Lindbladian dynamics:* completely positive trace-preserving semigroup evolution under jump operators L_a and rates γ_a, which can either damp correlations (via single-site decoherence) or drive dissipative state preparation and purification (for instance, single-qubit amplitude damping reduces the von Neumann entropy of $I/2$ from 1 bit to 0.68862 bits at $\gamma t = 1$); 3. *Classical statistical mixing:* convex combinations of density matrices or measurement-induced conditioning, which are static operations distinct from generator-driven evolution.

5.5 Edge-Normalized and Site-Normalized Density Corollaries

Define the active edge subset $A_{\nu_{\min}} := \{e \in E : \nu_e \ge \nu_{\min}\}$ with active fraction $|A_{\nu_{\min}}| \ge \eta|E|$, and let $C(t) := \sum_{e\in E} D_e(t)$ be the total edgewise correlation tally.

1. **Edge-Normalized Defect Density:**

$$\frac{C(t)}{|E|} \ge \eta(\nu_{\min}t - 24g^2t^2).$$

   For $0 < t \le \frac{\nu_{\min}}{48g^2}$, this yields $\frac{C(t)}{|E|} \ge \frac{\eta\nu_{\min}}{2}t$.
2. **Site-Normalized Defect Density:** If the graph family satisfies $|E| \ge aN$, then

$$\boxed{\frac{C(t)}{N} \ge a\eta(\nu_{\min}t - 24g^2t^2) \ge \frac{a\eta\nu_{\min}}{2}t}$$

   for $0 < t \le \frac{\nu_{\min}}{48g^2}$.

Remark 5.5.1 (Pairwise scope of the edge tally). The quantity $C(t) = \sum_{e\in E} D_e(t)$ is an overlapping sum of two-site product-retraction residuals. It is distinct from the global trace norm $\|\rho(t) - \bigotimes_{v\in V}\rho_v(t)\|_1$, from entanglement measures (as separable correlated states can have $D_e > 0$), from Bell nonlocality (§7.10), and from the cohomological quotient seminorm Φ.

5.6 Lipschitz Transfer Bound

When comparing two states ρ and σ under a retraction or projection map P that is L-Lipschitz with respect to a norm $\|\cdot\|$, the reverse triangle inequality gives the elementary transfer estimate

$$\|\rho - P(\rho)\| \ge \|\sigma - P(\sigma)\| - (1 + L)\|\rho - \sigma\|.$$

This abstract inequality is mechanized as `ProjectionTransfer.transfer` in `lean4/po000_l001_projection_transfer.lean` (along with its $K_\Pi = 1 + L = 3$ specialization `ProjectionTransfer.transfer_product_KPi_three` for the two-site product retraction, which has Lipschitz constant $L = 2$ on density operators by telescoping $\rho_a\otimes\rho_b - \sigma_a\otimes\sigma_b = (\rho_a - \sigma_a)\otimes\rho_b + \sigma_a\otimes(\rho_b - \sigma_b)$). Notice that Theorem 5.1 above differentiates the moving product residual $F_e(t) = \rho_e(t) - \rho_a(t)\otimes\rho_b(t)$ directly and therefore does not need this Lipschitz transfer step.

5.7 Non-Firing and Boundary Cases

We record seven boundary regimes where the first-order lower bound of Theorem 5.1 vanishes or becomes uninformative:

1. **Tangent Initial Directions ($\nu_e = 0$):** If $n_e = Q_e L_e = 0$, the first-order normal rate vanishes. The defect starts at $\mathcal{O}(t^2)$ and Theorem 5.1 gives the trivial bound $D_e(t) \geq -24g^2t^2$.
2. **Commutative or Product-Preserving Dynamics:** If $[h_{\{a,b\}}, \rho_a \otimes \rho_b] = 0$, no normal product defect is generated at first order on edge $\{a, b\}$.
3. **Sparse Active Edges:** If $|A_N|/|E_N| \to 0$, the active-set sum yields no positive uniform edge-normalized bound; if $|A_N|/N \to 0$, it yields no positive uniform site-normalized bound.
4. **Vanishing Rate-to-Curvature Ratio:** The guaranteed half-slope window shrinks to zero when $\nu_{\min}/g^2 \to 0$, including $\nu_{\min} \to 0$ at fixed $g > 0$. A vanishing rate alone does not decide this if the interactions also weaken.
5. **Growing Incident Strength:** If the full incident norm g grows while $\nu_{\min}$ stays bounded, the guaranteed window shrinks. Unbounded degree alone does not imply this: edge strengths can decrease so that their incident sums remain bounded.
6. **Higher-Order Defect Onset:** When $\nu_e = 0$ because of initial-state alignment, correlation production begins at order t^2 or higher; Section 5.8 provides higher-order Taylor bounds and §6.3 gives explicit quadratic and cubic onset examples.
7. **Purely Multipartite Correlations:** Because $C(t)$ sums two-site defects $D_e(t)$, it vanishes on states whose two-site marginals are all factorized even when k-body connected cumulants ($k \geq 3$) are nonzero (§5.12–§5.17).

5.8 Higher-Order Short-Time Bounds

The first-order estimate can be sharpened without changing the hypotheses of Theorem 5.1. Put $F_e(t) = \rho_e(t) - \rho_a(t) \otimes \rho_b(t)$, as above. All derivatives below are derivatives of this full residual, including both moving marginals.

Lemma 5.4 (uniform derivative bounds). For a nonempty set S of k sites and any integer $j \geq 0$,

$$\|\rho_S^{(j)}(t)\|_1 \leq (2g)^j k(k+1)\cdots(k+j-1)$$

for all real t, with empty product one. In particular,

$$\boxed{\|F_e^{(j)}(t)\|_1 \leq 2^{j+1}(j+1)!\, g^j.}$$

Proof. Expand $\mathrm{ad}_H^j(A_S)$ as a sum of nested commutators. At step $r+1$, a surviving term has support on at most $k+r$ sites, so the sum of norms of Hamiltonian terms meeting that support is at most $(k+r)g$. Each commutator contributes a factor two. This bound is applied to each branch of the expansion before summing; the full commutator need not have a single small support. The identity

$$\mathrm{Tr}(A_S \rho_S^{(j)}(t)) = i^j \,\mathrm{Tr}(\mathrm{ad}_H^j(A_S)\rho(t))$$

and trace-norm/operator-norm duality give the first assertion. For the product of the two one-site

marginals, the Leibniz rule gives

$$\left\|\frac{d^j}{dt^j}(\rho_a \otimes \rho_b)\right\|_1 \le \sum_{m=0}^{j} \binom{j}{m} (2g)^m m!\,(2g)^{j-m}(j-m)! = (2g)^j (j+1)!.$$

Adding the two-site bound gives the stated estimate. □

Theorem 5.5 (finite Taylor approximation). Define

$$P_{e,J}(t) = \sum_{j=1}^{J} \frac{t^j}{j!} F_e^{(j)}(0), \qquad R_J(t) = 2^{J+2}(J+2) g^{J+1} t^{J+1}.$$

For every integer $J \ge 1$ and every $t \ge 0$,

$$\|F_e(t) - P_{e,J}(t)\|_1 \le R_J(t), \qquad \boxed{\left|D_e(t) - \|P_{e,J}(t)\|_1\right| \le R_J(t).}$$

Consequently, for any Hermitian W with $\|W\|_\infty \le 1$,

$$D_e(t) \ge \sum_{j=1}^{J} \frac{t^j}{j!} \operatorname{Tr}(W F_e^{(j)}(0)) - R_J(t).$$

Proof. Taylor's integral remainder and Lemma 5.4 give

$$\|F_e(t) - P_{e,J}(t)\|_1 \le \frac{t^{J+1}}{(J+1)!} 2^{J+2}(J+2)!\, g^{J+1} = R_J(t).$$

Apply the reverse triangle inequality for the second conclusion and trace-norm duality for the witness inequality. □

Taking $J = 1$ gives the two-sided refinement of Theorem 5.1:

$$|D_e(t) - \nu_e t| \le 24 g^2 t^2, \qquad \lim_{t \downarrow 0} \frac{D_e(t)}{t} = \nu_e.$$

In particular, $\nu_e = 0$ implies $D_e(t) \le 24 g^2 t^2$.

More generally, if $m \ge 1$ is the first index with $A_m := F_e^{(m)}(0) \ne 0$, then

$$\left| D_e(t) - \frac{\|A_m\|_1}{m!} t^m \right| \le 2^{m+2}(m+2) g^{m+1} t^{m+1}, \qquad \lim_{t \downarrow 0} \frac{D_e(t)}{t^m} = \frac{\|A_m\|_1}{m!}.$$

For $g > 0$, a half-leading-term bound holds on

$$0 < t \le \frac{\|A_m\|_1}{2^{m+3}(m+2) m!\, g^{m+1}}.$$

For example, if $F_e'(0) = 0$ and $\mu_e = \|F_e''(0)\|_1 > 0$,

$$D_e(t) \ge \frac{\mu_e}{2} t^2 - 64 g^3 t^3 \ge \frac{\mu_e}{4} t^2 \quad \text{when} \quad 0 < t \le \frac{\mu_e}{256 g^3}.$$

The $g = 0$ case has constant product evolution and zero defect. A family version follows by replacing g by a uniform upper bound and $\|A_m\|_1/m!$ by a positive uniform lower bound on a declared positive density of edges, all with the same first nonzero order m.

These are bounds on the polynomial as a whole; different coefficients can cancel. Computing more derivatives does not by itself promise a longer interval on which the lower bound is positive. The constants also show convergence: the remainder is $2(J+2)(2gt)^{J+1}$, which tends to zero when $2g|t| < 1$. Each fixed finite system actually has an entire matrix-valued residual, since matrix exponentials and products are entire. Thus vanishing of every derivative at zero is equivalent to $F_e(t) = 0$ for all real t. The norm D_e itself need not be analytic. Section 5.16 replaces the all-derivatives test by a finite certificate for each fixed polynomial model. Neither assertion establishes an infinite-volume limit.

Schatten norms. For $1 \le p \le \infty$, the same remainder satisfies

$$\big|\|F_e(t)\|_p - \|P_{e,J}(t)\|_p\big| \le \|F_e(t) - P_{e,J}(t)\|_p \le R_J(t).$$

The first-order rate is then $\|n_e\|_p$. No dimension factor is needed when this rate is computed in the chosen norm. Converting only a known trace-norm rate to another norm still incurs the dimension factor stated in §5.4.

Proposition 5.6 (locality of the Taylor coefficients). Under the product initial condition and two-site interaction assumption, $F_e^{(j)}(0)$, for $j \ge 1$, is unchanged when the system is restricted to the induced subgraph on sites at graph distance at most $j-1$ from $\{a, b\}$. The restricted Hamiltonian keeps exactly those terms supported there, and the initial state is the corresponding product.

Proof. Give each Hamiltonian term its own scalar parameter. The j-th derivative of the residual is a homogeneous polynomial of degree j in these parameters. Consider one of its monomials and set all parameters absent from that monomial to zero. If its interaction supports do not connect a to b, the reduced state remains a product and that monomial's coefficient vanishes. If an interaction component does not meet a or b, its evolution disappears under the partial trace; the residual is independent of its parameters, so a monomial containing such a parameter also has zero coefficient. Hence every surviving monomial uses a connected interaction graph containing both endpoints, with at most j edges (one-site terms add no edges). Any vertex in such a graph is at distance at most $j-1$ from the endpoint set: connecting both endpoints requires at least one edge in addition to a shortest path from that vertex to the nearer endpoint. Every contributing term and initial factor therefore lies in the declared neighborhood. □

This gives a finite local computation for each fixed order. Its cost can grow exponentially with the neighborhood size; it is not a polynomial-time algorithm in unrestricted order, degree or local dimension.

5.9 Finite Regions and Larger Interaction Supports

The second-order estimate also extends to an explicitly changed interaction assumption. Let every Hamiltonian term have support of size at most r, let $g = \max_v \sum_{X \ni v} \|h_X\|_\infty$, and let S contain $k \ge 1$ sites. For a global product initial state, define

$$F_S(t) = \rho_S(t) - \bigotimes_{v \in S} \rho_v(t), \qquad \nu_S = \|F_S'(0)\|_1.$$

Then, for $t \ge 0$,

$$\boxed{\big|\|F_S(t)\|_1 - \nu_S t\big| \le 4k(k+r-1)g^2t^2.}$$

Proof. The first commutator has incident norm sum at most kg; each branch then has at most $k + r - 1$ sites. Thus $\|\rho_S''(t)\|_1 \le 4k(k + r - 1)g^2$. Each one-site second derivative is bounded by $4rg^2$, while each one-site first derivative is bounded by $2g$. The second derivative of the product has k second-derivative terms and $k(k-1)$ ordered cross terms, giving

$$4krg^2 + 4k(k-1)g^2 = 4k(k + r - 1)g^2.$$

The sum of the two bounds, followed by the integral remainder factor $1/2$, gives the result. □

Only Hamiltonian terms meeting S in at least two sites can contribute to $F_S'(0)$; terms meeting a single site give a product tangent. For $r = 2$ these are internal edges. For $r > 2$ a contributing term can also meet the complement, so its contribution may depend on initial states outside S. For $k = 1$, the residual is identically zero.

5.10 The Infinite-Volume Dynamics and a Defect-Density Limit

Here a thermodynamic limit means convergence of finite-volume evolution on every fixed local observable, uniformly on compact time intervals. It does not mean convergence of a global density matrix in trace norm. The construction below uses standard quantum-spin dynamics on the norm-completed algebra of local observables; existence results in this setting are established in the literature, including [9, Theorem 2.2]. The purpose here is to connect the elementary derivative estimate to an all-time limit and to the particular defect studied in this paper.

Assume a fixed countable, locally finite graph G, a finite-dimensional Hilbert space at each vertex, fixed Hermitian one-site and edge terms, and

$$g_* := \sup_v \sum_{X \ni v} \|h_X\|_\infty < \infty.$$

Let $\mathcal{A}$ be the operator-norm completion of the union of the finite tensor-product observable algebras, embedded by tensoring with identities. For finite $\Lambda \subset G$, put

$$H_\Lambda = \sum_{X \subseteq \Lambda} h_X, \qquad \alpha_t^\Lambda(A) = e^{itH_\Lambda} A e^{-itH_\Lambda}.$$

These are isometric automorphisms of the same algebra $\mathcal{A}$, including when A is not supported inside Λ.

Theorem 5.7 (all-time local thermodynamic limit). There is a strongly continuous one-parameter group of isometric $*$-automorphisms α_t of $\mathcal{A}$ such that, for every $A \in \mathcal{A}$ and $T < \infty$,

$$\lim_{\Lambda \uparrow G} \sup_{|t| \le T} \|\alpha_t^\Lambda(A) - \alpha_t(A)\|_\infty = 0.$$

The limit is independent of the finite exhaustion. Every initial state ω on $\mathcal{A}$ therefore has a limiting evolution $\omega_t = \omega \circ \alpha_t$; its finite-region density matrices are the trace-norm limits of those obtained by evolving $\omega|_{\mathcal{A}_\Lambda}$. These local limits are uniform for $|t| \le T$. No product-state assumption is needed for existence.

Proof. The case $g_* = 0$ is constant evolution. Suppose $g_* > 0$. For a local A supported on $k \ge 1$ vertices, the branchwise commutator count of Lemma 5.4 gives

$$\|(\mathrm{ad}_{H_\Lambda})^j A\|_\infty \le \|A\|_\infty (2g_*)^j k(k+1)\cdots(k+j-1).$$

Each nonzero branch adds at most one vertex at each commutator. Consequently, the order-j coefficient is independent of Λ once it contains the radius-j neighborhood of $\operatorname{supp} A$. That neighborhood is finite by local finiteness. Unitary conjugation preserves the norm of every derivative, so Taylor's integral remainder gives, for $0 \le \tau < 1/(2g_*)$,

$$\sup_{|t|\le\tau} \left\| \alpha_t^\Lambda(A) - \sum_{j=0}^{J} \frac{(it)^j}{j!} (\mathrm{ad}_{H_\Lambda})^j A \right\|_\infty \le \|A\|_\infty \binom{k+J}{J+1} (2g_*\tau)^{J+1}. \tag{5.10a}$$

For fixed k the right side tends to zero. Coefficient stabilization therefore makes the finite-volume evolutions uniformly Cauchy on this interval, with the same limit for every exhaustion. In fact the Cauchy estimate holds for all finite volumes containing a sufficiently large fixed neighborhood, not only for a selected sequence.

For $|t| < 1/(2g_*)$, take this limit on local observables. The limits preserve norms, products, adjoints and the identity, so they extend to isometric unital $*$-endomorphisms on $\mathcal{A}$. Approximating any $B \in \mathcal{A}$ by a local observable extends the uniform-in-t convergence to B: the two approximation errors are at most $2\|B - A\|_\infty$. It also gives norm continuity in t for each B.

For $s, t, s+t$ in this short interval, finite-volume composition passes to the limit because

$$\begin{aligned} \|\alpha_s^\Lambda(\alpha_t^\Lambda B) - \alpha_s(\alpha_t B)\|_\infty \le{}& \|\alpha_t^\Lambda B - \alpha_t B\|_\infty \\ &+ \|\alpha_s^\Lambda(\alpha_t B) - \alpha_s(\alpha_t B)\|_\infty \longrightarrow 0. \end{aligned}$$

Thus $\alpha_s\alpha_t = \alpha_{s+t}$ there. Taking $s = -t$ shows that these maps are automorphisms.

For an arbitrary real t, choose an integer n with $|t|/n < 1/(2g_*)$ and define $\alpha_t = (\alpha_{t/n})^n$. A common integer refinement of two subdivisions and the short-interval composition law show that this definition is independent of n. For $s, t \in \mathbb{R}$, choose n so that $s/n, t/n, (s+t)/n$ lie in the short interval. Their short-time maps commute and multiply to $\alpha_{(s+t)/n}$; taking n-th powers gives the group law. Continuity near zero and the group law give strong continuity for every real time.

It remains to show convergence of the original finite-volume evolution, rather than just to define a candidate extension. Fix $T < \infty$ and n with $\tau = T/n < 1/(2g_*)$. A telescoping identity and isometry give, with $u = t/n$,

$$\|\alpha_t^\Lambda B - \alpha_t B\|_\infty \le \sum_{\ell=0}^{n-1} \|(\alpha_u^\Lambda - \alpha_u)(\alpha_u^\ell B)\|_\infty. \tag{5.10b}$$

The set $\{\alpha_u^\ell B : |u| \le \tau,\ 0 \le \ell < n\}$ is compact, by norm continuity. Short-time convergence is uniform on this compact set: take a finite norm net and use the operator-norm bound $\|\alpha_u^\Lambda - \alpha_u\| \le 2$. The right side of (5.10b) therefore tends to zero uniformly for $|t| \le T$, for all sufficiently large finite volumes.

For a state ω, applying it to the observable convergence gives convergence of each matrix coefficient of every finite-region density matrix. All norms on that fixed finite-dimensional matrix space are equivalent, yielding uniform trace-norm convergence. Positivity, unit trace and consistency under partial trace pass to the limit. □

Product-state defect estimates in the limit. For fixed compatible product initial data and an edge $e = \{a, b\}$, put

$$F_e^\infty(t) = \rho_e^\infty(t) - \rho_a^\infty(t) \otimes \rho_b^\infty(t).$$

Every finite-order estimate in §5.8 passes to the limit. In particular, for $J \geq 1$ and every real t,

$$\|F_e^\infty(t) - P_{e,J}(t)\|_1 \leq 2(J+2)(2g_*|t|)^{J+1}. \tag{5.10c}$$

The coefficients are the stabilized finite-volume derivatives. For $2g_*|t| < 1$ the remainder tends to zero with J, so the identification with the derivatives of F_e^∞ at zero follows from its convergent power series. For any finite Λ containing the radius-$(J-1)$ neighborhood of e,

$$\|F_e^\Lambda(t) - F_e^\infty(t)\|_1 \leq 4(J+2)(2g_*|t|)^{J+1}. \tag{5.10d}$$

This explicit estimate is useful near zero; all-time convergence comes from (5.10b), not from extending the convergence radius of (5.10a).

An actual limit of the spatially averaged diagnostic needs additional spatial assumptions. The following elementary case complements the uniform family lower bounds in §5.5.

Corollary 5.8 (translation-invariant defect density). Suppose $G = \mathbb{Z}^d$ with nearest-neighbor edges, the on-site spaces, interactions and product initial state are translation invariant, and $\Lambda_L = \{-L, \dots, L\}^d$ has open boundaries. Write E_L for its internal nearest-neighbor edges and

$$d_L(t) = \frac{1}{|E_L|} \sum_{e \in E_L} D_e^{\Lambda_L}(t), \qquad D_r^\infty(t) = \|F_{\{0,\mathbf{e}_r\}}^\infty(t)\|_1.$$

Then, uniformly on every compact time interval,

$$d_L(t) \longrightarrow d_\infty(t) := \frac{1}{d} \sum_{r=1}^{d} D_r^\infty(t).$$

If $\bar{\nu} = d^{-1} \sum_{r=1}^{d} \nu_{\{0,\mathbf{e}_r\}}$, then

$$d_\infty(t) \geq \bar{\nu}\, t - 24 g_*^2 t^2 \qquad (t \geq 0).$$

In particular $\bar{\nu} > 0$ gives a strictly positive limiting defect density for $0 < t < \bar{\nu}/(24g_*^2)$.

Proof. Translating finite volumes and using exhaustion independence shows that the limiting dynamics, and hence the product-state evolution, is translation invariant. Given T and $\varepsilon > 0$, Theorem 5.7 gives a radius R such that every edge whose radius-R neighborhood is contained in Λ_L has defect within ε of the corresponding D_r^∞, uniformly for $|t| \leq T$. To obtain one R, use a finite matrix basis for each of the d edge orientations and the net convergence proved above; tensor-product continuity transfers the estimate to F_e. The fraction of remaining edges tends to zero as $L \to \infty$, and both finite and limiting defects lie in $[0, 2]$. Each orientation has exactly the same number of internal edges in the cube. Averaging establishes the limit. Averaging the limiting version of Theorem 5.1 establishes the lower bound. □

Theorem 5.7 supplies the all-time dynamical thermodynamic limit under the stated bounded-interaction assumptions. Corollary 5.8 additionally supplies a spatial density limit under translation invariance. Neither establishes an equilibrium free-energy limit, selection or uniqueness of a thermodynamic phase, a continuum scaling limit, or nonzero defect at arbitrarily late times.

5.10.1 Locality-controlled Duhamel bounds and numerical enclosures

To turn defect diagnostics into certified predictive bounds for physical observables, consider the expectation $\langle O\rangle_{\rho(t)}$ of an observable O along the true trajectory $\rho(t)$ compared with a reduced or model trajectory $\tilde{\rho}(t)$. The variation-of-constants formula rigorously distinguishes three separate diagnostic quantities: 1. *Equation residual:* $R(s) := \dot{\tilde{\rho}}(s) - \mathcal{L}\tilde{\rho}(s)$, which measures the failure of the model trajectory to satisfy the true physical master equation; 2. *Normal velocity defect:* $\nu_m := \mathrm{dist}_1(\mathcal{L}m, T_m\mathcal{M})$, the instantaneous rate of escaping the model manifold $\mathcal{M}$; 3. *State defect:* $D(m) := \mathrm{dist}_1(m, \mathcal{M})$, the instantaneous trace-norm distance to the model manifold.

Integrating the Heisenberg operator evolution or the Schrödinger equation residual yields the unweighted global Duhamel bound:

$$|\langle O\rangle_{\rho(t)} - \langle O\rangle_{\tilde{\rho}(t)}| \le \|O\|_\infty \int_0^t \|\mathcal{L}\tilde{\rho}(s) - \dot{\tilde{\rho}}(s)\|_1 \, ds + E_{\text{initial}} + E_{\text{numerical}}.$$

By trace-operator norm Hölder duality, $|\operatorname{Tr}(O\Delta\rho)| \le \|O\|_\infty \|\Delta\rho\|_1$. The factor of 2 in conventional literature appears only when converting raw trace norm $\|\cdot\|_1$ to trace distance $T = \frac{1}{2}\|\cdot\|_1$, since $|\operatorname{Tr}(O\Delta\rho)| \le 2\|O\|_\infty T(\rho,\tilde{\rho}) = \|O\|_\infty \|\Delta\rho\|_1$.

For a local observable O supported on a region X, the spatial influence of an interaction term on edge e is controlled by the locality weights:

$$w_{O,e}(\tau) \le \min\left(\|O\|_\infty,\ C_0\|O\|_\infty e^{v_{\text{LR}}\tau - \mu d(e,X)}\right),$$

where v_{LR} is the Lieb–Robinson velocity and $\mu > 0$ is the spatial decay rate. Notice that the operator-norm upper cap is $\|O\|_\infty$; omitting a redundant factor of $\|h_e\|_\infty$ is essential to prevent underestimating physical prediction errors in weakly coupled systems. Indeed, in a two-qubit entangling pair with $H = \varepsilon X \otimes X$ ($\varepsilon = 0.1$) starting from $\rho(0) = |00\rangle\langle 00|$ over duration $t = 0.1$, the exact physical prediction error for $O = Y \otimes X$ is $|\sin(2\varepsilon t)| = |\sin(0.02)| \approx 0.01999866... \le 0.02$. The equation residual has trace norm $\|\mathcal{L}\tilde{\rho}(s)\|_1 = 2\varepsilon = 0.2$, so $\int_0^t \|\mathcal{L}\tilde{\rho}(s)\|_1 \, ds = 2\varepsilon t = 0.02$, which is enclosed by the corrected unweighted bound $\|O\|_\infty(2\varepsilon t) = 0.02$.

In computational simulations, initial state uncertainty $\delta\rho_0$ and time-integration discretization errors $\delta_{\text{step},k}$ are enclosed by:

$$E_{\text{model}} = \|O\|_\infty \int_0^t \|\mathcal{L}\tilde{\rho}(s) - \dot{\tilde{\rho}}(s)\|_1 \, ds, \qquad E_{\text{total}} = E_{\text{model}} + E_{\text{initial}} + E_{\text{numerical}},$$

where $E_{\text{initial}} = \|O\|_\infty \|\delta\rho_0\|_1$ and $E_{\text{numerical}}$ must bound the propagated integration, quadrature and rounding errors in the prediction and residual integral. Certified upper bounds for these three terms enclose the exact physical expectation in $[y_{\text{pred}} - E_{\text{total}}, y_{\text{pred}} + E_{\text{total}}]$. The model term cannot be dropped even with exact initialization and integration: the stationary weak-coupling example above has nonzero error. Sampled trapezoidal quadrature without a proved remainder and freely chosen light-cone constants produce numerical estimates, not certified enclosures.

5.10.2 The Edge-Decomposed TDVP Duhamel Master Theorem

Global and local-in-time error bounds for variational quantum dynamics via standard Duhamel variation of constants are established in Martinazzo and Burghardt [10] and Zoufal, Sutter, and Woerner [11]. Theorem 5.10.A contributes the exact local decomposition of the generator residual

into partial-trace-annihilating edge normal operators and couples them with incident Lieb–Robinson spatial suppression weights.

Theorem 5.10.A (Edge-Decomposed TDVP Duhamel Master Theorem). Let $G = (V, E)$ be a finite graph with finite-dimensional local Hilbert spaces $\mathcal{H}_v$ ($v \in V$) and Hamiltonian $H = \sum_{v\in V} h_v + \sum_{e=\{u,w\}\in E} h_e$, where each h_v is 1-local and each h_e is 2-local on edge e. Work in the real affine space $\mathcal{A}_1 = \{X = X^\dagger : \operatorname{Tr} X = 1\}$ with variation space $\mathcal{A}_0 = \{Z = Z^\dagger : \operatorname{Tr} Z = 0\}$. The map $\Pi(X) = \bigotimes_v \operatorname{Tr}_{V\setminus v} X$ retracts $\mathcal{A}_1$ smoothly onto the products of Hermitian trace-one factors. Write $\mathcal{M}$ for the positive product subset and $T_m^{\mathrm{aff}}\mathcal{M}$ for the tangent of this smooth affine product extension. On $\mathcal{A}_0$, its derivative

$$P_m Z = \sum_v \rho_{<v} \otimes \operatorname{Tr}_{V\setminus v} Z \otimes \rho_{>v}$$

is a projection onto $T_m^{\mathrm{aff}}\mathcal{M}$; set $Q_m = I - P_m$ and $N_m = \ker P_m$. At full rank this is the physical product-manifold tangent. At rank boundaries it is a linear extension, not the feasible tangent cone of §5.11.1. These domain restrictions matter: differentiating the same unnormalized marginal-product formula on all operators gives $D\Pi_m(m) = |V|m$, which is not an idempotent tangent projection for $|V| > 1$. Let $\tilde\rho(t) = \bigotimes_{v\in V} \rho_v(t) \in \mathcal{M}$ be a product trajectory evolving under the projected (time-dependent variational principle / mean-field) generator:

$$\dot{\tilde\rho}(s) = P_{\tilde\rho(s)}(\mathcal{L}\tilde\rho(s)) = -iP_{\tilde\rho(s)}([H, \tilde\rho(s)]).$$

Then: 1. *Residual Locality & Single-Site Vanishing:* The Schrödinger equation residual $R(s) := \mathcal{L}\tilde\rho(s) - \dot{\tilde\rho}(s) = Q_{\tilde\rho(s)}(\mathcal{L}\tilde\rho(s))$ vanishes identically on all 1-local Hamiltonian terms $\sum_v h_v$, and decomposes into an exact sum of partial-trace-free 2-site edge normal contributions:

$$R(s) = \sum_{e=\{u,w\}\in E} n_e(s) \otimes \tilde\rho_{V\setminus e}(s),$$

where each edge normal defect operator

$$n_e(s) = Q_{e,\tilde\rho_e(s)}(-i[h_e, \tilde\rho_e(s)])$$

is supported strictly on edge e and satisfies the partial-trace annihilation identities

$$\operatorname{Tr}_u n_e(s) = 0, \qquad \operatorname{Tr}_w n_e(s) = 0.$$

(In particular, when $\tilde\rho(s)$ is full-rank, distinct edge summands $n_e(s) \otimes \tilde\rho_{V\setminus e}(s)$ are orthogonal with respect to the $\tilde\rho(s)^{-1}$-weighted inner product $\langle A, B\rangle_{\tilde\rho(s)^{-1}} = \operatorname{Tr}(A\tilde\rho(s)^{-1}B)$, and for pure product states they are Hilbert–Schmidt orthogonal as in Proposition 5.11.) 2. *Edgewise Weighted Duhamel Bound:* Suppose the Heisenberg evolution $\tau_\tau(O) = e^{i\tau H}Oe^{-i\tau H}$ of a local observable O supported on $X \subset V$ has a certified nonnegative residual bound $b_{O,e}(\tau)$ satisfying

$$\left\|\tau_\tau(O) - \mathbb{E}_{V\setminus e}(\tau_\tau(O)) \otimes I_e\right\|_\infty \le b_{O,e}(\tau), \qquad w_{O,e}(\tau) = \min\{\|O\|_\infty, b_{O,e}(\tau)\},$$

where $\mathbb{E}_{V\setminus e} : \mathfrak{B}(\mathcal{H}_V) \to \mathfrak{B}(\mathcal{H}_{V\setminus e})$ is the normalized partial trace over $e = \{u, w\}$. A proved Lieb–Robinson estimate may supply $b_{O,e}(\tau) = C_0\|O\|_\infty e^{v_{\mathrm{LR}}\tau - \mu d(e,X)}$; the constants must apply to the specified Hamiltonian and geometry. The residual norm itself need not be at most $\|O\|_\infty$: for $O = \operatorname{diag}(1, 1, 1, -1)$ and $e = V$, its normalized-trace residual at time zero has norm $3/2$, whereas

$\|O\|_\infty = 1$. The minimum bounds the trace pairing, as proved below. Then the expectation error between $\rho(t) = e^{t\mathcal{L}}\rho(0)$ and the TDVP trajectory satisfies:

$$\left|\langle O\rangle_{\rho(t)} - \langle O\rangle_{\tilde\rho(t)}\right| \le \|O\|_\infty \|\delta\rho_0\|_1 + \sum_{e\in E}\int_0^t w_{O,e}(t-s)\nu_e(s)\,ds,$$

where $\nu_e(s) = \|n_e(s)\|_1$ is the instantaneous edge normal leakage rate.

Proof. First, let $H_1 = \sum_{v\in V} h_v$. Then $-i[H_1, \tilde\rho(s)] = \sum_{v\in V} \tilde\rho_{<v} \otimes (-i[h_v, \rho_v(s)]) \otimes \tilde\rho_{>v}$. Each $-i[h_v, \rho_v(s)]$ is Hermitian and traceless and is the velocity of a local unitary orbit, including at rank boundaries. Thus $-i[H_1, \tilde\rho(s)] \in T^{\mathrm{aff}}_{\tilde\rho(s)}\mathcal{M}$, and $Q_{\tilde\rho(s)}(-i[H_1, \tilde\rho(s)]) = 0$.

Next, for each 2-local term $h_e = h_{\{u,w\}}$, $-i[h_e, \tilde\rho(s)] = (-i[h_e, \tilde\rho_e(s)]) \otimes \tilde\rho_{V\setminus e}(s)$. For this traceless edge operator $L = -i[h_e, \tilde\rho_e(s)]$, the sites outside e have zero marginal velocity. Consequently

$$P_{\tilde\rho(s)}(L \otimes \tilde\rho_{V\setminus e}(s)) = (\mathrm{Tr}_w L \otimes \rho_w + \rho_u \otimes \mathrm{Tr}_u L) \otimes \tilde\rho_{V\setminus e}(s).$$

The parentheses are $P_{e,\tilde\rho_e(s)}L$ on the Hermitian trace-zero edge space. This is an identity on the displayed edge-supported variations, not an operator factorization of the full projector on arbitrary variations. Consequently, $Q_{\tilde\rho(s)}(-i[h_e, \tilde\rho(s)]) = n_e(s) \otimes \tilde\rho_{V\setminus e}(s)$ with $n_e(s) = (I - P_{e,\tilde\rho_e})(-i[h_e, \tilde\rho_e(s)])$. Since $\mathrm{Tr}_w(P_{e,\tilde\rho_e}L) = \mathrm{Tr}_w L$, the normal operator satisfies $\mathrm{Tr}_w n_e(s) = \mathrm{Tr}_w(I - P_{e,\tilde\rho_e})L = 0$ and similarly $\mathrm{Tr}_u n_e(s) = 0$. Summing over all edges $e \in E$ gives $R(s) = \sum_{e\in E} n_e(s) \otimes \tilde\rho_{V\setminus e}(s)$.

Finally, substituting this decomposition into the variation-of-constants formula

$$\rho(t) - \tilde\rho(t) = e^{t\mathcal{L}}(\rho_0 - \tilde\rho_0) + \int_0^t e^{(t-s)\mathcal{L}} R(s)\,ds$$

and taking the trace pairing with O gives, for each edge e and lag $\tau = t - s$, the Heisenberg pairing $\mathrm{Tr}(\tau_\tau(O)(n_e(s) \otimes \tilde\rho_{V\setminus e}(s)))$. Because $\mathrm{Tr}_e n_e(s) = \mathrm{Tr}_u(\mathrm{Tr}_w n_e(s)) = 0$, any operator of the form $B_{V\setminus e} \otimes I_e$ has zero trace against $n_e(s) \otimes \tilde\rho_{V\setminus e}(s)$:

$$\mathrm{Tr}((B_{V\setminus e} \otimes I_e)(n_e(s) \otimes \tilde\rho_{V\setminus e}(s))) = \mathrm{Tr}_{V\setminus e}(B_{V\setminus e}\tilde\rho_{V\setminus e}(s))\,\mathrm{Tr}_e(n_e(s)) = 0.$$

Choosing $B_{V\setminus e} = \mathbb{E}_{V\setminus e}(\tau_\tau(O))$ and applying trace–operator Hölder duality yields

$$\begin{aligned}\left|\mathrm{Tr}(\tau_\tau(O)(n_e(s) \otimes \tilde\rho_{V\setminus e}(s)))\right| &= \left|\mathrm{Tr}((\tau_\tau(O) - \mathbb{E}_{V\setminus e}(\tau_\tau(O)) \otimes I_e)(n_e(s) \otimes \tilde\rho_{V\setminus e}(s)))\right| \\ &\le \left\|\tau_\tau(O) - \mathbb{E}_{V\setminus e}(\tau_\tau(O)) \otimes I_e\right\|_\infty \|n_e(s)\|_1 \|\tilde\rho_{V\setminus e}(s)\|_1 \\ &\le b_{O,e}(\tau)\,\nu_e(s).\end{aligned}$$

Direct Hölder duality also bounds the same pairing by $\|\tau_\tau(O)\|_\infty \nu_e(s) = \|O\|_\infty \nu_e(s)$. Taking the minimum of these two bounds gives $w_{O,e}(\tau)\nu_e(s)$. Integrating over $s \in [0,t]$ and summing over $e \in E$ yields the claim. □

Numerical comparison on the declared eight-site cluster. Theorem 5.10.A is compared with floating-point Krylov evolution (`scipy.sparse.linalg.expm_multiply`) on an 8-qubit cluster of a hexagon with an attached square in `code/tests/test_honeycomb_tdvp_duhamel_integration.py`: 1. For the central bond observable $O = Z_2Z_3$, the test compares the numerically evolved expectation with the mean-field prediction and a sampled unweighted residual integral on its declared time grid. 2. For the boundary observable $O = Z_0$ with longitudinal field $h_z = 0.2$, the test evaluates an exponentially weighted integral and compares it with the numerical expectation error at the sampled times.

These sampled comparisons illustrate the bound on a finite cluster. Companion package **PO-033** (*Certified Adaptive Quantum Dynamics from Projection Residuals*, v1.0.1) analyzes conditional analytic bounds and supplies an adaptive cluster prototype. However, as documented in its release manifest, that prototype is uncertified (heuristic_uncertified / `HEURISTIC_UNCERTIFIED`): it samples ideal tangent residuals at recorded times without bounding continuous-time integration, numerical quadrature, or floating-point error, and its formal Lean 4 mechanization formalizes conditional finite-sum inequalities over an abstract discrete carrier (`Nat`/`Int`), not continuous-time density-matrix dynamics. Validated continuous-time ODE integration and conservative quadrature enclosures on honeycomb lattices or general graphs remain open work.

5.10.3 Cut-Edge Pythagorean Cluster Rate Law

Theorem 5.10.B (Cut-Edge Pythagorean Cluster Rate Law for Site-Factorized Pure States). Let $G = (V, E)$ be a finite graph with 2-local Hamiltonian $H = \sum_{e\in E} h_e$. Let $\mathcal{C} = \{V_1, V_2, \dots, V_K\}$ be a partition of V into disjoint clusters ($V = \bigsqcup_{k=1}^{K} V_k$). Let $\mathcal{M}_{\mathcal{C}} = \{\bigotimes_{k=1}^{K} \rho_{V_k} : \rho_{V_k} \in \mathcal{S}(\bigotimes_{v\in V_k} \mathcal{H}_v)\}$ be the cluster-product state family, using the affine product-extension tangent of §5.10.2, and let $E_{\mathrm{cut}}(\mathcal{C}) = \{\{u, w\} \in E : u \in V_k, w \in V_l, k \neq l\}$ be the set of cut edges crossing between distinct clusters. Then for any **fully site-factorized pure state** $m = \bigotimes_{v\in V} |u_v\rangle\langle u_v| \in \mathcal{M} \subseteq \mathcal{M}_{\mathcal{C}}$ (with cluster factors $|u_{V_k}\rangle := \bigotimes_{v\in V_k} |u_v\rangle$): 1. Every internal edge $e \in E(V_k)$ within a cluster V_k generates a purely tangential direction in $T_m\mathcal{M}_{\mathcal{C}}$:

$$-i[h_e, m] \in T_m\mathcal{M}_{\mathcal{C}}, \qquad Q_{m,\mathcal{C}}(-i[h_e, m]) = 0.$$

2. The global cluster normal leakage rate $\nu_{\mathrm{global},\mathcal{C}}(m) = \|Q_{m,\mathcal{C}}(-i[H, m])\|_1$ satisfies the exact Pythagorean cut-edge sum:

$$\nu^2_{\mathrm{global},\mathcal{C}}(m) = \sum_{e\in E_{\mathrm{cut}}(\mathcal{C})} \nu^2_e(m),$$

where $\nu_e(m) = \|Q_{e,m_e}(-i[h_e, m_e])\|_1$ is the two-site normal rate on edge e. Consequently, at any site-factorized pure state $m \in \mathcal{M}$, selecting a K-cluster partition $\mathcal{C}$ that minimizes instantaneous cluster normal leakage $\nu_{\mathrm{global},\mathcal{C}}(m)$ is equivalent to the weighted minimum K-cut problem on G with edge capacities $c(e) = \nu^2_e(m)$, with the same declared nonempty-cluster, connectivity or balance constraints, if any. For $K = 2$ without extra constraints this is the usual minimum cut.

Proof. Decompose each single-site Hilbert space as $\mathcal{H}_v = \mathbb{C}|u_v\rangle \oplus |u_v\rangle^{\perp}$ and each cluster Hilbert space as $\mathcal{H}_{V_k} = \mathbb{C}|u_{V_k}\rangle \oplus |u_{V_k}\rangle^{\perp}$. An internal edge $e = \{u, w\} \subset V_k$ acts solely within $\mathcal{H}_{V_k}$, so $-i[h_e, m] = (-i[h_e, |u_{V_k}\rangle\langle u_{V_k}|]) \otimes \bigotimes_{j\neq k} |u_{V_j}\rangle\langle u_{V_j}| \in T_m\mathcal{M}_{\mathcal{C}}$ and $Q_{m,\mathcal{C}}(-i[h_e, m]) = 0$.

Now let $e = \{u, w\} \in E_{\mathrm{cut}}(\mathcal{C})$ be a cut edge with $u \in V_k$ and $w \in V_l$ ($k \neq l$). Because $m = \bigotimes_{v\in V} |u_v\rangle\langle u_v|$ is site-factorized and h_e acts as the identity on $(V_k \setminus \{u\}) \cup (V_l \setminus \{w\})$, the cluster normal projection on $V_k \cup V_l$ acts strictly on the two endpoint factors $\{u, w\}$:

$$\begin{aligned}&(I_{V_k} - |u_{V_k}\rangle\langle u_{V_k}|) \otimes (I_{V_l} - |u_{V_l}\rangle\langle u_{V_l}|)(h_e|u_{V_k} u_{V_l}\rangle)\\ &\quad = \big((I_u - |u_u\rangle\langle u_u|) \otimes (I_w - |u_w\rangle\langle u_w|)h_e|u_u u_w\rangle\big) \otimes |u_{(V_k\cup V_l)\setminus\{u,w\}}\rangle.\end{aligned}$$

Embedding into the full Hilbert space $\mathcal{H}_V$, the excitation vector $\chi_e = \tilde{\chi}_e \otimes |u_{V\setminus\{u,w\}}\rangle$ has both endpoints u, w in their orthogonal complements $|u_u\rangle^{\perp} \otimes |u_w\rangle^{\perp}$ and every other site $v \notin \{u, w\}$ in its unexcited state $|u_v\rangle$, with $2\|\chi_e\| = 2\|\tilde{\chi}_e\| = \nu_e(m)$. For any two distinct cut edges $e \neq e'$ in $E_{\mathrm{cut}}(\mathcal{C})$—including two cut edges that cross between the *same* pair of clusters (V_k, V_l)—the two-element endpoint sets $e = \{u, w\}$ and $e' = \{u', w'\}$ are distinct, so there is at least one site $v \in e \triangle e'$

at which one excitation vector lies in $|u_v\rangle^\perp$ while the other lies in $\mathbb{C}|u_v\rangle$. Hence $\langle \chi_e, \chi_{e'} \rangle = 0$ for all $e \neq e' \in E_{\mathrm{cut}}(\mathcal{C})$. By the rank-2 pure-state normal decomposition of Proposition 5.11, $Q_{m,\mathcal{C}}(-i[H,m]) = -i(|\chi_{\mathrm{global}}\rangle\langle u_V| - |u_V\rangle\langle \chi_{\mathrm{global}}|)$ where $\chi_{\mathrm{global}} = \sum_{e\in E_{\mathrm{cut}}(\mathcal{C})} \chi_e$, so $\nu^2_{\mathrm{global},\mathcal{C}}(m) = 4\|\chi_{\mathrm{global}}\|^2 = 4\sum_{e\in E_{\mathrm{cut}}(\mathcal{C})}\|\chi_e\|^2 = \sum_{e\in E_{\mathrm{cut}}(\mathcal{C})} \nu_e^2(m)$. □

Remark 5.10.B.1 (Counterexample for intra-cluster entangled states). The hypothesis that m is site-factorized ($m \in \mathcal{M}$) cannot be weakened to arbitrary pure cluster-product states $m = \bigotimes_{k=1}^K |u_{V_k}\rangle\langle u_{V_k}| \in \mathcal{M}_{\mathcal{C}}$ when two or more cut edges connect the same pair of clusters. Indeed, take four qubits partitioned into two clusters $V_1 = \{1,2\}$ and $V_2 = \{3,4\}$ in the intra-cluster Bell state $|u_{V_1}\rangle = |\Phi^+\rangle_{12} = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ and $|u_{V_2}\rangle = |\Phi^+\rangle_{34} = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$, with two cut edges $e_1 = \{1,3\}$ ($h_{e_1} = Z_1 \otimes Z_3$) and $e_2 = \{2,4\}$ ($h_{e_2} = Z_2 \otimes Z_4$). Because $Z_1|\Phi^+\rangle_{12} = Z_2|\Phi^+\rangle_{12} = |\Phi^-\rangle_{12} \in |u_{V_1}\rangle^\perp$ and $Z_3|\Phi^+\rangle_{34} = Z_4|\Phi^+\rangle_{34} = |\Phi^-\rangle_{34} \in |u_{V_2}\rangle^\perp$, the two distinct cut edges produce the **exact same** cluster excitation vector:

$$\chi_{e_1} = \chi_{e_2} = |\Phi^-\rangle_{12} \otimes |\Phi^-\rangle_{34}.$$

Thus $\chi_{\mathrm{global}} = \chi_{e_1} + \chi_{e_2} = 2|\Phi^-\rangle_{12} \otimes |\Phi^-\rangle_{34}$, yielding $\nu^2_{\mathrm{global},\mathcal{C}} = 4\|\chi_{\mathrm{global}}\|^2 = 16$, whereas $4(\|\chi_{e_1}\|^2 + \|\chi_{e_2}\|^2) = 8$ (and if $h_{e_2} = -Z_2 \otimes Z_4$, the two cut edges destructively interfere to give $\nu^2_{\mathrm{global},\mathcal{C}} = 0 \neq 8$; moreover, on the maximally mixed two-site marginals $\rho_{13} = \rho_{24} = I_4/4$, the two-site rates $\nu_{e_i}(\rho_{e_i})$ vanish). Site factorization eliminates this intra-cluster entanglement transfer channel and ensures that distinct cut edges excite orthogonal two-site subspaces.

5.11 Retraction residual and distance to the nearest product

The product of the marginals is a canonical retraction, but need not be a nearest product in trace norm. On a finite tensor product of k local Hilbert spaces let $\mathcal{M}$ be the compact set of product density matrices and set

$$d_{\mathcal{M}}(\rho) = \min_{\sigma\in\mathcal{M}} \|\rho - \sigma\|_1, \qquad D_\Pi(\rho) = \|\rho - \Pi\rho\|_1.$$

All distances here use the raw trace norm; divide both by two for trace distance. The following comparison does not assert a sharp constant.

Theorem 5.9 (metric comparison and first-order optimality). For every density matrix on k factors,

$$d_{\mathcal{M}}(\rho) \le D_\Pi(\rho) \le (k+1) d_{\mathcal{M}}(\rho).$$

At a full-rank product $m = \bigotimes_j m_j$, suppose $\rho(t) = m + tL + O(t^2)$ in trace norm, with L Hermitian and traceless and $t \downarrow 0$. Define the real tangent space and its retraction projection by

$$T_m = \left\{ \sum_j m_{<j} \otimes v_j \otimes m_{>j} : v_j = v_j^*,\ \mathrm{Tr}\, v_j = 0 \right\},$$
$$PL = \sum_j m_{<j} \otimes \mathrm{Tr}_{\neq j} L \otimes m_{>j}, \qquad Q = I - P.$$

Then

$$d_{\mathcal{M}}(\rho(t)) = t\mu + O(t^2), \qquad \mu = \inf_{v\in T_m} \|L - v\|_1,$$
$$D_\Pi(\rho(t)) = t\nu + O(t^2), \qquad \nu = \|QL\|_1,$$
$$\mu \le \nu \le \|Q\|_{1\to 1}\mu.$$

The induced norm of Q is taken on the Hermitian trace-zero space. The optimal coefficient has the dual certificate

$$\mu = \max\{\mathrm{Tr}(WL) : W = W^*,\ \|W\|_\infty \le 1,\ \mathrm{Tr}(Wv) = 0 \text{ for every } v \in T_m\}.$$

Proof. For any product $\sigma = \bigotimes_j \sigma_j$, telescoping the tensor product and using marginal contraction gives

$$\|\Pi\rho - \sigma\|_1 \le \sum_j \|\rho_j - \sigma_j\|_1 \le k\|\rho - \sigma\|_1.$$

The triangle inequality and minimization establish the finite comparison; the other direction uses $\Pi\rho \in \mathcal{M}$.

A minimizer $\sigma(t) = \bigotimes_j \sigma_j(t)$ exists by compactness. Since m is an admissible competitor, $\|\sigma(t) - m\|_1 = O(t)$. Thus every $\Delta_j(t) = \sigma_j(t) - m_j$ is $O(t)$, and the finite product expansion is

$$\sigma(t) = m + \sum_j m_{<j} \otimes \Delta_j(t) \otimes m_{>j} + O(t^2).$$

The linear term lies in T_m, uniformly giving $d_{\mathcal{M}}(\rho(t)) \ge t\mu - O(t^2)$. This lower argument works at boundary products too. At full rank, for any fixed $v \in T_m$ represented by v_j, all $m_j + tv_j$ are densities for sufficiently small t. Their product is $m + tv + O(t^2)$. A minimizing tangent vector exists in this finite-dimensional closed subspace, yielding the matching upper bound. Expanding $\Pi\rho(t)$ gives the formula for ν. Since $PL \in T_m$ and $Qv = 0$ on T_m, $\mu \le \nu$ and $\|QL\|_1 \le \|Q\|\|L - v\|_1$. Finally, the finite-dimensional Hahn–Banach distance formula and Hermitian trace/operator norm duality give the displayed maximum. □

Full rank is used for the physical upper tangent curve. An arbitrary rank-deficient mixed product requires its feasible tangent cone; the theorem must not be transferred to that boundary by dropping this premise. The pure Hamiltonian case has a separate certificate below.

Proposition 5.10 (a strict mixed-state distortion family). Put $r_p = \mathrm{diag}(p, 1-p)$, $r_q = \mathrm{diag}(q, 1-q)$, with $0 < p, q < 1$, and evolve $r_p \otimes r_q$ under $H = X \otimes Z$. With $a = 1 - 2p$ and $b = 2q - 1$,

$$L = aY \otimes Zr_q, \quad PL = abY \otimes r_q, \quad QL = 2aq(1-q)Y \otimes Z.$$

Consequently

$$\mu = 4|a| \min(q, 1-q), \qquad \nu = 8|a|q(1-q).$$

For $p \neq 1/2$ their ratio is $2\max(q, 1-q)$, approaching 2 within the full-rank family. This is an example family, not a sharp bound over all states.

Proof. For $q \ge 1/2$ take the primal tangent vector $v = aY \otimes r_q$ and the dual witness

$$W = \mathrm{sign}(a)Y \otimes \mathrm{diag}((1-q)/q, -1).$$

It has operator norm at most one. It annihilates $v_A \otimes r_q$ because the second-factor pairing is zero, and $r_p \otimes v_B$ because $\mathrm{Tr}(Yr_p) = 0$. Direct calculation gives $\|L - v\|_1 = \mathrm{Tr}(WL) = 4|a|(1-q)$. For $q < 1/2$ use $v = -aY \otimes r_q$ and $W = \mathrm{sign}(a)Y \otimes \mathrm{diag}(1, -q/(1-q))$. Theorem 5.9 and the spectrum of $Y \otimes Z$ establish the assertion. If $p = 1/2$ both rates vanish and the ratio is undefined. At $p = q = 3/4$ the certificate gives $\mu = 1/2$ but $\nu = 3/4$. □

5.11.1 Feasible tangent cones at rank-deficient boundaries and cluster refinement monotonicity

When analyzing rank-deficient quantum states at the boundary of the state space, two distinct tangent cones must be carefully distinguished: 1. *Ambient state-space tangent cone $T_{\mathcal{D}}^{\text{feas}}(m)$:* For an arbitrary rank-deficient mixed state m on the boundary of the ambient state space $\mathcal{D} := \mathcal{S}(\mathcal{H}) = \{\rho \geq 0, \operatorname{Tr}\rho = 1\}$, the cone of first-order variations preserving positivity is

$$T_{\mathcal{D}}^{\text{feas}}(m) = \left\{v = v^\dagger,\ \operatorname{Tr} v = 0 : P_0 v P_0 \geq 0\right\},$$

where $P_0 = I - \operatorname{supp}(m)$ projects onto $\ker(m)$. This is the closed first-order (Bouligand) tangent cone: a direction may require a curve with a second-order correction, not the straight ray $m + tv$. For an arbitrary traceless Hermitian variation L, its trace-norm distance to $T_{\mathcal{D}}^{\text{feas}}(m)$ is given by the exact spectral formula:

$$\operatorname{dist}_1(L, T_{\mathcal{D}}^{\text{feas}}(m)) = 2\left\|(P_0 L P_0)^-\right\|_1,$$

where $(A)^- = \frac{|A|-A}{2}$ denotes the negative part of self-adjoint operator A. To prove the distance, put $N = (P_0 L P_0)^-$ and $\beta = \operatorname{Tr} N$. Then $v = L + N - \beta m$ lies in the cone, and $\|L - v\|_1 = 2\beta$ because N and m have orthogonal supports. If P_- is the negative spectral projection of $P_0 L P_0$, the contraction $W = I - 2P_-$ has $\operatorname{Tr}(Wv') \leq 0$ for every cone element v' and $\operatorname{Tr}(WL) = 2\beta$, proving the matching lower bound. This is first-order positivity feasibility in the ambient state space; it does *not* measure preservation of factorization.

2. *Product manifold tangent cone $T_{\mathcal{M}}^{\text{feas}}(m)$:* On the product manifold $\mathcal{M}$, first-order variations are constrained to local product tangent directions:

$$T_{\mathcal{M}}^{\text{feas}}(m) = \left\{v = \sum_j m_{<j} \otimes v_j \otimes m_{>j} : v_j = v_j^\dagger,\ \operatorname{Tr} v_j = 0,\ P_0^{(j)} v_j P_0^{(j)} \geq 0\right\}.$$

   In particular, $\operatorname{dist}_1(L, T_{\mathcal{M}}^{\text{feas}}(m))$ *cannot* be evaluated by summing local marginal projections $P_0^{(j)} \operatorname{Tr}_{\neq j}(L) P_0^{(j)}$. A marginal-sum formula examines only local partial traces and entirely misses emerging correlations.

Exact two-qubit entangling counterexample. Consider the pure product state $m = |00\rangle\langle 00|$, entangling Hamiltonian $H = X \otimes X$, and dynamical velocity $L = -i[H, m] = -i(|11\rangle\langle 00| - |00\rangle\langle 11|)$. Both local partial traces vanish identically: $\operatorname{Tr}_A(L) = 0$ and $\operatorname{Tr}_B(L) = 0$. The ambient cone distance is $\operatorname{dist}_1(L, T_{\mathcal{D}}^{\text{feas}}(m)) = 0$ because $P_0 = I - |00\rangle\langle 00|$ gives $P_0 L P_0 = 0$ (L connects $\operatorname{supp}(m)$ exclusively to the orthogonal subspace $|11\rangle$). However, every product tangent has the form $v = v_A \otimes |0\rangle\langle 0| + |0\rangle\langle 0| \otimes v_B$, for which the matrix element $\langle 11|v|00\rangle = 0$. The actual distance to the product tangent cone is $\operatorname{dist}_1(L, T_{\mathcal{M}}^{\text{feas}}(m)) = 2$. This distance is certified with zero duality gap by the dual witness $W = L$, which satisfies $\|W\|_\infty = 1$, $\operatorname{Tr}(Wv) = 0$ for all $v \in T_{\mathcal{M}}$, and $\operatorname{Tr}(WL) = \operatorname{Tr}(L^2) = 2$. The finite instance above has an exact written primal/dual certificate. While numerical algorithms (such as the optimization routines in `code/feasible_tangent_cone.py`) compute candidate product-cone projections, exact lower and upper bounds are certified through independent primal and dual feasibility verifiers.

Cluster product manifolds and refinement monotonicity. Let $V = \bigsqcup_{k=1}^K V_k$ be a partition $\mathcal{C}$ of the lattice vertices into disjoint clusters, and define the cluster product manifold:

$$\mathcal{M}_{\mathcal{C}} = \left\{\bigotimes_{k=1}^K \rho_{V_k} : \rho_{V_k} \in \mathcal{S}(\mathcal{H}_{V_k})\right\}.$$

If partition $\mathcal{C}_2$ is a refinement of partition $\mathcal{C}_1$ (denoted $\mathcal{C}_1 \preceq \mathcal{C}_2$, meaning every cluster in $\mathcal{C}_2$ is contained within a cluster in $\mathcal{C}_1$), then $\mathcal{M}_{\mathcal{C}_2} \subseteq \mathcal{M}_{\mathcal{C}_1}$. Consequently, the feasible trace-norm distance satisfies refinement monotonicity:

$$\mathrm{dist}_1(L, T^{\mathrm{feas}}_{\mathcal{M}_{\mathcal{C}_1}}(m)) \le \mathrm{dist}_1(L, T^{\mathrm{feas}}_{\mathcal{M}_{\mathcal{C}_2}}(m)).$$

Cluster merging cannot increase the distance, by inclusion of the admissible tangent cones. Absorbing an interaction does not imply a strict decrease: for $m = |00\rangle\langle 00|$ and $H = Z \otimes Z$, the velocity and both distances are zero before and after merging. Strictness requires an actual distance gap. For the site-factorized pure-state rate in Theorem 5.10.B, strictness is equivalent to a positive sum of squared rates on the absorbed cut edges; this criterion is not a theorem for arbitrary mixed-state cones.

5.12 Pure excitation sectors, optimality, and declared Born weights

Let $u = \bigotimes_v u_v$ be a normalized pure product, $m = |u\rangle\langle u|$, and H a finite one-/two-local Hermitian Hamiltonian. Combine all terms on the same unordered pair into h_e before defining any sector mass. For $e = \{a, b\}$, let χ_e be the vector obtained by applying

$$((I - |u_a\rangle\langle u_a|) \otimes (I - |u_b\rangle\langle u_b|))h_e$$

to $u_a \otimes u_b$ and tensoring back the unchanged factors. Put $\chi = \sum_e \chi_e$.

Theorem 5.11 (orthogonal sectors and pure metric optimality). Distinct pairs have orthogonal χ_e, even when the pairs share a vertex. The global product-retraction normal vector and rates satisfy

$$n_{\mathrm{global}} = -i(|\chi\rangle\langle u| - |u\rangle\langle \chi|),$$
$$\nu_{\mathrm{global}} = 2\|\chi\|, \qquad \nu_e = 2\|\chi_e\|, \qquad \nu^2_{\mathrm{global}} = \sum_e \nu_e^2.$$

For this Hamiltonian curve,

$$d_{\mathcal{M}}(\rho(t)) = D_\Pi(\rho(t)) + O(t^2) = t\nu_{\mathrm{global}} + O(t^2), \qquad t \downarrow 0.$$

These global $O(t^2)$ constants are for a fixed finite volume; no uniform in-volume time window follows from this identity.

Proof. Decompose each factor as $\mathbb{C}u_v \oplus u_v^\perp$. The tensor space is an orthogonal direct sum labeled by its excited-site subset. A one-/two-local Hu has only empty, singleton and pair sectors. The empty sector cancels in $-i[H, m]$, and P removes exactly its singleton terms. Different pair labels give orthogonal sectors: a site in the symmetric difference supplies an orthogonal tensor factor. Only h_e reaches the pair-e sector, also establishing the edge rate using §5.3.1. On $\mathrm{span}\{u, \chi\}$ the normal vector has eigenvalues $\pm\|\chi\|$ and it is zero on the orthogonal complement.

If $\chi \neq 0$, put $\eta = \chi/\|\chi\|$ and $W = -i(|\eta\rangle\langle u| - |u\rangle\langle \eta|)$. Then $\|W\|_\infty = 1$, $\mathrm{Tr}(WL) = 2\|\chi\|$, and W annihilates every product tangent direction: such a direction changes at most one factor, whereas W couples the empty sector to pair sectors. The lower minimizer argument of Theorem 5.9 is valid at this boundary and yields $d_{\mathcal{M}} \ge t\nu_{\mathrm{global}} - O(t^2)$. The physical product $\Pi\rho(t)$ supplies the upper bound. If $\chi = 0$, that upper bound is $O(t^2)$ and the same conclusion follows. □

There is an exact arithmetic certificate for the squared rates without taking square roots. When the initial pure projectors and interactions have entries in $\mathbb{Q}(i)$, put $r_{ab} = |u_a u_b\rangle\langle u_a u_b|$ and $R_{ab} =$

$(I - r_a) \otimes (I - r_b)$. Then

$$w_e = \|\chi_e\|^2 = \mathrm{Tr}(r_{ab} h_e R_{ab} h_e) \in \mathbb{Q}_{\geq 0}, \qquad \nu_e^2 = 4w_e, \quad \nu_{\mathrm{global}}^2 = 4 \sum_e w_e.$$

Hermiticity and positivity identify the rational trace as a nonnegative squared norm. A negative value in such a replay is a failed input or calculation, not a physical rate. Terms on the same edge can interfere and cancel, so their individual squared masses cannot be summed before their operators are combined. For $|000\rangle$ and $H = X_1X_2 + 2X_1X_3 + 3X_2X_3$, the sector Gram matrix is $\mathrm{diag}(1, 4, 9)$, local rates are $(2, 4, 6)$ and the global squared rate is 56.

To connect this result to the supplied normalization construction [18], the weight assignment must be declared: use $w_e = \nu_e^2/4$, and when $W_0 = \sum_e w_e > 0$ set

$$p_e = \frac{w_e}{W_0} = \frac{\nu_e^2}{\sum_f \nu_f^2}.$$

This is a normalized profile of the orthogonal excitation masses in the already imported Hilbert space. It need not be the profile obtained by normalizing the original additive tally: $C'(0+) = \sum_e \nu_e = 2\sum_e \sqrt{w_e}$. The two profiles coincide when all nonzero rates are equal. The triangle example gives $p = (1/14, 2/7, 9/14)$ and additive slope 12. When $W_0 = 0$, the rate identity still holds but the profile is undefined. Normalization alone neither selects these weights nor establishes Born uniqueness, a Hilbert lift, a measurement interpretation, or a Gleason premise. This is consistent with the normalization nonselection result of the witnessed difference programme [12].

Nor do the masses determine the normal vector. For fixed u and vectors $\chi, \chi' \perp u$,

$$\|n(\chi) - n(\chi')\|_1 = 2\|\chi - \chi'\|.$$

Changing a sector's physical relative phase can preserve every w_e while changing the normal direction. Rephasing a basis vector together with its coordinate is only a representation change and does not supply such a witness.

5.13 Limits of a global trace-norm versus edge-sum comparison

Write $D_{\mathrm{global}} = \|\rho - \bigotimes_v \rho_v\|_1$ and $C = \sum_{e \in E} D_e$. Marginal contraction always gives

$$\max_e D_e \leq D_{\mathrm{global}} \leq 2, \qquad C \leq |E| D_{\mathrm{global}}.$$

The size factor cannot be removed without additional assumptions, and even vanishing pair defects can miss the global defect.

Proposition 5.12 (two failures of unrestricted equivalence). There is no positive size-independent c with $D_{\mathrm{global}} \geq cC$ for all finite product-state one-/two-local evolutions, even with $g = 1$ and a fixed arbitrarily small positive time. There is also no finite K with $D_{\mathrm{global}} \leq KC$ at unrestricted times in that class.

Proof. Take M disjoint qubit pairs, initial state $|00\rangle$ per pair, and $H = \sum_e X_a X_b$. For $0 < t < \pi/4$, the exact two-qubit computation of §6.1 gives

$$D_e(t) = \sin(2t) + \tfrac{1}{2}\sin^2(2t) > 0.$$

Thus $C = MD_e(t)$ while $D_{\text{global}} \le 2$. This establishes the first failure at any such fixed time. At zero the corresponding slopes are $2M$ and $2\sqrt{M}$, by Theorem 5.11.

For the second, take the five-cycle, $H = \sum_{j \bmod 5} Z_j Z_{j+1}$, initial state $|+\rangle^{\otimes 5}$ and $t = \pi/4$. Up to a common phase the state is

$$|\psi\rangle = 2^{-5/2} \sum_{b \in \{0,1\}^5} (-1)^{\sum_j b_j + \sum_j b_j b_{j+1}} |b\rangle.$$

Every two-site marginal is $I_4/4$. Here is a direct cancellation check: for two distinct retained strings let d be their binary difference, supported on at most two sites. In the partial-trace sum the phase difference has, at each discarded vertex v, linear coefficient $d_{v-1} + d_{v+1}$ modulo two. A single changed vertex in a five-cycle has a neighbor outside the retained pair with coefficient one; two changed vertices, whether adjacent or nonadjacent, also have an outside neighbor adjacent to exactly one of them. Summing that bit cancels the off-diagonal entry. Diagonal entries are $1/4$. One-site marginals are therefore $I_2/2$, and

$$C = 0, \qquad D_{\text{global}} = \big\| |\psi\rangle\langle\psi| - I_{32}/32 \big\|_1 = 31/16.$$

The last spectrum consists of $31/32$ once and $-1/32$ thirty-one times. All entries of this density and its marginals are rational. □

This settles the unrestricted equivalence question between D_{global} and C negatively. The cycle time is outside the certified small-time window of Theorem 5.1; it does not refute every additional near-zero upper bound, nor the matching and total-correlation density result of [5]. An intensive edge average and a bounded global distinguishability are different quantities.

5.14 Support components and connected cumulants

Fix reference densities r_j and let E_j trace out site j and reinsert r_j in its position. These are commuting idempotent linear maps. Define

$$T_S = \prod_{j \in S} (I - E_j) \prod_{j \notin S} E_j.$$

Proposition 5.13 (reference-support decomposition). The T_S are idempotents, $T_S T_R = 0$ for $S \neq R$, and $\sum_S T_S = I$. On Hermitian trace-zero directions at $m = \bigotimes_j r_j$,

$$P = \sum_{|S|=1} T_S, \qquad Q = \sum_{|S| \ge 2} T_S.$$

If $r_j = \rho_j$, then

$$\rho - \Pi\rho = \sum_{|S| \ge 2} T_S \rho.$$

Proof. Expand $\prod_j (E_j + (I - E_j))$; unequal subsets have a factor $E_j(I - E_j) = 0$. Moreover $T_\emptyset Z = \operatorname{Tr}(Z) m$, and $T_{\{j\}} Z$ inserts $\operatorname{Tr}_{\neq j} Z - \operatorname{Tr}(Z) r_j$ into factor j. These formulas give every assertion. The references remain fixed when composing the projectors. □

These projectors are algebraically complementary; they need not be orthogonal or make the trace norm additive. Their highest component need not represent a connected correlation. For independent Bell pairs $\rho = B_{12} \otimes B_{34}$,

$$T_{\{1,2,3,4\}} \rho = (B_{12} - I_4/4) \otimes (B_{34} - I_4/4) \neq 0, \qquad \|T_{\{1,2,3,4\}} \rho\|_1 = 9/4.$$

The connected four-site cumulant instead vanishes.

For a nonempty site set S define the operator cumulant

$$\kappa_S(\rho) = \sum_{\pi\in\mathrm{Part}(S)} (-1)^{|\pi|-1}(|\pi|-1)! \bigotimes_{B\in\pi} \rho_B,$$

where tensors are always restored to the original site order. For a pair this is F_{ab}; for three sites it is

$$\rho_{abc} - \rho_{ab}\otimes\rho_c - \rho_{ac}\otimes\rho_b - \rho_{bc}\otimes\rho_a + 2\rho_a\otimes\rho_b\otimes\rho_c.$$

It measures connected correlation, including classical correlation; nonzero cumulant is not by itself an entanglement witness.

To establish cancellation under independent splitting without assuming commuting matrices, introduce a square-free support algebra. Its coefficients on A are operators on A; multiply coefficients by their reordered tensor product if supports are disjoint and by zero if supports overlap. This is a commutative algebra because actual products use disjoint factors. Its positive-support ideal is nilpotent. In $M = \sum_{A\subseteq S} \rho_A z_A$, $\rho_\emptyset = 1$, the coefficient of z_S in the finite series $\log M$ is precisely κ_S: a partition with ℓ blocks occurs $\ell!$ times in the ℓth logarithm term. If ρ_S splits across a nontrivial partition, M factors into the component series. The logarithm is their sum and has no term crossing those components. Thus $\kappa_S = 0$ whenever S crosses independent components.

Let G be the graph of allowed two-site interactions, and let $\tau_G(S)$ be the minimum number of edges in a connected subgraph containing S, allowing Steiner vertices. Set $\tau_G(S) = \infty$ if no such subgraph exists. Let $(k)_j = k(k+1)\cdots(k+j-1)$, with $(k)_0 = 1$, and set

$$a_k = \sum_{\pi\in\mathrm{Part}(k)} (|\pi|-1)!, \qquad b_k = \sum_{\pi\in\mathrm{Part}(k)} |\pi|!.$$

For $k = 1,\dots,6$ these are respectively $(1,2,6,26,150,1082)$ and $(1,3,13,75,541,4683)$.

Theorem 5.14 (connected onset, locality and all-order remainder). For a finite one-/two-local Hamiltonian, a product initial state, and $|S| = k \geq 2$,

$$\kappa_S^{(j)}(0) = 0 \quad (j < \tau_G(S)), \qquad \tau_G(S) \geq k-1.$$

If $j \geq k-1$, its jth derivative depends only on the initial factors and Hamiltonian terms in the induced radius-$(j-k+1)$ neighborhood of S. At every real time,

$$\|\kappa_S^{(j)}(t)\|_1 \leq a_k (2g)^j (k)_j.$$

Writing $P_{S,J}(t) = \sum_{j=0}^{J} t^j \kappa_S^{(j)}(0)/j!$, for $J \geq 0$,

$$\|\kappa_S(t) - P_{S,J}(t)\|_1 \leq a_k \binom{k+J}{J+1} (2g|t|)^{J+1}.$$

The same bound controls the difference of their trace norms. In particular, for finite $\tau = \tau_G(S)$,

$$\|\kappa_S(t)\|_1 \leq a_k \binom{k+\tau-1}{\tau} (2g|t|)^{\tau}.$$

If $\tau = \infty$, the cumulant is identically zero at every time.

Proof. Give each Hamiltonian term its own real parameter. The jth state jet, and hence the jth cumulant jet, is a homogeneous degree-j polynomial in those parameters. For a particular monomial set every absent parameter to zero; its coefficient is unchanged. If the remaining interaction graph splits S, the corresponding evolution factorizes and the cumulant vanishes for all parameter choices. Its monomial coefficient is therefore zero. If S is in one component but the monomial also uses a parameter in another component, the state reduced to S is independent of that parameter, again giving zero. Thus any surviving monomial uses one connected component containing all of S, with at most j distinct two-site edges. This establishes onset. That component has at most $j+1$ vertices. A vertex at distance r from S in it requires r distinct vertices outside the k sites of S, so $k+r \le j+1$. The ambient distance is no larger. This establishes the stated induced-neighborhood locality, including one-local terms, which consume degree without adding edges. Connectivity is necessary, not sufficient: commutation or cancellation can postpone onset further.

For any partition π, Lemma 5.4 and the multinomial product rule bound its jth derivative by

$$(2g)^j \sum_{\sum_B j_B = j} \frac{j!}{\prod_B j_B!} \prod_{B\in\pi} (|B|)_{j_B} = (2g)^j (k)_j.$$

The equality is coefficient extraction from $\prod_B (1-z)^{-|B|} = (1-z)^{-k}$. Sum the absolute partition coefficients to obtain a_k. Integral Taylor remainder gives the displayed estimate. Set $J = \tau - 1$ and use the zero jets for the onset bound. Disconnected graph components factorize at every time, establishing the infinite-τ case. □

At $k=2$, $a_2 = 2$ and the remainder is exactly the coefficient in Theorem 5.5. For the three-site chain $H = X_1X_2 + Y_2X_3$ and $|000\rangle$, the pair and triple first jets vanish, whereas

$$\kappa''_{13}(0) = 2i|01\rangle\langle 10| - 2i|10\rangle\langle 01|, \qquad \|\kappa''_{13}(0)\|_1 = 4, \quad \|\kappa''_{123}(0)\|_1 = 8.$$

Both have Steiner length two and attain the permitted quadratic onset.

For fixed k the onset estimate supplies short-time spatial decay: if $q = 2g|t| < r < 1$, the generating-function bound gives

$$\|\kappa_S(t)\|_1 \le a_k (1-q/r)^{-k} r^{\tau_G(S)}.$$

Indeed $\binom{k+\tau-1}{\tau}(q/r)^\tau \le (1-q/r)^{-k}$. One can replace $\tau_G(S)$ by a smaller distance in this exponential bound, but not automatically in the preceding polynomial-times-power bound, which need not be monotone at small τ. This is a short-time propagation bound. Subdividing time does not extend its zero-jet proof to arbitrary times: the state after the first interval is generally no longer product.

5.15 Fixed-support limits and connected-cumulant densities

For two densities on S, telescoping each partition product and contracting marginals establishes the useful global Lipschitz estimate

$$\|\kappa_S(\rho) - \kappa_S(\sigma)\|_1 \le b_k \|\rho - \sigma\|_1, \qquad \|\kappa_S(\rho)\|_1 \le a_k.$$

For a partition with ℓ blocks, the product difference has norm at most $\ell\|\rho-\sigma\|_1$; multiplying by $(\ell-1)!$ and summing gives b_k. The second inequality uses norm one for each density factor. Consequently the fixed-support local thermodynamic limit of Theorem 5.7 immediately yields a cumulant limit uniformly on compact real-time intervals. For $J \ge k-1$ and a finite volume Λ

containing the induced radius-$(J-k+1)$ neighborhood of S, locality and the two Taylor remainders also give the explicit estimate

$$\|\kappa_S^{\Lambda}(t) - \kappa_S^{\infty}(t)\|_1 \le 2a_k \binom{k+J}{J+1} (2g_*|t|)^{J+1}.$$

The stabilized coefficients pass to the limit. For $2g_*|t| < 1$ the remainders tend to zero and identify these coefficients as derivatives of the limit. At arbitrary times convergence instead uses Theorem 5.7 and the Lipschitz estimate, not extrapolation of the series beyond its useful radius.

Theorem 5.15 (density for finitely many fixed shapes). Assume a translation-invariant locally finite graph on $\mathbb{Z}^d$, finite local dimensions identified by translation, translation-covariant one-/two-site interactions with $g_* < \infty$, and a translation-invariant product initial state. Use compatible open-boundary cubes $\Lambda_L = \{-L, \dots, L\}^d$. Fix finitely many finite shapes S_α, each of size at least two, and nonnegative fixed weights w_α. Then, uniformly on every $|t| \le T$,

$$\begin{aligned} \mathcal{C}_L(t) &= \frac{1}{|\Lambda_L|} \sum_\alpha w_\alpha \sum_{x : x+S_\alpha \subseteq \Lambda_L} \|\kappa_{x+S_\alpha}^{\Lambda_L}(t)\|_1 \\ &\longrightarrow \sum_\alpha w_\alpha \|\kappa_{S_\alpha}^{\infty}(t)\|_1. \end{aligned}$$

Proof. Exhaustion independence in Theorem 5.7 makes the limiting dynamics translation covariant. Given an error tolerance and a compact time interval, choose a finite neighborhood of each shape so that every volume containing that neighborhood approximates its limit within the tolerance. The preceding Lipschitz estimate transfers this to cumulant norms. Translation gives the same bound for all translates whose chosen neighborhoods lie inside Λ_L. Local finiteness and translation invariance give finitely many edge displacements, and these neighborhoods are fixed finite sets. The fraction of their translates crossing the cubic boundary tends to zero, while the number of retained translates of each shape divided by $|\Lambda_L|$ tends to one. On the exceptional boundary translates use the uniform bound $a_{|S_\alpha|}$. Sum the finitely many weighted errors. □

This completes a fixed-shape density extension of §5.10 and a short-time connected propagation estimate. An all-time spatial propagation estimate, densities summing over unbounded support orders, and observer budgets growing with volume require additional arguments and are not consequences of this finite-shape theorem.

5.16 Finite onset or persistent zero for polynomial defects

The all-derivatives criterion in §5.8 has a finite replacement for each fixed finite system. The object differentiated is the vector-valued residual, not its norm, which need not be differentiable at zero.

Theorem 5.16 (finite polynomial onset-or-invariance certificate). Let V and W be finite-dimensional real or complex vector spaces, $\dim V = n$, let $A : V \to V$ be linear, and let $f : V \to W$ be a polynomial of degree at most $p \ge 0$. Fix $m \in V$ with $f(m) = 0$. Define

$$R(t) = f(e^{tA}m), \qquad \mathcal{W}_p = \bigoplus_{r=0}^{p} \operatorname{Sym}^r V, \qquad z_m = (1, m, m^{\odot 2}, \dots, m^{\odot p}).$$

On degree zero put $\mathcal{L}_0 = 0$, and on degree $r \ge 1$ put

$$\mathcal{L}_r(v_1 \odot \cdots \odot v_r) = \sum_{i=1}^{r} v_1 \odot \cdots \odot Av_i \odot \cdots \odot v_r, \qquad \mathcal{L} = \bigoplus_{r=0}^{p} \mathcal{L}_r.$$

There is a linear readout $O : \mathcal{W}_p \to W$ such that $R(t) = Oe^{t\mathcal{L}}z_m$. Let

$$\mathcal{C}_m = \operatorname{span}\{\mathcal{L}^j z_m : j \geq 0\}, \qquad q(s) = s^\kappa + \sum_{i=0}^{\kappa-1} q_i s^i$$

be the minimal polynomial of $\mathcal{L}|_{\mathcal{C}_m}$. Then

$$1 \leq \kappa = \dim \mathcal{C}_m \leq \dim \mathcal{W}_p = \binom{n+p}{p}, \qquad \boxed{R(t) \equiv 0 \iff R^{(j)}(0) = 0 \quad (0 \leq j < \kappa).}$$

If R is not identically zero, its first nonzero derivative has order

$$\boxed{1 \leq m_{\text{onset}} \leq \kappa - 1.}$$

In particular, $\kappa = 1$ and $f(m) = 0$ force persistent zero.

Proof. Decompose $f = \sum_{r=0}^{p} f_r$ into homogeneous terms. In characteristic zero, polarization gives linear maps $O_r : \operatorname{Sym}^r V \to W$ with $f_r(v) = O_r(v^{\odot r})$. The tensor product rule gives $(e^{tA}m)^{\odot r} = e^{t\mathcal{L}_r}m^{\odot r}$; summing these readouts gives O and the displayed representation of R.

Write $z = z_m$. Its degree-zero coordinate is one, so $z \neq 0$. The first linear dependence in the Krylov sequence $z, \mathcal{L}z, \ldots$ has the form $q(\mathcal{L})z = 0$, with all earlier vectors independent. Multiplying this relation by every power of $\mathcal{L}$ shows that these earlier vectors span an invariant space containing the whole sequence. Thus their number is $\dim \mathcal{C}_m = \kappa$. A polynomial annihilates z if and only if it annihilates $\mathcal{C}_m$, since z is cyclic, establishing the stated minimal-polynomial interpretation and dimension bound.

Put $b_j = R^{(j)}(0) = O\mathcal{L}^j z$. The full vector relation gives, for every $j \geq 0$,

$$b_{j+\kappa} = -\sum_{i=0}^{\kappa-1} q_i b_{j+i}.$$

If $b_0, \ldots, b_{\kappa-1}$ vanish, induction makes every b_j vanish. The finite matrix exponential is entire, so $R(t) = \sum_{j\geq 0} b_j t^j/j!$ then vanishes for all real t (and for all complex t after complexification). The converse follows by differentiation. If R is nonzero, one of these first κ derivatives must be nonzero; $f(m) = 0$ excludes index zero. □

Certificate form and sharpness. Minimality is useful for the shortest state-space cutoff, but is not necessary for a valid finite test. Any supplied monic polynomial a of degree $K \geq 1$ with $a(\mathcal{L})z_m = 0$ gives the sufficient-and-necessary zero test on $0 \leq j < K$ and, for nonzero R, the upper bound $K - 1$. Checking only a recurrence among reported zero outputs does not establish that vector relation. An exact certificate contains the model, polynomial readout, annihilator coefficients, and the jets $O\mathcal{L}^j z_m$; the checker reconstructs the lift and replays the relation and jets. To claim $K = \kappa$, it also checks independence of the first K cyclic vectors. The cutoff may exceed the minimal recurrence degree of the output because O can annihilate part of the cyclic space.

The upper bound is attained: with $V = \mathbb{R}$, $A = 1$, $m = 1$ and $f(x) = (x-1)^p$ for $p \geq 1$, $R(t) = (e^t - 1)^p$ has onset p and $\kappa = p + 1$. The unlifted generator has minimal polynomial of degree one, so substituting that degree would give a false zero certificate. Even for fixed degree $p = 1$, nilpotent chains of growing dimension, observed at the last coordinate, give arbitrarily late onset. No size-independent cutoff follows from polynomial degree alone.

Corollary 5.17 (pair defects and connected cumulants). For a fixed finite quantum system, use the real vector space of Hermitian matrices, of dimension $n = d^2$ for global Hilbert dimension d, and the linear Liouvillian $A(X) = -i[H, X]$. The pair readout

$$f_e(X) = \mathrm{Tr}_{e^c} X - (\mathrm{Tr}_{a^c} X) \otimes (\mathrm{Tr}_{b^c} X)$$

is a polynomial of degree at most two on that vector space. For product initial density m, $f_e(m) = 0$. Consequently its residual $F_e(t)$ is either identically zero or has onset at most $\kappa_e - 1$, where $\kappa_e \le \binom{d^2+2}{2}$. Under the one-/two-local interaction hypotheses of §5.8, a first nonzero jet supplied by this finite test can be inserted directly into that section's remainder and leading-rate bounds.

For a fixed set S of $k \ge 2$ sites, the connected cumulant of §5.14 is

$$f_S(X) = \sum_{\pi \in \mathrm{Part}(S)} (-1)^{|\pi|-1} (|\pi| - 1)! \bigotimes_{B \in \pi} \mathrm{Tr}_{B^c} X.$$

Tensor factors are placed in the fixed site order. Each reduced operator is linear in X, so this is a polynomial of degree at most k, and it vanishes on the normalized product initial state. Theorem 5.16 applies with $\kappa_S \le \binom{d^2+k}{k}$. Under the additional product and one-/two-local interaction hypotheses of Theorem 5.14, every nonzero cumulant therefore has

$$\boxed{\tau_G(S) \le m_{\mathrm{onset}} \le \kappa_S - 1.}$$

Here $\tau_G(S)$ is the minimum number of edges of a connected interaction subgraph spanning S. If no such subgraph exists, the independent-component cancellation gives persistent zero. If finite $\tau_G(S) \ge \kappa_S$, the lower-order vanishing already covers the entire cyclic cutoff and again forces persistent zero. These conclusions concern the finite residual operator; a norm or incomplete observer report cannot replace its jets.

Proof. The polynomial degree and initial-zero facts just established allow Theorem 5.16. The lower bound and disconnected case are precisely the cancellation statement of Theorem 5.14; combining the two bounds gives the remaining assertions. □

The cutoff depends on the declared finite model and can grow with global Hilbert dimension. Finite coefficient locality in §5.14 does not supply an annihilator uniform across growing systems, and the local limits of §§5.10 and 5.15 do not change this restriction. The exact rational implementation `code/polynomial_onset.py` checks finite instances and rejects inexact scalars, invalid annihilators and forged jets. It has a declared lift-dimension budget; its execution is separate from this general written proof, which is not newly mechanized in Lean.

Prior art, positioning, and non-claim of lifting priority. Polynomial lifting and finite linear observability have established antecedents across dynamical systems and mathematical control theory: 1. Classical Carleman linearization embeddings embed finite-dimensional polynomial differential equations into infinite-dimensional or truncated linear operator systems (Carleman [22]). 2. In data-driven dynamical systems and Koopman operator theory, exact finite linear representations and Koopman invariant subspaces are constructed via dictionary and polynomial lifting functions (Brunton et al. [21]). 3. Differential algebraic observability and Lie-derivative rank tests for polynomial and rational differential systems similarly provide finite criteria for state reconstructibility (e.g., Sedoglavic [23]; Villaverde et al. [24]).

Non-claim of priority: This monograph does not claim priority for the general algebraic mechanism of symmetric-power / polynomial lifting or the finite Krylov/cyclic observability of linear representations.

Specific contribution: The contribution of Theorem 5.16 and Corollary 5.17 consists in: 1. The application to non-product projection defects, non-equilibrium defect fields, and reduced state retractions under Hamiltonian and Liouvillian dynamics. 2. Reconstructible, machine-checkable certificates: Theorem 5.16 requires an annihilator of the full lifted state z_m under $\mathcal{L}$, and derives the residual recurrence from this state-space identity. This distinction separates a vanishing truncated jet (which may vanish through order $\kappa-2$ before exhibiting nonzero onset) from persistent zero. 3. Integration with interaction locality: Corollary 5.17 binds the finite upper cyclic cutoff (κ_S-1) to the graph-theoretic Steiner-tree lower bound $(\tau_G(S))$, keeping the finite state-space dimension limit separate from interaction locality. 4. Explicit finite model-specific bounds, rational test verification, and fail-closed certificate refusal contracts.

Roadmap: Minimal observer for future defect prediction. A direct consequence of the lifted cyclic structure in §5.16 is the characterization of the smallest observer sufficient to predict the future defect trajectory.

Using §5.16's lifted cyclic space $\mathcal{C}_m = \operatorname{span}\{\mathcal{L}^j z_m : j \geq 0\}$, define the unobservable subspace on $\mathcal{C}_m$:

$$N = \bigcap_{j=0}^{\kappa-1} \ker\left(O\mathcal{L}^j|_{\mathcal{C}_m}\right).$$

For a linear report Q on $\mathcal{C}_m$, the criterion is uniform over **all** $z, z' \in \mathcal{C}_m$: for every such pair,

$$Qz = Qz' \implies Oe^{t\mathcal{L}}z = Oe^{t\mathcal{L}}z' \quad \text{for every } t$$

if and only if $\ker(Q|_{\mathcal{C}_m}) \subseteq N$. For a fixed, known z_m, its trajectory is already determined by the model; the criterion concerns distinguishing possible lifted initial states.

On the full linear carrier $\mathcal{C}_m$, the minimum number of additional unrestricted linear measurements needed is

$$\dim\ker Q - \dim(\ker Q \cap N),$$

with both kernels taken on $\mathcal{C}_m$. The annihilator recurrence of Theorem 5.16 identifies the common kernel of all future readouts with N; Proposition 4.7's rank-nullity argument gives the measurement count.

For a restricted admitted subset $Z \subseteq \mathcal{C}_m$, the exact criterion instead is

$$(Z - Z) \cap \ker Q \subseteq N.$$

The ambient kernel inclusion is sufficient but need not be necessary on Z; the unrestricted linear-carrier measurement count is not asserted to be minimal there. Conditioning constants for imperfect reports require the singular values of the relevant restricted observability map.

5.17 Sharp onset, algebraic cancellation, and local zero certificates

Fix the finite tensor factors, a normalized product initial density m, and an affine real family of one-/two-local Hermitian Hamiltonians $H(\lambda) = H_0 + \sum_{a=1}^{r} \lambda_a H_a$. Fix S, $|S| = k \geq 2$. The allowed graph includes every edge occurring in this family; it is not recomputed after setting a coupling to zero. Write d for the global Hilbert dimension and use $K = \binom{d^2+k}{k}$ as a common, possibly large cutoff. In a fixed real Hermitian basis define the vector polynomials

$$b_j(\lambda) = \kappa_S^{(j)}(0;\lambda), \qquad E_j = \{\lambda : b_0(\lambda) = \cdots = b_j(\lambda) = 0\}.$$

Each coordinate of b_j has degree at most j. A parameter-dependent initial state is not included in this declaration.

Theorem 5.18 (finite cancellation stratification). If the allowed graph connects S, put $\tau = \tau_G(S)$. The onset-j stratum is exactly $E_{j-1} \setminus E_j$ for $\tau \le j < K$; the persistent-zero stratum is exactly E_{K-1}. All E_j are real algebraic sets. If b_τ is not the zero polynomial on the declared parameter space, then onset equals τ outside the proper algebraic exceptional set E_τ. This complement is open, dense and of full Lebesgue measure. If b_τ vanishes identically, no such generic-equality conclusion is licensed. Disconnected S gives persistent zero throughout the family.

Proof. Nested commutators give polynomial state jets of degree at most j. The partition/product formula for cumulants preserves that degree, since the derivative orders in every product sum to j. Theorem 5.14 makes all orders below τ zero. For each fixed parameter, Corollary 5.17 bounds its cyclic dimension by K, so vanishing of orders $0, \dots, K-1$ is equivalent to persistent zero. The first nonzero vector polynomial value therefore gives precisely the displayed stratum. A nonzero coordinate of b_τ has a zero set of empty interior and Lebesgue measure zero; E_τ is contained in it and is closed. Disconnected components factorize. □

This is a finite algebraic classification for a *specified* family, not an efficient algorithm in global volume. Different edge monomials can cancel at a particular parameter value. Exhibiting one nonzero monomial coefficient establishes that the polynomial is nonzero, not that every nonzero-coupling Hamiltonian attains the graph bound. The locality radius in Theorem 5.14 applies to each b_j, but not to the common global cutoff K.

Proposition 5.19 (a constructive sharp family for every finite tree). Let T be any finite tree with $k \ge 2$ vertices, put $S = V(T)$ and $r = k-1$. At vertex v take one qubit for each incident edge, so $\mathcal{H}_v = (\mathbb{C}^2)^{\otimes \deg(v)}$. Initialize all qubits in $|0\rangle$. Orient each edge $e = (u, v)$ once and set

$$H = \sum_{e=(u,v)} \lambda_e X_{u,e} X_{v,e}.$$

Define W_v as the product of Y on outgoing incidence qubits and X on incoming incidence qubits. Then the connected scalar readout is exactly

$$\chi_T(t) := \operatorname{Tr}\left[\left(\bigotimes_v W_v\right)\kappa_S(t)\right] = (-1)^r \prod_{e\in T} \sin(2\lambda_e t).$$

Consequently

$$\chi_T^{(r)}(0) = r!(-2)^r \prod_e \lambda_e, \qquad m_{\text{onset}} = r = \tau_T(S) \quad \text{if every } \lambda_e \neq 0.$$

If any coupling is zero the tree splits and the full cumulant stays zero. Thus this family's exceptions are exactly the coordinate hyperplanes.

Proof. Edge terms act on disjoint incidence qubits. On each edge the state is $\cos(\lambda_e t)|00\rangle - i\sin(\lambda_e t)|11\rangle$; its single-qubit X, Y means vanish and its $Y \otimes X$ mean is $-\sin(2\lambda_e t)$. Every nonempty proper vertex subset has a cut edge, so the moment of its W_v product contains a zero single-qubit factor. All proper-partition terms in the scalar cumulant vanish. The full moment is the product of the edge moments. Its nonzero order-r derivative, together with Theorem 5.14, establishes the full operator's onset. □

This supplies a nonzero point for every admitted tree in the full affine space of Hermitian edge interactions on these local dimensions, with this fixed initial state. Theorem 5.18 therefore gives generic equality there. Bounded tree degree keeps local dimension bounded. It does **not** establish generic equality for every qubit-only tree or every proper terminal subset with Steiner vertices: those families require their own nonzero coefficient. The three-qubit chain in §5.14 already supplies a sharp proper-terminal example. No uniform lower rate follows as a coupling tends to zero or as the number of tree edges grows.

A genuine delay with the same graph bound. For two qubits set $H = aX_1 + bZ_1X_2$ and $m = |00\rangle\langle 00|$. Direct differentiation gives

$$\kappa'_{12}(0) = 0, \qquad \kappa''_{12}(0) = ab(X \otimes X - Y \otimes Y).$$

Thus $ab \neq 0$ has onset two although $\tau = 1$. If $a = 0$, the first spin stays a Z eigenstate and the evolution is product; if $b = 0$ the evolution is local. These are all persistent-zero cases in this family. In particular, the order-one polynomial can vanish on an entire admitted family without forcing persistent zero.

Symmetry-protected exceptions, with an exact scope. Specify a finite unitary symmetry group Γ, require $U_g m U_g^* = m$ and $[U_g, H(\lambda)] = 0$, and suppose the chosen vector readout is equivariant into a real representation D_g. Its trajectory lies in $\mathrm{Fix}(D)$. If the projector $P_\Gamma = |\Gamma|^{-1} \sum_g D_g$ is zero, that readout stays zero. Indeed the evolved density remains invariant, and equivariance gives $R(t) = D_g R(t)$; averaging gives $R(t) = P_\Gamma R(t)$. For a nonzero fixed subspace this selection rule alone decides nothing. Parameterize the symmetry-allowed affine Hamiltonian space and restrict the polynomials b_j to it. Theorem 5.18 still gives its exact onset strata; the entire family has persistent zero if and only if every restricted b_j, $0 \leq j < K$, is the zero polynomial. Necessity follows from vanishing at every real parameter; sufficiency follows pointwise from the cutoff. This classifies exceptions relative to the declared symmetry family, not all possible symmetry groups. An odd scalar observable can be zero while the full cumulant is nonzero; its selection rule must not be promoted to an operator-zero certificate.

Proposition 5.20 (boundary-checked local persistent-zero certificate). Let $R \supseteq S$ be a finite region. Supply a finite-dimensional real space $\mathcal{A}_R$ of Hermitian operators, containing identity, with an explicit basis such that

$$i[H_R, \mathcal{A}_R] \subseteq \mathcal{A}_R, \qquad [h_Z, X] = 0.$$

The second identity is required for every boundary-crossing term h_Z and every basis vector X. Suppose the desired readout is a polynomial of degree at most p in the expectations of this basis. If $n_R = \dim \mathcal{A}_R$, vanishing of its first $\binom{n_R+p}{p}$ jets, including order zero, certifies persistent zero in every finite ambient volume with the same local and boundary data. A smaller verified cyclic annihilator is sufficient as in Theorem 5.16.

Proof. Terms outside R commute with supported operators. The boundary test removes every crossing contribution. Invariance under $i[H_R, \cdot]$ makes the expectation vector solve a closed finite linear ODE with initial values computed from m. Apply Theorem 5.16 to that vector and the declared polynomial. Its equality with the ambient expectations follows from uniqueness of the linear ODE. To certify the full κ_S, include a complete Hermitian operator basis on S and use its component polynomials, not only one measured contraction. □

The invariant region and crossing-commutator tests are substantive premises; a radius sufficient for a few jets is not a substitute. For example, a missing symmetry or one noncommuting boundary term invalidates this local certificate, even when several initial jets happen to vanish.

The finite checker in `code/local_persistent_zero.py` verifies the supplied Hermitian basis, identity containment, independence, local generator, every declared crossing commutator, initial density, polynomial degree and full cyclic annihilator. The certificate is bound to the entire declared model, including all Hamiltonian terms. The negative control changes a commuting $Z \otimes X$ boundary to $X \otimes X$: its first jet vanishes but its second is nonzero, and the certificate is rejected. These are finite model checks; completeness of a laboratory's Hamiltonian description is not inferred.

A measurement protocol for the leading coefficient. Fix Hermitian contractions M_i on S and measure the $2^k - 1$ subset moments $\langle \prod_{i \in B} M_i \rangle$. Their partition polynomial gives $\chi_S(t) = \mathrm{Tr}[(\bigotimes_i M_i)\kappa_S(t)]$. These measurements are supported on S; for Pauli M_i, one joint local basis setting supplies all subset moments. If every moment estimate, clipped to $[-1, 1]$, has error at most ϵ, telescoping partition products gives error at most $b_k\epsilon$, with b_k as in §5.14. Given verified zero lower jets and $a = \chi_S^{(m)}(0)/m!$, Theorem 5.14 gives

$$\left| \frac{\hat{\chi}_S(t)}{t^m} - a \right| \le B_{k,m}|t| + \frac{b_k\epsilon}{|t|^m},$$

$$B_{k,m} = a_k \binom{k+m}{m+1} (2g)^{m+1}.$$

It also gives $|\chi_S(t)| \ge |a||t|^m - B_{k,m}|t|^{m+1}$. Independent repetitions of bounded outcomes suffice: Hoeffding's inequality and a union bound give simultaneous moment error ϵ with probability at least $1 - \delta$ after $N \ge 2\epsilon^{-2} \log(2(2^k - 1)/\delta)$ shots of that setting. This does not assume independence between moments extracted from the same shot. Resolving higher onset orders costs precision; the error amplified by $|t|^{-m}$ is not suppressed by taking arbitrarily small times. Measuring one contraction certifies nonzero onset when it is nonzero, not persistent zero of the operator. Fixed-support tomography is an alternative, not global tomography. Bounds uniform in volume require fixed k, m, bounded g, and a positive lower bound on $|a|$.

The role of these results differs from Lieb–Robinson propagation bounds [1,3,4,9]: they identify exact initial cancellations and finite-model zero certificates, whereas those bounds constrain propagation. No priority for connected expansions, symmetry selection rules, or polynomial observability is claimed.

Finite-horizon observation trade-offs and marginal-indistinguishable state pairs. In experimental observation protocols with a finite total shot budget M_{total} distributed across M_{settings} measurement settings ($M_s = M_{\text{total}}/M_{\text{settings}}$ shots per setting), two distinct estimation targets can be compared: 1. *Leading Taylor/jet coefficient estimation:* For a k-site cumulant with verified zero jets below order m, the preceding estimate controls $\chi_S^{(m)}(0)/m!$ using a sampled cumulant, the analytic remainder, and measurement error amplified by $|t|^{-m}$. The ordered-Bell constant b_k controls conversion of subset moments into a support-k cumulant at each fixed time. It does not control the separate finite-difference or interpolation problem in time; that conditioning depends on the sampling times, derivative order and remainder assumptions. Finitely many samples alone do not determine an exact derivative. 2. *Future trajectory prediction:* Predicting the trajectory $\langle O \rangle_{\rho(t)}$ across a finite horizon $t \in [0, T]$ via the Duhamel convolution bound.

For one declared $[-1, 1]$-valued observable per setting and independent repetitions, Hoeffding's inequality and a union bound give simultaneous absolute estimation errors at most the following

value with probability at least $1-\delta$:

$$\epsilon_{\text{shot}} = \sqrt{\frac{2\ln(2M_{\text{settings}}/\delta)}{M_s}} = \sqrt{\frac{2M_{\text{settings}}\ln(2M_{\text{settings}}/\delta)}{M_{\text{total}}}},$$

On that probability event, a bound ϵ_{pred} for the same observable combines as $\epsilon_{\text{total}} \le \epsilon_{\text{pred}} + \epsilon_{\text{shot}}$. If each setting supplies multiple reported observables, the union bound counts those observables; a nonlinear cumulant requires its additional b_k error factor.

To illustrate how all lower-order marginals can agree identically while the k-body connected cumulant differs macroscopically, consider for $k=6$ the opposite-parity six-qubit GHZ states

$$|\text{GHZ}_6^{\pm}\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes 6} \pm |1\rangle^{\otimes 6}), \qquad \rho_{\pm} = |\text{GHZ}_6^{\pm}\rangle\langle\text{GHZ}_6^{\pm}|,$$

and the interpolated family $\rho(\epsilon) = \rho_{\text{cl}} + \frac{\epsilon}{2}(|0\rangle^{\otimes 6}\langle 1|^{\otimes 6} + |1\rangle^{\otimes 6}\langle 0|^{\otimes 6})$ for $\epsilon \in [-1,1]$, where $\rho_{\text{cl}} = \frac{1}{2}(|0\rangle^{\otimes 6}\langle 0|^{\otimes 6} + |1\rangle^{\otimes 6}\langle 1|^{\otimes 6})$ is the decohered classical mixture ($\rho(1)=\rho_+$ and $\rho(-1)=\rho_-$). For every proper subset $A \subsetneq \{1,\dots,6\}$ ($1 \le |A| \le 5$), tracing out the complement $V \setminus A$ annihilates the six-site off-diagonal coherence, so all proper reduced states coincide identically:

$$\text{Tr}_{V\setminus A}\,\rho(\epsilon) = \text{Tr}_{V\setminus A}\,\rho_{\text{cl}} = \frac{1}{2}(|0\rangle^{\otimes|A|}\langle 0|^{\otimes|A|} + |1\rangle^{\otimes|A|}\langle 1|^{\otimes|A|}) \qquad (1 \le |A| \le 5).$$

Because every proper marginal of order ≤ 5 is independent of ϵ, all lower-order marginal differences and all lower-order connected-cumulant differences vanish identically between $\rho(\epsilon)$ and ρ_{cl}, while the 6-site connected cumulant difference is exact:

$$\kappa_6(\rho(\epsilon)) - \kappa_6(\rho_{\text{cl}}) = \rho(\epsilon) - \rho_{\text{cl}} = \frac{\epsilon}{2}(|0\rangle^{\otimes 6}\langle 1|^{\otimes 6} + |1\rangle^{\otimes 6}\langle 0|^{\otimes 6}),$$

with trace norm $\|\kappa_6(\rho(\epsilon)) - \kappa_6(\rho_{\text{cl}})\|_1 = |\epsilon|$ and $\|\kappa_6(\rho_+) - \kappa_6(\rho_-)\|_1 = 2$ (witnessed by the 6-body Pauli observable $X^{\otimes 6}$, for which $\langle X^{\otimes 6}\rangle_{\rho_+} - \langle X^{\otimes 6}\rangle_{\rho_-} = 2$). These results complement multipartite Lieb–Robinson bounds (Tran et al. [25]): whereas power-law Lieb–Robinson bounds constrain spatial propagation velocity in long-range architectures, the algebraic cancellation strata and parity constructions above identify exact structural blind spots where few-body marginals carry zero signal despite order-unity k-body connected correlations.

5.18 Residual uncertainty and the time available for correction (NUM1)

A numerical residual and an error enclosure have different roles. Suppose the hypotheses of a short-time bound in this paper give

$$D(t) \ge \nu t - Kt^2 \qquad (0 \le t \le T),$$

where $K \ge 0$, and suppose an independently validated computation gives $|\hat{\nu} - \nu| \le \eta$ with $\hat{\nu}, \eta \ge 0$. Set $\nu_- := \max(0, \hat{\nu} - \eta)$. Then

$$D(t) \ge \max\{0, \nu_- t - Kt^2\}.$$

Proof. The enclosure implies $\nu \ge \nu_-$; substitution and $D(t) \ge 0$ give the claim. If $K > 0$ and $\nu_- > 0$, then $D(t) \ge \nu_- t/2$ on $0 \le t \le \min\{T, \nu_-/(2K)\}$. If $K = 0$, the linear bound holds throughout the declared interval. If $\hat{\nu} \le \eta$, this observation supplies no positive onset certificate. □

This is a boundary on what the recorded residual establishes, not a claim that the true residual vanishes when its uncertainty interval includes zero. For example, $\hat{\nu} = \eta = 1/50$ yields zero guaranteed margin. The selected rational arithmetic is replayed by `code/residual_error_budget.py`; that calculator does not establish the input enclosure or the physical remainder bound.

For upper error control, the actual-defect estimate of §5.10 extends across finitely many correction jumps. If e bounds the current trace-norm error, $r \geq 0$ bounds the actual defect rate on a horizon H, and $j \geq 0$ reserves the integration and correction-jump allowances, then

$$E(h) \leq e + rh + j \qquad (0 \leq h \leq H).$$

For a trace-norm target ε, with $e + j \leq \varepsilon$, the largest interval licensed by this scalar bound is

$$h_{\text{safe}} = \begin{cases} \min\{H, (\varepsilon - e - j)/r\}, & r > 0, \\ H, & r = 0. \end{cases}$$

If $e + j > \varepsilon$, the bound licenses no interval, including duration zero. Exhausting this sufficient budget does not prove that the actual error has crossed the target: cancellations may make the bound loose. PO-033 §3.5 gives the piecewise Duhamel proof and accounts for each jump. PO-034 §4.3 propagates state uncertainty through the product retraction and its declared observation map.

These distinctions matter for the accompanying RSI runtime experiment. A computed tiny negative Choi eigenvalue can be an eigensolver artifact, whereas a genuinely negative binary-rational Choi direction must still be refused. Likewise, an approximate Hamiltonian propagator may be useful as a numerical diagnostic while failing exact unitary admission. An accepted diagnostic, a state enclosure and a bound on the actual trajectory defect remain separate inputs. The extension uses the established a posteriori perspective of Zoufal, Sutter and Woerner; the present jump and scalar-budget consequences are derived above and in PO-033.

6. Worked Examples

6.1 Example 1: Two Qubits with Exact Trace-Norm Defect

Consider two qubits with Hamiltonian and initial state

$$H = J(X \otimes X), \qquad \rho(0) = |00\rangle\langle 00|.$$

The evolved state is $|\psi(t)\rangle = \cos(Jt)|00\rangle - i\sin(Jt)|11\rangle$. Write $c = \cos(Jt)$ and $s = \sin(Jt)$. The product retraction is the product of single-site marginals $\Pi\rho(t) = \rho_A(t) \otimes \rho_B(t)$ with $\rho_A(t) = \rho_B(t) = c^2|0\rangle\langle 0| + s^2|1\rangle\langle 1|$. In the ordered basis $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$,

$$\rho(t) - \Pi\rho(t) = \begin{pmatrix} c^2s^2 & 0 & 0 & i\,cs \\ 0 & -c^2s^2 & 0 & 0 \\ 0 & 0 & -c^2s^2 & 0 \\ -i\,cs & 0 & 0 & c^2s^2 \end{pmatrix}.$$

This Hermitian matrix has eigenvalues $\{c^2s^2 + cs,\ c^2s^2 - cs,\ -c^2s^2,\ -c^2s^2\}$, so for $|Jt| \leq \pi/4$ (where $|cs| \geq c^2s^2$) its trace norm is

$$D_{12}(t) = \|\rho(t) - \Pi\rho(t)\|_1 = 2|cs| + 2c^2s^2.$$

The absolute value covers both signs of J: for $J < 0$ we have $s = \sin(Jt) < 0$, so $cs < 0$ and the two eigenvalues $c^2s^2 \pm cs$ exchange roles, leaving $|c^2s^2 + cs| + |c^2s^2 - cs| = 2|cs|$ unchanged whenever $|cs| \geq c^2s^2$. At $t = \pi/(8|J|)$ we get $|Jt| = \pi/8$, so $|cs| = \frac{1}{2\sqrt{2}}$ and $c^2s^2 = \frac{1}{8}$, hence

$$D_{12}\left(\tfrac{\pi}{8|J|}\right) = \tfrac{1}{\sqrt{2}} + \tfrac{1}{4} \approx 0.9571.$$

The calculation is verified by the following script:

```
import numpy as np
for theta in (np.pi / 8, -np.pi / 8):  # Jt > 0 and Jt < 0
    c, s = np.cos(theta), np.sin(theta)
    a, b = c * c * s * s, c * s
    Delta = np.array(
        [
            [a, 0, 0, 1j * b],
            [0, -a, 0, 0],
            [0, 0, -a, 0],
            [-1j * b, 0, 0, a],
        ],
        dtype=complex,
    )
    # both signs: 0.9571067811865476
    print(np.sum(np.abs(np.linalg.eigvalsh(Delta))))
    print(2 * abs(b) + 2 * a)
```

The defect is positive and computed in closed form, witnessing non-product structure that the marginal retraction cannot represent.

6.2 Example 2: Transverse-Field Ising Model

Consider the 1D transverse-field Ising model $H = -J\sum_i Z_iZ_{i+1} - h\sum_i X_i$ with initial state $|+\rangle^{\otimes N}$ and bond term $H_e = -J\,Z_aZ_b$. We compute both the raw rate ν_e and the dimension-normalized rate γ_e in closed form. Because $|+\rangle$ is an eigenstate of X, the transverse field commutes with the initial one-site state and contributes nothing to the first-order normal rate. With $\rho_a = \rho_b = |+\rangle\langle+|$ and $(Z \otimes Z)|{+}{+}\rangle = |{-}{-}\rangle$,

$$-i[H_e, \rho_a \otimes \rho_b] = iJ(|{-}{-}\rangle\langle{+}{+}| - |{+}{+}\rangle\langle{-}{-}|).$$

Both one-site marginals of this operator vanish (each term carries a factor $\langle+|-\rangle = 0$), so P_e annihilates it and Q_e acts as the identity: $n_e = -i[H_e, \rho_a \otimes \rho_b]$. In the two-dimensional span of $\{|{+}{+}\rangle, |{-}{-}\rangle\}$ this is J times a Pauli-Y, with eigenvalues $\pm J$, hence

$$\nu_e = \|n_e\|_1 = 2|J|, \qquad \gamma_e = \frac{\nu_e}{d^2} = \frac{|J|}{2}.$$

For $J \neq 0$ this holds on every bond, for every field strength h, independently of N. Thus the active-edge hypothesis holds with $\eta = 1$ and raw $\nu_{\min} = 2|J|$. Lemma 5.3 covers the one-local field terms, and with $g \leq 2|J| + |h|$, Theorem 5.1 gives the half-slope window

$$0 < t \leq \frac{|J|}{24(2|J| + |h|)^2}.$$

The calculation is verified by the following script:

```python
import numpy as np
plus = np.array([1, 1]) / np.sqrt(2)
rp = np.outer(plus, plus)  # |+><+|
Z = np.array([[1, 0], [0, -1]])
He = -np.kron(Z, Z)  # J = 1 bond term
rab = np.kron(rp, rp)
rhodot = -1j * (He @ rab - rab @ He)

def ptr(M, keep):
    # partial trace of a 2-qubit operator
    M = M.reshape(2, 2, 2, 2)
    if keep == 0:
        return np.einsum("ijkj->ik", M)
    return np.einsum("ijil->jl", M)

Ne = (
    rhodot
    - np.kron(ptr(rhodot, 0), rp)
    - np.kron(rp, ptr(rhodot, 1))
)
# 2.0 => nu_e = 2, gamma_e = 1/2
print(np.sum(np.abs(np.linalg.eigvalsh(Ne))))
```

The rate is positive and computed in closed form on every bond, so the active-edge density hypothesis holds with $\eta = 1$.

6.3 Quadratic and Cubic Onset

A zero first-order rate does not determine the first nonzero order. Two small systems illustrate what the Taylor bounds add.

For $H = Z \otimes Z + X \otimes I$ and initial state $|0+\rangle$, direct differentiation gives

$$F'(0) = 0, \qquad F''(0) = \begin{pmatrix} 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix}.$$

Its eigenvalues are $2, -2, 0, 0$, so $D(t) = 2t^2 + O(t^3)$. Here $g = 2$, and Theorem 5.5 gives the explicit bound $D(t) \geq 2t^2 - 512t^3$.

For the two-qubit Ising Hamiltonian

$$H = -JZ \otimes Z - h(X \otimes I + I \otimes X), \qquad \rho(0) = |00\rangle\langle 00|,$$

the first *two* residual derivatives vanish:

$$F'(0) = F''(0) = 0, \qquad F'''(0) = 8iJh^2(|00\rangle\langle 11| - |11\rangle\langle 00|).$$

Thus, when $Jh \neq 0$,

$$\|F'''(0)\|_1 = 16|J|h^2, \qquad D(t) = \frac{8}{3}|J|h^2t^3 + O(t^4).$$

With $g = |J| + |h|$, the quantitative bound is

$$D(t) \geq \frac{8}{3}|J|h^2t^3 - 160g^4t^4.$$

It retains half the leading term for $0 < t \leq |J|h^2/(120g^4)$.

These derivatives can be checked by expanding the state amplitudes:

$$\begin{aligned} c_{00}(t) &= 1 + iJt - \tfrac{1}{2}(J^2 + 2h^2)t^2 - \tfrac{i}{6}J(J^2 + 2h^2)t^3 + O(t^4), \\ c_{01}(t) = c_{10}(t) &= iht - \tfrac{i}{6}h(J^2 + 4h^2)t^3 + O(t^4), \\ c_{11}(t) &= -h^2t^2 - \tfrac{i}{3}Jh^2t^3 + O(t^4). \end{aligned}$$

Forming $|\psi(t)\rangle\langle\psi(t)|$, tracing its marginals and subtracting their product gives the displayed residual derivatives. In particular, the isolated term $-2\dot{\rho}_a(0) \otimes \dot{\rho}_b(0) = -2h^2Y \otimes Y$ is canceled by the other terms in $F''(0)$. Keeping it alone would incorrectly predict correlation under independent local fields even at $J = 0$.

The exact symbolic implementation in `code/short_time_jets.py` and its tests compare nested-commutator derivatives with this separate amplitude expansion. They also retain the tangent-commutator example, independent local evolution, mixed-state centering, the pure-state rate identity and a four-site locality example. The Lean file `lean4/po000_short_time_jets.lean` checks the support-factor and factorial-normalization arithmetic, including the constants 24, 64 and 160. The analytic and matrix-norm arguments remain written proofs.

---

7. Observer Visibility and the Local Green / Global Red Gap

This section separates three distinct mathematical objects: a direction-valued projection residual, a declared 1-cochain obtained by applying linear observation maps to that residual, and a coboundary obtained only when an explicit 0-cochain potential is supplied. The finite quantum lower bounds in §5 concern the raw trace norm of the first object, whereas the characterization in §4 governs global aggregation once a local gauge family has been specified.

7.1 Relation to Hilbert extensions

The conditional ℓ^1 characterization of §4 does not require every downstream statistic to remain an ℓ^1 sum. Passing to an L^2_{RMS} or Hilbert-space envelope is an additional representation step that introduces inner-product structure and phase-symmetry assumptions not used in Theorems 3.1, 4.2, or 5.1.

7.2 Projection residual, induced cochain, and coboundary

Let R_e be a local residual space on edge e and let

$$q_e : R_e \longrightarrow V_e$$

be a bounded linear observation map into a normed vector space V_e. For a local normal residual $n_e \in R_e$, define

$$\omega_e := q_e(n_e).$$

The family $\omega = (\omega_e)_e$ is an **induced 1-cochain** in $C^1 = \prod_e V_e$. Forming ω does not by itself make ω a coboundary.

It is useful to distinguish three structural levels:

1. **Induced cochain:** the observation maps q_e are specified and $\omega_e = q_e(n_e)$ is evaluated edgewise.
2. **Quotient obstruction:** a coboundary differential $d_0 : C^0 \to C^1$ and norm gauge are specified, so the quotient seminorm
$$\Phi([\omega]) = \inf_{\alpha \in C^0} \|\omega - d_0\alpha\|_1$$
is well defined.
3. **Coboundary (exactness):** an explicit 0-cochain $\alpha \in C^0$ is exhibited with $\omega = d_0\alpha$.

In finite-dimensional spaces, $\mathrm{im}(d_0)$ is automatically closed, so $\Phi([\omega]) = 0$ if and only if $\omega \in \mathrm{im}(d_0)$. In an infinite-dimensional Banach presentation, closedness of $\mathrm{im}(d_0)$ must be verified separately; otherwise $\Phi([\omega])$ measures only the distance to $\overline{\mathrm{im}(d_0)}$.

Identifying a physical residual with a coboundary requires level 3. The companion paper [6] develops the finite cochain-adapter schema and higher-cell extensions when a second differential $d_1 : C^1 \to C^2$ satisfying $d_1 d_0 = 0$ is present. At the covering-edge level of the present paper, C^2, d_1, and any descended curvature map $\overline{d}_1 : Q^1 \to C^2$ are not needed until §7.7–§7.9; even when $d_1 d_0 = 0$ holds, the cocycle condition $d_1\omega = 0$ is an independent closure property that does not follow solely from ω being induced by a projection residual.

7.3 Relation to higher-cell and representation extensions

Within the finite obstruction framework [8], a nonvanishing local residual or quotient class diagnoses an incompatibility of the chosen reduced representation; passing to a richer representation space or higher-order cell complex is an explicit modeling step with its own compatibility conditions. The companion work [7] studies the analogous projection-residual diagnostics in compressed evaluation pipelines.

7.3.3 Covering-Edge Diagrams and Poset Coboundaries Before analyzing observer visibility, we fix the compact algebraic setup for covering-edge descent complexes on posets (companion package **PO-001** develops the full saturated-chain presheaf theory; see also Mac Lane [2]). Let $(P, \leq)$ be a finite partially ordered set, and let $E_{\mathrm{cov}} = \{(a, b) \in P \times P : a \lessdot b\}$ denote its set of covering edges (b covers a). A **covering-edge Banach diagram** assigns to each element $a \in P$ a real or complex normed fiber space $(F(a), \|\cdot\|_{F(a)})$, and to each covering edge $(a, b) \in E_{\mathrm{cov}}$ a contractive linear transition map $r_{b,a} : F(b) \to F(a)$ ($\|r_{b,a}\| \leq 1$).

The corresponding 0-cochain space (space of global sections) and 1-cochain space are the direct product/sum spaces
$$C^0 = \bigoplus_{a \in P} F(a), \qquad C^1 = \bigoplus_{e=(a,b) \in E_{\mathrm{cov}}} F(a),$$
equipped with their ℓ^1 fiber sums. The **canonical covering-edge coboundary operator** $d_0 : C^0 \to C^1$ assigns to each section $s = (s(a))_{a \in P} \in C^0$ the edgewise compatibility defect
$$(d_0 s)_e = \Delta(s)_e := r_{b,a} s(b) - s(a) \ \in \ F(a) \qquad \text{for } e = (a, b) \in E_{\mathrm{cov}}.$$
A section $s \in C^0$ is a global section (or compatible) if and only if $d_0 s = 0$; the global compatibility defect norm is $I(s) := \|d_0 s\|_1 = \sum_{e \in E_{\mathrm{cov}}} \|\Delta(s)_e\|_{F(a)}$.

7.4 Best-Single-Observer Visibility Gap and the Local Green / Global Red Bound

While §7.2 distinguishes an induced cochain from a quotient class and a coboundary, a further distinction arises when a glued section is monitored by a *family of projection-limited local observers*. Each observer inspects only a subset of covering edges through contractive linear observation maps. When the kernels of these observation maps jointly cover the subspace supporting the edgewise compatibility defect, every individual observer reports zero local defect ("Local Green") even though the global compatibility defect norm is strictly positive ("Global Red").

Here "local" specifies which coordinates an observer can inspect. Bell local causality instead constrains conditional response probabilities, and local tomography (Finite QM [20, §13.8]) constrains determination of composite states by product effects. Section 7.10 separates these conditions.

7.4.1 Definition (local observer carrier family) Let F be a covering-edge Banach diagram over the finite poset $(P, \leq)$ as defined in §7.3.3, and for an edge $e = (a, b) \in E_{\mathrm{cov}}$ write $a_e := a$ for its lower endpoint. A **local observer carrier** is a pair

$$\mathcal{C}^\alpha \;=\; (E^\alpha,\ (q_e^\alpha)_{e\in E^\alpha}),$$

where $E^\alpha \subseteq E_{\mathrm{cov}}$ is the observer's edge domain and each

$$q_e^\alpha : F(a_e) \longrightarrow V_e^\alpha$$

is a contractive linear observation map into a normed space V_e^α. A **local observer carrier family** is a finite, nonempty collection $\{\mathcal{C}^\alpha\}_{\alpha\in A}$ $(A \neq \emptyset)$. For an edge e, its **watcher set** is $W(e) := \{\alpha \in A : e \in E^\alpha\}$, and its **joint watcher kernel** is

$$K_e \;:=\; \begin{cases} \bigcap_{\alpha\in W(e)} \ker q_e^\alpha, & W(e) \neq \emptyset, \\ F(a_e), & W(e) = \emptyset. \end{cases}$$

The family has **full edge coverage** when $\bigcup_{\alpha\in A} E^\alpha = E_{\mathrm{cov}}$, i.e., $W(e) \neq \emptyset$ for every edge $e \in E_{\mathrm{cov}}$.

7.4.2 Definition (local tallies, Local Green, Global Red, visibility gap) For a section $s \in C^0$, with edgewise compatibility defect $\Delta(s)_e = (d_0 s)_e = r_{b,a}(s(b)) - s(a)$ as in §7.3.3, define the **observed local tally** of observer α:

$$J^\alpha(s) \;:=\; \sum_{e\in E^\alpha} \big\| q_e^\alpha(\Delta(s)_e) \big\|_{V_e^\alpha}.$$

We say that observer α reports **Local Green** at s when $J^\alpha(s) = 0$, and that the family is **All-Green** at s when $J^\alpha(s) = 0$ for every $\alpha \in A$. We say that s is **Global Red** when $I(s) = \sum_{e\in E_{\mathrm{cov}}} \|\Delta(s)_e\| > 0$. The **best-single-observer visibility gap** of the family at s is

$$\Lambda_{\max}(s) \;:=\; I(s) \;-\; \max_{\alpha\in A} J^\alpha(s) \;\geq\; 0 \qquad \text{(nonnegativity by Proposition 7.4.3).}$$

When no ambiguity can arise, we write Λ for $\Lambda_{\max}$.

Remark 7.4.2.1 (Unfaithfulness of observed local gauges). Each observer's induced local gauge $j_e^\alpha(v) := \|q_e^\alpha v\|_{V_e^\alpha}$ is a seminorm, so it is convex and positively homogeneous, but it is not

faithful whenever $\ker q_e^\alpha \neq \{0\}$: the implication $J_{\{e\}}(x) = 0 \Rightarrow x = 0$ of Axiom RE1 (§4.1) fails on $\ker q_e^\alpha$. Defining $p_e := \max_{\alpha \in W(e)} \|q_e^\alpha \Delta(s)_e\|$ (with $p_e := 0$ if $W(e) = \emptyset$) decomposes the visibility gap into two nonnegative contributions:

$$\Lambda_{\max}(s) = \underbrace{\sum_e (\|\Delta(s)_e\| - p_e)}_{\text{edgewise observation/contraction loss}} + \underbrace{\left(\sum_e p_e - \max_\alpha J^\alpha(s)\right)}_{\text{single-observer domain fragmentation}} .$$

Thus a positive visibility gap $\Lambda_{\max}(s) > 0$ can arise either from nontrivial observer kernels, from no single observer watching all edges carrying visible defect, or from both mechanisms simultaneously.

7.4.3 Proposition (passivity and Green propagation) For every section $s \in C^0$ and every observer $\alpha \in A$,

$$J^\alpha(s) \;\leq\; \sum_{e \in E^\alpha} \|\Delta(s)_e\| \;\leq\; I(s).$$

In particular, $I(s) = 0$ implies $J^\alpha(s) = 0$ for every $\alpha \in A$: global compatibility implies Local Green for every observer.

Proof. Each q_e^α is contractive, so $\|q_e^\alpha(\Delta(s)_e)\|_{V_e^\alpha} \leq \|\Delta(s)_e\|_{F(a_e)}$ termwise. Summing over $E^\alpha \subseteq E_{\text{cov}}$ and enlarging the domain of summation to E_{cov} (all summands being nonnegative) yields both inequalities. □

Contractivity ensures that passive linear observations can attenuate or annihilate defect norm, but cannot manufacture it. The converse fails whenever the defect lies inside the joint watcher kernels.

7.4.4 Proposition (kernel-cover characterization of All-Green) Fix a section $s \in C^0$. The observer family is All-Green at s **if and only if**

$$\Delta(s)_e \in K_e \qquad \text{for every } e \in E_{\text{cov}}.$$

Consequently:

1. *(Local Green / Global Red coexistence.)* If $I(s) > 0$ and $\Delta(s)_e \in \ker q_e^\alpha$ for every edge e and every watcher $\alpha \in W(e)$, then every observer reports Local Green while s is Global Red, and $\Lambda(s) = I(s)$.
2. *(Sufficiency versus necessity.)* All-Green implies $I(s) = 0$ under the joint-kernel triviality hypothesis of Corollary 7.4.7; without that hypothesis the implication can fail even under full edge coverage (Proposition 7.4.5). Conversely, edgewise joint-kernel triviality is not necessary for local-to-global soundness on coupled carriers, because coboundary coupling across shared fibers can force invisible edge defects to vanish (Proposition 7.4.7-B).

Proof. Because $J^\alpha(s)$ is a finite sum of nonnegative norms, $J^\alpha(s) = 0$ if and only if $q_e^\alpha(\Delta(s)_e) = 0$, i.e., $\Delta(s)_e \in \ker q_e^\alpha$, for all $e \in E^\alpha$. Quantifying over all $\alpha \in A$ shows that All-Green holds if and only if $\Delta(s)_e \in \bigcap_{\alpha \in W(e)} \ker q_e^\alpha = K_e$ for every watched edge e, while for unwatched edges ($W(e) = \emptyset$) the condition $\Delta(s)_e \in F(a_e) = K_e$ is vacuous. Claim 1 and Claim 2 follow immediately. □

7.4.5 Proposition (three-patch Local Green / Global Red witness) There exists a covering-edge Banach diagram, a section s, and a local observer carrier family with **full edge coverage** such that the family is All-Green at s while $I(s) = 6 > 0$.

Construction and proof. Take the finite poset

$$P = \{x_{12}, x_{23}, x_{31}, u_1, u_2, u_3\}, \qquad x_{ij} < u_i, \quad x_{ij} < u_j,$$

consisting of three patches u_1, u_2, u_3 and three pairwise overlaps x_{ij}; every displayed relation is a covering relation, so

$$E_{\mathrm{cov}} = \{(x_{ij}, u_i),\ (x_{ij}, u_j) \ : \ \{i,j\} \subset \{1,2,3\}\}, \qquad |E_{\mathrm{cov}}| = 6.$$

Assign to every element the fiber $F(x) = (\mathbb{R}^3, \|\cdot\|_1)$, and let every restriction map $r_{u,x}$ be the identity map. Let (e_1, e_2, e_3) be the standard basis of $\mathbb{R}^3$ and define the section

$$s(u_i) = e_i \ \ (i = 1,2,3), \qquad s(x_{ij}) = 0.$$

Then on each covering edge (x_{ij}, u_i),

$$\Delta(s)_{(x_{ij},u_i)} = r_{u_i,x_{ij}}(s(u_i)) - s(x_{ij}) = e_i,$$

so each of the six edge defects is a standard basis vector of ℓ^1 norm 1, giving

$$I(s) = \sum_{e \in E_{\mathrm{cov}}} \|\Delta(s)_e\|_1 = 6 > 0.$$

Define three observers $\mathcal{C}^1, \mathcal{C}^2, \mathcal{C}^3$, where observer α watches the two edges whose upper element is u_α,

$$E^\alpha = \{(x_{\alpha\beta}, u_\alpha) \ : \ \beta \neq \alpha\},$$

through the coordinate-deletion projection

$$q_e^\alpha = \pi_\alpha : \mathbb{R}^3 \to \mathbb{R}^2, \qquad \pi_\alpha \text{ deletes coordinate } \alpha, \qquad \ker \pi_\alpha = \mathbb{R}\, e_\alpha,$$

which is ℓ^1-contractive. Every edge (x_{ij}, u_i) belongs to E^i, so $\bigcup_\alpha E^\alpha = E_{\mathrm{cov}}$ (full edge coverage) with $W((x_{ij}, u_i)) = \{i\}$ and $K_{(x_{ij},u_i)} = \mathbb{R}\, e_i$. On each edge watched by α, the defect is $e_\alpha \in \ker \pi_\alpha$, so

$$J^\alpha(s) = \sum_{e \in E^\alpha} \|\pi_\alpha(\Delta(s)_e)\|_1 = 0 \qquad (\alpha = 1,2,3).$$

Thus the family is All-Green while $\Lambda(s) = I(s) = 6$. □

The construction is directly verified by the following script:

```
import numpy as np

e = np.eye(3)
s = {
    "u1": e[0],
    "u2": e[1],
    "u3": e[2],
    "x12": np.zeros(3),
    "x23": np.zeros(3),
    "x31": np.zeros(3),
```

```
}
edges = [
    ("x12", "u1"),
    ("x12", "u2"),
    ("x23", "u2"),
    ("x23", "u3"),
    ("x31", "u3"),
    ("x31", "u1"),
]
watcher = {
    ("x12", "u1"): 0,
    ("x12", "u2"): 1,
    ("x23", "u2"): 1,
    ("x23", "u3"): 2,
    ("x31", "u3"): 2,
    ("x31", "u1"): 0,
}
# identity restrictions
d = {ed: s[ed[1]] - s[ed[0]] for ed in edges}
I = sum(np.abs(v).sum() for v in d.values())
J = [
    sum(
        np.abs(np.delete(d[ed], a)).sum()
        for ed in edges
        if watcher[ed] == a
    )
    for a in range(3)
]
print(I, J)  # 6.0 [0.0, 0.0, 0.0]
```

7.4.6 Corollary (unbounded visibility gap; absence of a recovery modulus) With the diagram and observers of Proposition 7.4.5 held fixed, the scaled sections s_λ ($s_\lambda(u_i) = \lambda e_i$, $s_\lambda(x_{ij}) = 0$, $\lambda > 0$) satisfy $J^\alpha(s_\lambda) = 0$ for all α and $I(s_\lambda) = 6\lambda$. Consequently, for every $R > 0$ there exists an All-Green section with $I(s) = R$, and there is **no** function $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ satisfying

$$I(s) \;\leq\; f\Big(\max_{\alpha \in A} J^\alpha(s)\Big) \qquad \text{for all sections } s \in C^0.$$

Proof. By linearity of Δ (§2.2) and positive homogeneity of the norms, $I(s_\lambda) = \lambda I(s_1) = 6\lambda$ and $J^\alpha(s_\lambda) = \lambda J^\alpha(s_1) = 0$. Any such function f would have to satisfy $f(0) \geq 6\lambda$ for all $\lambda > 0$, which contradicts that $f(0)$ is a fixed non-negative real scalar. □

7.4.7 Corollary (sufficient condition for local-to-global soundness) Suppose the observer family has full edge coverage and edgewise joint-kernel triviality: $K_e = \{0\}$ for every $e \in E_{\text{cov}}$, so that the combined watcher map $\mathcal{Q}_e : v \longmapsto (q_e^\alpha v)_{\alpha \in W(e)}$ is injective on every edge fiber. Then All-Green at s implies $I(s) = 0$.

Proof. By Proposition 7.4.4, All-Green implies $\Delta(s)_e \in K_e = \{0\}$ for every $e \in E_{\text{cov}}$, so $\Delta(s)_e = 0$ for all e and $I(s) = 0$. □

Proposition 7.4.5 shows that full edge coverage alone does not suffice when $K_e \neq \{0\}$. Injectivity of the combined watcher map $\mathcal{Q}_e$ does not require any single watcher q_e^α to be faithful on its own.

Moreover, as the next proposition shows, edgewise joint-kernel triviality is a sufficient condition on arbitrary 1-cochains, but is not necessary for section-realized defects $\Delta(s)$, because shared lower fibers couple the defects across adjacent covering edges.

7.4.7-B Proposition (non-necessity of joint-kernel triviality on coupled carriers) There exists a covering-edge Banach diagram and a local observer carrier family with **full edge coverage** and a **nontrivial** joint watcher kernel $K_e \neq \{0\}$ on *every* edge, such that All-Green at any section $s \in C^0$ nevertheless forces $I(s) = 0$.

Construction and proof. Take the poset $P = \{x, u_1, u_2\}$ with covering relations $x < u_1$ and $x < u_2$, fibers $F(\cdot) = (\mathbb{R}^2, \|\cdot\|_1)$, and zero restriction maps $r_{u_i,x} := 0$ $(i = 1, 2)$. Then for every section $s \in C^0$,

$$\Delta(s)_{(x,u_i)} = r_{u_i,x}(s(u_i)) - s(x) = -s(x) \qquad (i = 1, 2).$$

Thus the realizable defect cochains lie on the diagonal of C^1. Define two observers: observer $i \in \{1, 2\}$ watches only the edge (x, u_i) through the contractive coordinate functional $q^i(v_1, v_2) := v_{3-i}$. Edge coverage is full, each edge has the single watcher i, and the joint watcher kernels are nontrivial on both edges:

$$K_{(x,u_1)} = \mathbb{R}\, e_1 \neq \{0\}, \qquad K_{(x,u_2)} = \mathbb{R}\, e_2 \neq \{0\}.$$

If the family is All-Green at s, Proposition 7.4.4 gives $\Delta(s)_{(x,u_i)} = -s(x) \in K_{(x,u_i)}$ for $i = 1, 2$, so $-s(x) \in \mathbb{R}e_1 \cap \mathbb{R}e_2 = \{0\}$. Hence $s(x) = 0$, both edge defects vanish, and $I(s) = 0$. □

```
# coupled two-edge carrier: both restrictions are the zero map, so both
# edge defects equal -s(x); observer i sees the OTHER coordinate.
for sx in ((0.0, 0.0), (1.0, 0.0), (0.0, -2.0), (3.0, 4.0)):
    d = (-sx[0], -sx[1])            # both edges carry this defect
    J1, J2 = abs(d[1]), abs(d[0])   # observer 1 reads coord 2, and vice versa
    I = 2 * (abs(d[0]) + abs(d[1]))
    all_green = J1 == 0 and J2 == 0
    print(sx, all_green, I)         # all_green is True only when I == 0
```

7.4.8 Remarks on represented defect norms versus quotient cohomology

1. **Represented defect norm versus quotient obstruction.** Every quantity $I(s)$ in §7.4 is the ℓ^1 compatibility defect norm of a declared section $s \in C^0$ (§2.2). Because $\Delta(s) = d_0 s$ is a coboundary by definition (level 3 of §7.2), its quotient semi-norm vanishes identically: $\Phi([\Delta(s)]) = 0$. Thus the three-patch example of Proposition 7.4.5 exhibits a section with $I(s) > 0$ and $\Phi([\Delta(s)]) = 0$. Section 7.4.10 below complements this by exhibiting an explicit non-exact 1-cochain $\omega \in C^1$ on the two-edge carrier with strictly positive quotient norm $\Phi([\omega]) = 2 > 0$.
2. **Notation for sheaf Laplacians.** Whenever a Hilbert-space completion is introduced on a finite carrier, we write $L_0 := d_0^* d_0$ for the 0-cochain Hodge Laplacian so as not to collide with the covering-edge coboundary operator Δ of §2.2.

7.4.9 Lean 4 formalization of the visibility-gap propositions The results of §7.4 are mechanized in `lean4/po000_l004_observer_visibility.lean`:

- Proposition 7.4.3: `PO000.ObserverVisibility.tally_le_rawNorm` and `localGreen_of_globalGreen`.
- Proposition 7.4.4: `PO000.ObserverVisibility.allGreen_iff_kernel`.
- Proposition 7.4.5: `PO000.ObserverVisibility.ThreePatch.three_patch_witness` and `ThreePatch.scaledDefect_is_section_defect`.

- Corollary 7.4.6: `PO000.ObserverVisibility.ThreePatch.rawNorm_scaled` and `ThreePatch.no_recovery_modulus`.
- Corollary 7.4.7: `PO000.ObserverVisibility.rawNorm_eq_zero_of_allGreen` and `coverage_of_jointTrivial`.
- Proposition 7.4.7-B: `PO000.ObserverVisibility.Coupled.full_coverage`, `Coupled.joint_kernel_nontrivial`, and `Coupled.allGreen_forces_zero`.
- §7.4.10: `PO000.ObserverVisibility.QuotientWitness.repair_mass_ge_two`, `QuotientWitness.repair_mass_attained`, and `QuotientWitness.omega_not_exact`.

7.4.10 A fiber-level quotient witness with $\Phi([\omega]) > 0$ On the two-edge carrier of Proposition 7.4.7-B with scalar fibers $F(\cdot) = \mathbb{R}$ and zero restriction maps $r_{u_i,x} = 0$, the coboundary image is the diagonal

$$\operatorname{im}(d_0) = \{(-\alpha, -\alpha) : \alpha \in \mathbb{R}\} \subset \mathbb{R}^2.$$

For the 1-cochain $\omega = (1, -1) \in C^1$, the quotient obstruction norm is

$$\Phi([\omega]) = \inf_{\alpha \in C^0} \|\omega - d_0\alpha\|_1 = \inf_{\alpha \in \mathbb{R}} (|1 + \alpha| + |\alpha - 1|) = 2 > 0,$$

where the lower bound $|1 + \alpha| + |1 - \alpha| \geq 2$ follows from the triangle inequality and is attained at $\alpha = 0$. In particular, $\omega \notin \operatorname{im}(d_0)$.

```
# quotient mass of omega = (1, -1) against the diagonal im(d0)
masses = [
    abs(1 + a / 8) + abs(a / 8 - 1)
    for a in range(-64, 65)  # alpha = -8 .. 8 in steps of 1/8
]
print(min(masses))  # 2.0, attained at alpha = 0; never below 2
```

7.5 Phase information discarded by normalized weights

The excitation profile in §5.12 forgets relative directions. A second, finite Hilbert-space question makes that loss quantitative. Fix weights $p_j \geq 0$, $\sum_j p_j = 1$, and declare the **full complex phase fiber**

$$\mathcal{F}_p = \{c : c_j = \sqrt{p_j} e^{i\theta_j},\ \theta_j \in \mathbb{R}\}.$$

These are hypotheses about the admitted carrier. A finite sign carrier, a list of Gaussian vectors, or an independently phase-complete Hilbert carrier need not have the same attainable outputs. None is selected by normalization.

Theorem 7.5 (phase interval and scalar minimax). For a fixed complex row a, put $r_j = |a_j|\sqrt{p_j}$, $R = \sum_j r_j$, and $\ell = \max(0, 2\max_j r_j - R)$. Then

$$\{|ac|^2 : c \in \mathcal{F}_p\} = [\ell^2, R^2].$$

The best deterministic scalar prediction from p alone, under worst absolute error on this fiber, is

$$m_a(p) = \tfrac{1}{2}(R^2 + \ell^2), \qquad \varepsilon_a(p) = \tfrac{1}{2}(R^2 - \ell^2).$$

When $\|a\|_2 \leq 1$, a^*a is an effect and the interval lies in $[0, 1]$. For a general row the interval formula remains algebraic but is not necessarily a probability interval.

Proof. Absorb the arguments of a_j into the independent phases. The triangle and reverse-triangle inequalities give the proposed endpoints. To establish that all moduli between them are attainable, induct on the lengths. Suppose previous resultants have moduli $[L, U]$. Adding a vector of length s gives moduli $[|x - s|, x + s]$ for $x \in [L, U]$. The total image is compact and connected, as the continuous image of $[L, U] \times S^1$. Its maximum is $U + s$ and its minimum is $\max(0, L - s, s - U)$. Induction gives exactly $[\ell, R]$, including zero lengths. Squaring preserves this interval. Any prediction z has worst error at least half the endpoint separation, since $R^2 - \ell^2 \le |R^2 - z| + |z - \ell^2|$. The midpoint attains that bound. Cauchy–Schwarz gives $R \le \|a\|_2$. □

The minimax midpoint argument applies to any nonempty finite attained set, using its actual minimum and maximum; the interval assertion requires the full phase fiber. Supremum and infimum give the same minimax value on a dense phase subgroup, but the endpoints need not be attained there.

Independent uniform phases, signs, or quarter turns have vanishing cross-term means, so the ensemble mean is $\bar{f} = \sum_j r_j^2$. This imports a probability law on phase choices. It does not define a deterministic answer on the fiber. In fact, on the full fiber,

$$\bar{f} \le m_a(p), \qquad \bar{f} = m_a(p) \iff \text{at most two of the } r_j \text{ are positive.}$$

To see this, let $r = \max_j r_j$ and let S be the sum of the others. If $r \ge S$, the midpoint is $r^2 + S^2$; subtracting $r^2 + \sum_{j \ne j_*} r_j^2$ leaves the nonnegative cross products among the other lengths, zero exactly when at most one other is positive. If $r \le S$, $\ell = 0$ and $\bar{f} \le rR \le R^2/2 = m_a(p)$. Equality requires exactly two equal positive lengths, apart from the zero case. Thus the phase-mean predictor has worst error $R^2 - \bar{f}$, exceeding minimax by $m_a(p) - \bar{f}$ whenever three lengths are active. For $p_j = 1/n$, $a_j = 1/\sqrt{n}$ and $n \ge 2$, the full range is $[0, 1]$, mean $1/n$, minimax radius $1/2$, and mean worst error $1 - 1/n$.

A finite sign example makes the carrier distinction exact. With three equal input weights and the uniform normalized row, the attained sign values are $\{1/9, 1\}$, whose midpoint and radius are $5/9$ and $4/9$. Full complex phases attain $[0, 1]$. A different, explicitly specified sign fixture is

$$z = (2s_1, s_2, s_3), \quad s_j \in \{-1, 1\}, \quad a = (1/2, 1/4, 1/8), \quad \|z\|^2 = 6.$$

It gives

$$\frac{|az|^2}{6} = \frac{(8s_1 + 2s_2 + s_3)^2}{384} \in \frac{\{25, 49, 81, 121\}}{384}.$$

The mean is $23/128$, midpoint $73/384$, radius $1/8$, and mean worst error $13/96$. Each value occurs twice among the eight sign choices. This is a reconstructed fixture for the supplied fractions, not a claim to have recovered unsupplied original input data. Here the largest length dominates, so the full phase endpoints happen to agree with the sign endpoints. The preceding equal length example supplies the strict endpoint gap. The phrase "Gaussian host" alone specifies neither example: Gaussian-integer modulus fibers can contain more than four phases, and Gaussian-rational unit phases form an infinite dense subgroup. An exact finite-host claim must provide its actual admitted list or group and its arithmetic comparison rule.

Proposition 7.6 (effect-specific descent and resolution). For a Hermitian effect E, the readout $c^* E c$ is constant on $\mathcal{F}_p$ if and only if $E_{ij} = 0$ for distinct active coordinates i, j with $p_i p_j > 0$. It descends to weights on every profile if and only if E is diagonal.

Proof. Its phase expansion is

$$\sum_j p_j E_{jj} + \sum_{i\neq j} \sqrt{p_i p_j} E_{ij} e^{i(\theta_j - \theta_i)}.$$

The nontrivial characters are distinct on the full independent torus. Integrating against each character shows that constancy forces exactly the stated coefficients to vanish. The converse is direct. □

For an orthogonal block decomposition, the observer $\mathcal{D}(\rho) = \sum_b P_b \rho P_b$ retains each entire diagonal block. On all density matrices a Hermitian readout descends through $\mathcal{D}$ if and only if $E = \mathcal{D}(E)$. Indeed equality of the readouts on ρ and $\mathcal{D}(\rho)$ is equivalent to $\mathrm{Tr}((E-\mathcal{D}(E))\rho) = 0$ for every density. An observer retaining only block populations $\mathrm{Tr}(P_b\rho)$ is coarser: its criterion is $E \in \mathrm{span}_{\mathbb{R}}\{P_b\}$, so E must be scalar on each block. For blocks $\{1,2\} \mid \{3\}$, $E = \mathrm{diag}(1,0,0)$ is block diagonal, yet $|1\rangle$ and $|2\rangle$ have equal block populations and readouts one and zero.

A smaller phase host changes even the necessity direction. On independent real-sign lifts, constancy only forces $\mathrm{Re}\, E_{ij} = 0$ on active pairs. The effect $E = (I + Y)/2$ has readout $1/2$ on every $(\pm 1, \pm 1)/\sqrt{2}$, but gives one and zero on $(1, \pm i)/\sqrt{2}$. Independent quarter turns expose both off-diagonal components. For a correlated finite diagonal group, coefficients sharing the same character must first be summed; they may cancel. Similarly, coarse measurement outcomes can cancel phase sensitivity: two Hadamard rank-one effects are individually sensitive, whereas their sum I is constant. Every descent statement must therefore name both the phase carrier and the resolved readout.

7.6 Prediction under dynamics and completed linear observers

Let $H = H^*$, $U(t) = e^{-itH}$, and let the initial observer retain only the weights p. For one fixed output i,

$$\dot{p}_i(0) = 2\,\mathrm{Im}\left(\overline{c}_i \sum_j H_{ij} c_j\right).$$

On the full complex phase fiber this has the exact range

$$[-B_i(p), B_i(p)], \qquad B_i(p) = 2\sqrt{p_i} \sum_{j\neq i} |H_{ij}| \sqrt{p_j}.$$

The diagonal term is real; align each off-diagonal summand to the positive or negative imaginary axis to attain either endpoint. Connectedness supplies the intermediate values. The scalar derivative minimax radius is $B_i(p)$. This aligns one output at a time. Conservation of total probability prevents independent simultaneous endpoint choices for every coordinate.

Theorem 7.7 (exact all-time scalar prediction and a short-time floor). At any fixed real time, set

$$r_j(t) = |U(t)_{ij}| \sqrt{p_j}, \quad R_i(t) = \sum_j r_j(t), \quad \ell_i(t) = \max(0, 2 \max_j r_j(t) - R_i(t)).$$

The possible ith populations are exactly $[\ell_i(t)^2, R_i(t)^2]$, with minimax radius

$$\varepsilon_i(p,t) = \tfrac{1}{2}(R_i(t)^2 - \ell_i(t)^2)$$

and midpoint half the sum of those endpoints. If $h = \inf_{c\in\mathbb{R}}\|H - cI\|_\infty$ (the half spectral diameter), then for $t \geq 0$

$$\varepsilon_i(p,t) \geq \max\{0, tB_i(p) - 2h^2t^2\}.$$

If $p_i > 0$, $\varepsilon_i(p,t) = tB_i(p) + O(t^2)$ as $t \downarrow 0$.

Proof. Apply Theorem 7.5 to row i of $U(t)$. For the lower bound, the second derivative of the P_i expectation is bounded by $\|[H,[H,P_i]]\|_\infty \leq 4h^2$. Each of the two derivative-endpoint lifts therefore has Taylor remainder at most $2h^2t^2$, giving a separation at least $2tB_i - 4h^2t^2$. Half this separation is a prediction lower bound. For the asymptotic statement, $r_i(t) = \sqrt{p_i} + O(t^2)$ and $\sum_{j\neq i} r_j(t) = t\sum_{j\neq i}|H_{ij}|\sqrt{p_j} + O(t^2)$. The ith length dominates for small positive time, and half the difference of the squared sum and squared difference equals twice their product. □

If $p_i = 0$, every first derivative vanishes, but future ambiguity can start at second order. For

$$H = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad p = (0, 1/2, 1/2),$$

the bright lift $(0,1,1)/\sqrt{2}$ and dark lift $(0,1,-1)/\sqrt{2}$ give output zero populations $\sin^2(\sqrt{2}t)$ and 0. The full-phase range is $[0, \sin^2(\sqrt{2}t)]$ and radius half its upper endpoint. A vanishing first-order coefficient therefore does not establish future descent. For $H = X$, $p = (1/2, 1/2)$ the full-phase radius is $|\sin(2t)|/2$, but the real-sign lifts are eigenrays and have constant weights at every time.

All these midpoint assertions concern one scalar output. Componentwise scalar midpoints need not be a probability vector. For the three-dimensional Fourier unitary and uniform input weights, every output has interval $[0,1]$, so their midpoints sum to $3/2$. The fiber contains every output simplex vertex. If a predictor must itself be a probability vector, those vertices force worst ℓ^∞ error at least $1 - \min_i q_i \geq 2/3$; the uniform predictor attains $2/3$. The scalar radius $1/2$ is not a joint distribution certificate under general couplings.

The finite-QM observer-completion results [19] suggest a second, exactly soluble carrier. Here admit **all density matrices** on $\mathbb{C}^d$ and linear statistics $q_j(\rho) = \mathrm{Tr}(A_j\rho)$, with Hermitian A_j. Include the known trace coordinate by setting $\mathcal{S} = \mathrm{span}_{\mathbb{R}}\{I, A_1, \ldots, A_m\}$.

Theorem 7.8 (linear sufficiency and norm-distance minimax). A Hermitian readout $\mathrm{Tr}(E\rho)$ factors through q on the full density carrier if and only if $E \in \mathcal{S}$. More generally its global worst-case scalar prediction error is

$$\inf_f \sup_\rho \big|\mathrm{Tr}(E\rho) - f(q(\rho))\big| = \inf_{B\in\mathcal{S}} \|E - B\|_\infty.$$

The supremum is over density matrices, and arbitrary deterministic scalar functions are admitted. An affine estimator already attains this global optimum. This formula does not compute the radius of each restricted pure phase fiber separately.

Proof. Membership gives an affine formula using the trace. If $E \notin \mathcal{S}$, let $D \neq 0$ be its Hilbert–Schmidt projection onto $\mathcal{S}^\perp$. Then D is Hermitian, traceless, and orthogonal to each A_j, while $\mathrm{Tr}(ED) = \|D\|_{\mathrm{HS}}^2 > 0$. For sufficiently small $\epsilon > 0$, the densities $I/d \pm \epsilon D$ have equal reports and distinct readouts. This establishes necessity using actual physical states.

Write $\delta = \mathrm{dist}_\infty(E, \mathcal{S})$. Finite-dimensional norm duality yields

$$\delta = \max\{\mathrm{Tr}(ED) : D = D^* \in \mathcal{S}^\perp,\ \|D\|_1 \leq 1\}.$$

Because $I \in \mathcal{S}$, all these D are traceless. Every Hermitian traceless D with $\|D\|_1 \le 2$ is the difference of two densities: in its Jordan decomposition $D = D_+ - D_-$ the two traces equal some $a \le 1$; add $(1-a)\tau$ to both parts for any density τ. If $D \perp \mathcal{S}$, the resulting densities have equal reports. Conversely every equal-report difference is in $\mathcal{S}^\perp$ and has trace norm at most two. Thus the maximum readout width of an observer fiber is exactly 2δ. Each fiber is compact; its midpoint attains half its width and its endpoints force that error. A best $B \in \mathcal{S}$ exists and gives the affine estimator $\mathrm{Tr}(B\rho)$ with error at most δ, establishing attainment also in the affine class. □

A chosen retraction of E need not minimize the operator norm either. For $E = J_3/3$ and the diagonal observer, dephasing gives $I/3$ with error $2/3$, whereas $I/2$ gives error $1/2$. The uniform-weight phase fiber attains both zero and one, so Theorem 7.8 or its endpoint lower bound establishes $1/2$ optimal.

At time t, sufficiency replaces E by its Heisenberg evolution $U(t)^* E U(t)$. The smallest invariant real observation space containing $I, A_1, \dots, A_m$ is the span of

$$I, \quad (\mathcal{L}^*)^r A_j \quad (0 \le r < d^2), \qquad \mathcal{L}^*(A) = i[H, A].$$

Cayley–Hamilton gives invariance, and every invariant containing space contains all these iterates. This is the finite row-space completion principle of [19] on the admitted density carrier. It counts linear statistics, not samples, bits, or simultaneously measurable sharp observables. For example observing Z under $H = X$ requires Y for closure, and Y does not commute with Z. On a restricted carrier ambient span membership remains sufficient, but need not be necessary.

The same sufficiency question can be posed before choosing a Hilbert carrier: which admitted histories have equal answers after every available continuation? Section 7.10 relates this protected-history quotient to the linear completion above. The quantum calculation assumes density states and Hamiltonian dynamics; the native finite construction supplies neither assumption by itself.

7.7 Curvature together with retained cohomology

Return to a specified finite real cochain complex $C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2$, with $d_1 d_0 = 0$, $B = \mathrm{im}\, d_0$, and quotient norm $\Phi(\omega) = \mathrm{dist}(\omega, B)$. Let $\ell : C^1 \to Y$ be a linear report into a finite-dimensional normed space, and require $\ell B = 0$. This requirement makes it a report on repair classes.

Theorem 7.9 (augmented quotient observability). The map

$$[\omega] \longmapsto (d_1\omega, \ell\omega)$$

is injective on C^1/B if and only if the induced ℓ is injective on $H^1 = \ker d_1/B$. These conditions are equivalent to the existence of a finite constant K such that

$$\Phi(\omega) \le K(\|d_1\omega\| + \|\ell\omega\|) \quad \text{for every } \omega \in C^1.$$

Proof. The kernel consists exactly of classes represented in $\ker d_1 \cap \ker \ell$. It is zero precisely when ℓ separates H^1. An injective linear map from the finite-dimensional quotient has a bounded inverse on its image, which gives the estimate for the sum norm. Conversely the estimate forces every zero-report class to have zero quotient norm and hence to be zero. □

For an explicit constant, it suffices to exhibit linear lifts $R : C^2 \to C^1$ and $S : Y \to C^1$ satisfying

$$\omega - R d_1 \omega - S\ell\omega \in B.$$

Then $\Phi(\omega) \le \|R\|\|d_1\omega\| + \|S\|\|\ell\omega\|$. Such lifts exist in finite dimensions when the augmented map is injective: extend its inverse from its image and lift a basis of the quotient. Their norms, however, need not be uniform across a family.

For example, take coordinates (b, x, y) with ℓ^1 norm, repair line $B = \mathbb{R}(1, 0, 0)$, $d_{1,n}(b, x, y) = x/n$ and $\ell(b, x, y) = y$. Then $\Phi = |x| + |y|$, the augmented map is injective for every n, and H^1 retains the y coordinate, but its best K is n. Fixed dimension and algebraic observability alone give no uniform lower singular bound. There is also a positive uniform example retaining nonzero cohomology: take direct sums of N fixed blocks with $d_1(b, x, y) = x$ and $\ell(b, x, y) = y$, using sum norms. Then $\Phi = \|d_1\omega\|_1 + \|\ell\omega\|_1$ with $K = 1$ for every N, and $\dim H^1 = N$. This is an explicit linear-complex family; it does not supply a geometric projection-to-cochain adapter.

7.8 A physical cycle adapter and a measurable product-distance bound

We now construct an explicit cochain adapter on a quantum spin ring, providing concrete application data for the abstract framework of §7.2. Number $n \ge 5$ qubits cyclically and orient comparison edges $j \to j + 1$. Consider the real cellular cochain complex of the unfilled cycle:

$$C^0 = \mathbb{R}^n, \quad C^1 = \mathbb{R}^n, \quad C^2 = 0, \qquad (d_0\alpha)_j = \alpha_{j+1} - \alpha_j, \quad d_1 = 0.$$

The coboundary space $\operatorname{im} d_0$ consists of discrete cycle gradients, which model local additive calibration shifts across adjacent sites. Because the cycle is unfilled ($C^2 = 0$), the condition $d_1 d_0 = 0$ holds trivially.

Equip C^1 with the intensive ℓ^1 norm $\|\omega\|_{1,n} = n^{-1} \sum_j |\omega_j|$ and define the circulation report $\ell\omega = n^{-1} \sum_j \omega_j$. The coboundary space and first cohomology are

$$B = \operatorname{im} d_0 = \ker \ell, \qquad H^1 = C^1/B \simeq \mathbb{R}, \qquad \Phi(\omega) = \min_\alpha \|\omega - d_0\alpha\|_{1,n} = |\ell\omega|.$$

Indeed, telescoping around the cycle gives $B \subseteq \ker \ell$, and any zero-sum vector integrates around the cycle to a periodic potential in C^0. For an arbitrary 1-cochain ω, subtracting its mean $\ell\omega$ yields a zero-sum coboundary whose constant residual attains the triangle-inequality lower bound $|\ell\omega|$. Consequently, $\ker \ell = B$ and the quotient conditioning constant in Theorem 7.9 is $K = 1$ for every ring size $n \ge 5$.

Fix the norm-one observables

$$K_j = Z_{j-1}X_jZ_{j+1}, \qquad \mathcal{O}_n = \{K_j, X_j, Z_j : 0 \le j < n\}, \qquad d_{\mathcal{O}_n}(\rho, \sigma) = \max_{O \in \mathcal{O}_n} |\operatorname{Tr} O(\rho - \sigma)|.$$

Let $\mathcal{P}_n = \{\bigotimes_j \sigma_j\}$ denote the set of n-qubit product states, $\Pi\rho = \bigotimes_j \rho_j$ the marginal product retraction, and define the linear evaluation map $A_n(X)_j = \operatorname{Tr}(K_j X)$ together with the induced 1-cochain

$$\omega(\rho) = A_n(\rho - \Pi\rho), \qquad \omega_j(\rho) = \langle K_j \rangle_\rho - \langle Z_{j-1} \rangle_\rho \langle X_j \rangle_\rho \langle Z_{j+1} \rangle_\rho.$$

Each coordinate $\omega_j(\rho)$ depends only on the three-site marginal on $\{j-1, j, j+1\}$, and cyclic site translations intertwine A_n, Π, and the shift operators on C^0 and C^1.

Theorem 7.10 (physical obstruction and measurement error). For every n-qubit density matrix ρ,

$$\boxed{\inf_{\sigma \in \mathcal{P}_n} d_{\mathcal{O}_n}(\rho, \sigma) \ge \tfrac{1}{4}\Phi(A_n(\rho - \Pi\rho)).}$$

If all displayed moment estimates have simultaneous absolute error at most ϵ and are clipped to $[-1,1]$, let $\widehat{\omega}$ denote the resulting empirical cochain. Then

$$\inf_{\sigma\in\mathcal{P}_n} d_{\mathcal{O}_n}(\rho,\sigma) \geq \max\{0, \tfrac{1}{4}|\ell\widehat{\omega}| - \epsilon\}.$$

Adding any cycle gradient $d_0\alpha$ to ω or $\widehat{\omega}$ leaves both lower bounds invariant.

Proof. For any product state $\sigma \in \mathcal{P}_n$, the expectation $\langle K_j\rangle_\sigma$ factorizes into the product of the three single-site expectations. Setting $d = d_{\mathcal{O}_n}(\rho,\sigma)$ and telescoping the three-factor product difference inside $[-1,1]^3$ gives $|\omega_j(\rho)| \leq d + 3d = 4d$. Averaging over j, applying $|\ell\omega| \leq \|\omega\|_{1,n}$, and taking the infimum over $\sigma \in \mathcal{P}_n$ yields the first inequality. The same telescoping bound gives $\|\widehat{\omega} - \omega\|_{1,n} \leq 4\epsilon$ and hence $|\ell\widehat{\omega}| \leq \Phi(\omega) + 4\epsilon$. Finally, $\omega(\sigma) = 0$ for every $\sigma \in \mathcal{P}_n$ and $\ell d_0 = 0$. □

Note that Theorem 7.10 bounds distance to the product manifold $\mathcal{P}_n$ rather than to the convex hull of separable states; classically correlated mixtures can also carry a nonzero certificate (Hein, Eisert, and Briegel [20]). For product observables of maximum arity k, the telescoping constant 4 generalizes to $k+1$.

Corollary 7.11 (Ising rings, a uniform onset window, and pair blindness). Let

$$H_n = J\sum_{j \bmod n} Z_jZ_{j+1}, \qquad J \neq 0, \qquad \rho_n(0) = |+\rangle\langle+|^{\otimes n}.$$

For every $n \geq 5$ and every real time t,

$$\omega_j(\rho_n(t)) = -\sin^2(2Jt), \qquad \Phi(\omega(\rho_n(t))) = \sin^2(2Jt),$$

and therefore

$$\inf_{\sigma\in\mathcal{P}_n} d_{\mathcal{O}_n}(\rho_n(t),\sigma) \geq \tfrac{1}{4}\sin^2(2Jt),$$
$$\tfrac{1}{4}\sin^2(2Jt) \geq J^2t^2 - \tfrac{4}{3}J^4t^4 \geq \tfrac{2}{3}J^2t^2 \quad (|t| \leq (2|J|)^{-1}).$$

Both the onset window and the prefactors are independent of the ring size n. For inhomogeneous bond couplings J_j, the exact coordinate is $-\sin(2J_{j-1}t)\sin(2J_jt)$, whose sign-alternating terms can cancel in the cycle average $\ell\omega$.

Proof. All bond terms commute. In the Heisenberg evolution of K_j, only the two incident bond terms $Z_{j-1}Z_j$ and Z_jZ_{j+1} anticommute with X_j. Expanding each conjugation via $e^{i\theta P}Qe^{-i\theta P} = Q\cos(2\theta) + iPQ\sin(2\theta)$ for $P^2 = I$ and $\{P,Q\} = 0$ leaves $-X_j\sin^2(2Jt)$ as the unique term with nonzero expectation in $|+\rangle^{\otimes n}$. All single-site Z expectations vanish identically, so the product correction is zero and Theorem 7.10 yields the distance bound. On $0 \leq x \leq 1$, $\sin x \geq x - x^3/6 \geq 0$ implies $\sin^2 x \geq x^2 - x^4/3$; substituting $x = 2|Jt|$ completes the proof. □

Along this trajectory, all nonempty Z-only moments vanish, so every proper-partition product in the three-site connected scalar cumulant of (Z_{j-1}, X_j, Z_{j+1}) vanishes and the connected cumulant coincides with $\omega_j(\rho_n(t))$, with leading coefficient $-4J^2$ and Steiner length $\tau = 2$.

At the graph-state time $t_* = \pi/(4|J|)$, all pairwise product defects vanish while the three-body cochain obstruction attains its maximum $\Phi(\omega(\rho_n(t_*))) = 1$. This phenomenon extends to general higher-degree interaction graphs as follows.

Theorem 7.12 (Higher-Degree Graph-State Minimax Theorem). Let $G = (V,E)$ be a finite simple graph with minimum degree $d_{\min} \geq 2$. Evolve the initial product state $|+\rangle^{\otimes|V|}$ under

the Ising Hamiltonian $H = J\sum_{\{u,v\}\in E} Z_u Z_v$ ($J \neq 0$) to the graph-state time $t_* = \pi/(4|J|)$, with unitary $U(t_*) = e^{-it_* H}$. Then: 1. *Complete Two-Body Marginal Blindness:* - If G has girth at least 4 (triangle-free), then for every edge $e = \{u, v\} \in E$ the two-site reduced density matrix is maximally mixed,

$$\rho_{uv}(t_*) = \frac{I_4}{4},$$

and the pairwise product defect vanishes identically across E:

$$D_e(t_*) = \|\rho_e(t_*) - \rho_u(t_*) \otimes \rho_v(t_*)\|_1 = 0, \qquad C(t_*) = \sum_{e\in E} D_e(t_*) = 0.$$

- If G has girth at least 5, then $\rho_{uv}(t_*) = I_4/4$ holds for *every* pair of distinct vertices $\{u, v\} \subset V$, whether adjacent or non-adjacent. 2. *Degree-Parity Stabilizers and Minimax Product Separation:* Under $U(t_*)$, the Heisenberg evolution of X_v at a vertex v of degree d_v is

$$U(t_*)^\dagger X_v U(t_*) = \big(i\,\mathrm{sign}(J) Z_v\big)^{d_v} X_v \prod_{w\in N(v)} Z_w.$$

Define the local transverse Pauli operator $O_v := X_v$ when d_v is even and $O_v := Y_v$ when d_v is odd, and set $K_v := O_v \prod_{w\in N(v)} Z_w$, so that $|\langle K_v\rangle_{\rho(t_*)}| = 1$ for every $v \in V$. (Equivalently, applying the single-qubit Z-rotation gauge $V_{\mathrm{loc}} = \prod_{v\in V} \exp(i\frac{\pi}{4}\,\mathrm{sign}(J) d_v Z_v)$, which preserves the product manifold $\mathcal{P}_V$ and all computational-basis marginals, transforms $\rho(t_*)$ into the standard graph state $|G\rangle\langle G|$ with $O_v = X_v$ and $K_v = X_v \prod_{w\in N(v)} Z_w$ for all d_v [20].) Let $\mathcal{O}_G = \{K_v, O_v, Z_v : v \in V\}$. For any d-regular graph G ($d \geq 2$) of girth at least 4, every product state $\tau = \bigotimes_{v\in V} \tau_v \in \mathcal{P}_V$ satisfies the universal product lower bound

$$d_{\mathcal{O}_G}(\rho(t_*), \tau) = \max_{A\in\mathcal{O}_G} |\mathrm{Tr}(A\rho(t_*)) - \mathrm{Tr}(A\tau)| \geq r_d,$$

where $r_d \in (0, 1)$ is the unique positive real root of the polynomial master equation

$$r_d^{d+1} + r_d - 1 = 0.$$

Moreover, r_d is the exact minimax value over the relaxed box $(\langle O_v\rangle, \langle Z_v\rangle) \in [-1, 1]^{2|V|}$. On the true quantum product manifold $\mathcal{P}_V$, where each single-qubit factor obeys the Bloch-disk positivity constraint $\langle O_v\rangle_{\tau_v}^2 + \langle Z_v\rangle_{\tau_v}^2 \leq 1$, the exact minimax distance is

$$\inf_{\tau\in\mathcal{P}_V} d_{\mathcal{O}_G}(\rho(t_*), \tau) = r_d^{\mathrm{Q}} := \min_{\substack{x,z\in[0,1]\\ x^2+z^2\leq 1}} \max\{1 - xz^d,\ x,\ z\}.$$

For $d \in \{1, 2\}$, $r_d\sqrt{2} < 1$ so the Bloch constraint is inactive and $r_d^{\mathrm{Q}} = r_d$ (attained in the interior at $x = z = r_d$); for $d \geq 3$, $r_d > 1/\sqrt{2}$ so the Bloch boundary $x^2 + z^2 = 1$ is active and $r_d^{\mathrm{Q}} > r_d$ is the unique root $z \in (1/\sqrt{2}, 1)$ of $1 - z^d\sqrt{1 - z^2} = z$. 3. *Closed-Form and Numerical Minimax Constants:* Both r_d and r_d^{Q} increase strictly with degree d: - For $d = 1$ (scalar optimization reference, outside this theorem's $d_{\min} \geq 2$ premise): $r_1 = r_1^{\mathrm{Q}} = \frac{\sqrt{5}-1}{2} \approx 0.6180339887$; - For $d = 2$ (1D cycle graph, $n \geq 5$): $r_2 = r_2^{\mathrm{Q}} = \frac{(108+12\sqrt{93})^{1/3}}{6} - \frac{2}{(108+12\sqrt{93})^{1/3}} \approx 0.6823278038$; - For $d = 3$ (2D honeycomb lattice): polynomial lower bound $r_3 \approx 0.7244919590$, exact quantum minimax $r_3^{\mathrm{Q}} \approx 0.7324640561$; - For $d = 4$ (2D square lattice): polynomial lower bound $r_4 \approx 0.7548776662$, exact quantum minimax

$r_4^{\mathrm{Q}} \approx 0.7732827496$; - For $d \geq 3$, $1 - \frac{\ln(d+1)}{d+1} \leq r_d < r_d^{\mathrm{Q}} < 1$, so both tend to 1 as $d \to \infty$. 4. *Bounded Measurement Settings:* All observables in $\mathcal{O}_G$ can be measured in at most

$$M_{\text{settings}} \leq \chi(G^2) + 1 \leq \Delta(G)^2 + 2$$

joint product Pauli measurement settings, independent of $|V|$, where $\chi(G^2)$ is the chromatic number of the distance-2 square graph G^2.

Proof. 1. *Two-Body Marginal Blindness:* Expand $\rho_{uv}(t_*) = \mathrm{Tr}_{V\setminus\{u,v\}}\,\rho(t_*)$ in the computational basis $\{|x\rangle : x \in \{0,1\}^V\}$. Since $|+\rangle^{\otimes|V|} = 2^{-|V|/2}\sum_x |x\rangle$ and $Z_a|x\rangle = (-1)^{x_a}|x\rangle$, the matrix element $\langle x_u x_v|\rho_{uv}(t_*)|y_u y_v\rangle$ is $2^{-|V|}$ times the sum over $z \in \{0,1\}^{V\setminus\{u,v\}}$ of the phase

$$\exp\left(-i\frac{\pi}{4}\,\mathrm{sign}(J)\sum_{\{a,b\}\in E}\left((-1)^{\tilde{x}_a+\tilde{x}_b} - (-1)^{\tilde{y}_a+\tilde{y}_b}\right)\right),$$

where $\tilde{x}, \tilde{y}$ agree with z on $V \setminus \{u,v\}$ and take the values (x_u, x_v) and (y_u, y_v) on $\{u,v\}$. Terms in E disjoint from $\{u,v\}$ cancel identically between $\tilde{x}$ and $\tilde{y}$. For each $w \in V \setminus \{u,v\}$, write $s_w = (-1)^{z_w} \in \{\pm 1\}$; the factor depending on z_w is $(-i\,\mathrm{sign}(J))^{\alpha_w s_w}$ where

$$\alpha_w = \mathbf{1}_{\{u,w\}\in E}\big((-1)^{x_u} - (-1)^{y_u}\big)/2 + \mathbf{1}_{\{v,w\}\in E}\big((-1)^{x_v} - (-1)^{y_v}\big)/2 \in \{-2,-1,0,1,2\}.$$

Whenever $\alpha_w = \pm 1$, summing over $s_w \in \{+1,-1\}$ gives $(-i\,\mathrm{sign}(J)) + (i\,\mathrm{sign}(J)) = 0$. If $e = \{u,v\} \in E$ and G has girth ≥ 4, then $N(u) \cap N(v) = \emptyset$. Since $d_{\min} \geq 2$, if $x_u \neq y_u$ there exists $w \in N(u) \setminus \{v\}$, and because $w \notin N(v)$ one has $\alpha_w = \pm 1$, so the sum over z_w vanishes; symmetrically, if $x_v \neq y_v$ there exists $w' \in N(v)\setminus\{u\}$ with $\alpha_{w'} = \pm 1$. (Note that $d_{\min} \geq 2$ is sharp: if u were a leaf with $N(u) = \{v\}$, then $U(t_*)^\dagger(Y_uZ_v)U(t_*) = \pm X_u$ would give $\langle Y_uZ_v\rangle_{\rho(t_*)} = \pm 1 \neq 0$ and $\rho_{uv}(t_*) \neq I_4/4$.) If G has girth ≥ 5, any two non-adjacent vertices $u \neq v$ share at most one common neighbor, and since $d_u, d_v \geq 2$, each has at least one neighbor not adjacent to the other, again giving a vertex w with $\alpha_w = \pm 1$ whenever $(x_u, x_v) \neq (y_u, y_v)$. The diagonal entries $(x_u = y_u, x_v = y_v)$ equal $1/4$, so $\rho_{uv}(t_*) = I_4/4$ and $D_e(t_*) = 0$.

2. *Degree-Parity Stabilizers and Minimax Separation:* Since Z_uZ_w commutes with X_v unless $v \in \{u,w\}$ and $e^{i(\pi/4)\,\mathrm{sign}(J)Z_vZ_w}X_ve^{-i(\pi/4)\,\mathrm{sign}(J)Z_vZ_w} = (i\,\mathrm{sign}(J)Z_v)X_vZ_w$, multiplying over the d_v incident edges $w \in N(v)$ yields the displayed Heisenberg formula. Using $Z_vX_v = iY_v$, the factor $(i\,\mathrm{sign}(J)Z_v)^{d_v}X_v$ equals $(\mathrm{sign}(J))^{d_v}(-1)^{\lceil d_v/2\rceil}O_v = \pm O_v$, so $U(t_*)^\dagger K_v U(t_*) = \pm X_v$ has expectation $s_v := \langle K_v\rangle_{\rho(t_*)} \in \{+1,-1\}$, while $\langle O_v\rangle_{\rho(t_*)} = \langle Z_v\rangle_{\rho(t_*)} = 0$ because $d_v \geq 2$ and girth ≥ 4 make $\rho_v(t_*) = I_2/2$.

Now let $\tau = \bigotimes_{v\in V}\tau_v \in \mathcal{P}_V$ and put $\delta = d_{\mathcal{O}_G}(\rho(t_*), \tau)$, $x_v = \langle O_v\rangle_{\tau_v}$, and $z_v = \langle Z_v\rangle_{\tau_v}$. Then $|x_v| \leq \delta$ and $|z_w| \leq \delta$ for all $w \in N(v)$. Because $K_v = O_v\prod_{w\in N(v)}Z_w$ acts on distinct vertices $\{v\}\cup N(v)$, its expectation in τ factorizes:

$$|\langle K_v\rangle_\tau| = |x_v|\prod_{w\in N(v)}|z_w| \leq \delta\cdot\delta^d = \delta^{d+1}.$$

By the triangle inequality, $1 = |s_v| \leq |s_v - \langle K_v\rangle_\tau| + |\langle K_v\rangle_\tau| \leq \delta + \delta^{d+1}$. Since $f_d(\delta) = \delta^{d+1} + \delta$ is strictly increasing on $[0,\infty)$ with $f_d(r_d) = 1$, we obtain $\delta \geq r_d$. Over the relaxed box $(x_v, z_v) \in [-1,1]^{2|V|}$, setting $x_v = s_v r_d$ and $z_v = r_d$ attains $\delta = r_d$.

On the quantum product manifold $\mathcal{P}_V$, positivity of each single-qubit state τ_v requires $x_v^2 + z_v^2 \leq 1$. Choose a vertex $v_* \in V$ attaining $z := \max_{w\in V}|z_w| = |z_{v_*}| \in [0,1]$, and put $x := |x_{v_*}| \in [0,1]$. Then

$x^2 + z^2 \le 1$, every neighbor $w \in N(v_*)$ satisfies $|z_w| \le z$, and thus $|\langle K_{v_*}\rangle_\tau| \le xz^d$, which forces

$$\delta \ge \max\{1 - xz^d,\ x,\ z\} \ge r_d^{\mathrm{Q}}.$$

Conversely, for any $(x, z) \in [0, 1]^2$ with $x^2 + z^2 \le 1$, the site-homogeneous product state $\tau(x, z) = \bigotimes_{v\in V} \frac{1}{2}(I + s_v x\, O_v + z\, Z_v) \in \mathcal{P}_V$ attains $d_{\mathcal{O}_G}(\rho(t_*), \tau(x, z)) = \max\{1 - xz^d, x, z\}$. For $d \in \{1, 2\}$, $2r_d^2 < 1$, so $(x, z) = (r_d, r_d)$ lies strictly inside the Bloch disk and attains $r_d^{\mathrm{Q}} = r_d$. For $d \ge 3$, $2r_d^2 > 1$. To establish the constrained optimum, the equation $1 - z^d\sqrt{1 - z^2} = z$ is equivalent on $(0, 1)$ to

$$\frac{1 - z}{1 + z} = z^{2d}.$$

Its left side decreases and its right side increases, giving exactly one root r; at $z = 1/\sqrt{2}$ the left side is greater for $d \ge 3$, so $r > 1/\sqrt{2}$. Put $z_0 = \sqrt{d/(d + 1)}$. Bernoulli's inequality gives $z_0^{2d} \ge 1/(d+1)$ and hence $z_0^d\sqrt{1 - z_0^2} \ge 1/(d+1) > 1-z_0$, so $r < z_0$. For a proposed error $\delta \in [1/\sqrt{2}, r)$, maximizing xz^d subject to $x, z \le \delta$ and the disk gives $\delta^d\sqrt{1 - \delta^2}$: the circle function $z^d\sqrt{1 - z^2}$ increases up to z_0, and the remaining box segment increases up to that circle. This maximum is less than $1 - \delta$, precluding such an error value. Errors below $1/\sqrt{2}$ are already excluded by the box lower bound $r_d > 1/\sqrt{2}$. At $z = r$, $x = \sqrt{1 - r^2} \le r$ attains error r. Thus $r_d^{\mathrm{Q}} = r > r_d$. The defining root equations show strict increase with d.

3. *Measurement Settings:* Two star stabilizers K_u, K_v have disjoint supports whenever $d_G(u, v) \ge 3$. Thus any proper vertex coloring of the distance-2 square graph G^2 partitions V into at most $\chi(G^2) \le \Delta(G^2) + 1 \le \Delta(G)^2 + 1$ color classes, each of which can be measured in a single product Pauli setting (measuring O_u on the color class and Z_w on all other vertices, which also supplies all single-site O_u expectations). Adding one all-Z setting for $\{Z_v\}_{v\in V}$ gives at most $\chi(G^2) + 1 \le \Delta(G)^2 + 2$ product settings. □

Under observational perturbation or finite-sample estimation error bounded by η, the distance obeys the stability floor

$$\inf_{\tau\in\mathcal{P}_V} d_{\mathcal{O}_G}(\rho, \tau) \ge \max\{0, r_d^{\mathrm{Q}} - \eta\} \ge \max\{0, r_d - \eta\}.$$

Operational measurement procedure and nuisance parameter models. Under local additive calibration drift, the reported cochain transforms as $\tilde{\omega} = \omega + d_0\alpha + \xi$, where $d_0\alpha \in \mathrm{im}(d_0)$ represents localized gradient shifts and ξ represents zero-mean measurement noise with $\|\xi\|_{1,n} \le \eta$. Because $\ell d_0 = 0$, the quotient functional satisfies

$$\Phi([\tilde{\omega}]) = |\ell(\omega + d_0\alpha + \xi)| = |\ell(\omega + \xi)|,$$

exhibiting exact gauge invariance under additive potential coboundaries $d_0\alpha$. In contrast, multiplicative detector gain drift $\omega' = g_{\mathrm{det}}^3\omega$ scales the three-point correlator outside $\mathrm{im}(d_0)$ and satisfies

$$|\Phi(\omega') - \Phi(\omega)| \le |g_{\mathrm{det}}^3 - 1|\Phi(\omega),$$

preserving a strictly positive quotient whenever $|g_{\mathrm{det}}^3 - 1| < 1$. On the 1D cycle ($d = 2$), properly coloring G^2 requires at most $\chi(C_n^2) \le 5$ colors (and on C_n itself, coloring vertices at distance ≥ 3 requires at most 5 settings, or 3 settings when $3 \mid n$ by grouping non-overlapping three-site supports, plus 1 all-Z setting). With N independent shots per setting, Hoeffding's inequality and a union bound over the $3n$ single-site and three-site observables certify simultaneous estimation error at most ϵ with probability at least $1 - \delta$ whenever $N \ge 2\epsilon^{-2}\log(6n/\delta)$.

7.9 Selected finite calculations and their scope

The following calculations complement §§5–7, separating written finite identities from numerical demonstrations.

1. **Rational repair quotients and cumulant constants.** For a finite rational differential $d : C^{k-1} \to C^k$, finite linear-programming duality gives

$$\min_{\alpha} \|\omega - d\alpha\|_1 = \max_{\|y\|_\infty \le 1,\ d^T y = 0} \langle y, \omega \rangle.$$

   Rational data admit rational optimal witnesses; any proposed witness must still satisfy its constraints and matching objectives. The exact cycle certificates in `code/physical_obstruction.py` instantiate this argument, not every higher-dimensional complex. Partition enumeration gives $a_k = (1, 2, 6, 26, 150, 1082)$ and $b_k = (1, 3, 13, 75, 541, 4683)$ for $1 \le k \le 6$ under the definitions of §5.14. These support-size constants do not measure finite-difference conditioning in time.
2. **Marginal derivative versus residual.** On Hermitian trace-zero two-site variations at a density product, partial-trace contraction and tensor norm multiplicativity give $\|D\Pi_m Z\|_1 \le 2\|Z\|_1$. The constant 2 is sharp: at $m = |00\rangle\langle 00|$, take $Z = |11\rangle\langle 11| - |00\rangle\langle 00|$. Then $D\Pi_m Z = \mathrm{diag}(-2, 1, 1, 0)$, of norm 4, while $\|Z\|_1 = 2$. The triangle inequality gives $\|(I - D\Pi_m)Z\|_1 \le 3\|Z\|_1$; it does not establish sharpness of 3. Likewise the product retraction is 2-Lipschitz on density states and its residual is at most 3-Lipschitz. Neither statement says $\|D\Pi_m\| = 3$.
3. **Commuting next-nearest-neighbor perturbation.** For $H = J_1 \sum_j Z_j Z_{j+1} + J_2 \sum_j Z_j Z_{j+2}$ on C_n, $n \ge 5$, all terms commute. At $t_* = \pi/(4|J_1|)$, $J_1 \ne 0$, conjugating the cycle stabilizer by the two additional incident ZZ rotations gives

$$\Phi(\omega(t_*)) = \cos^2(2J_2 t_*) = 1 - \frac{\pi^2}{4}(J_2/J_1)^2 + O((J_2/J_1)^4).$$

   Terms with an extra nonempty Z string have zero expectation in the unperturbed graph state. The quotient is positive precisely when this cosine is nonzero; $|J_2/J_1| < 1$ suffices, whereas $J_2 = J_1$ gives zero. This is a commuting perturbation, not evidence of integrability breaking. A transverse X field is noncommuting and requires a separate Duhamel bound or an explicit evolution calculation.
4. **Plaquettes and retained cohomology.** On C_4, the opposite-site marginal at graph-state time is $(I + X \otimes X)/4$, with product defect 1, while every edge marginal is $I_4/4$. Filling plaquettes does not imply that every non-exact 1-cochain has nonzero curvature: on a square torus, the cochain equal to 1 on positively oriented horizontal edges and 0 on vertical edges is closed and has a nonzero horizontal period. Curvature detects nonclosedness. It detects every non-exact class only when $H^1 = 0$; otherwise §7.7 requires the additional cohomology report.
5. **Boundary tangents.** The written two-qubit witness in §5.11.1 separates the ambient positivity cone from the product tangent cone. Floating-point rank thresholds and unverified polar candidates in `code/feasible_tangent_cone.py` remain numerical diagnostics. They do not certify either the exact rank stratum or an optimum.
6. **Local noise with an explicit convention.** Use $\mathcal{L} = -i[H, \cdot] + \gamma_z \sum_j \mathcal{D}_{Z_j}$, $\gamma_z \ge 0$, with the dissipator convention of §5.4.1. Since $\mathcal{D}_Z^\dagger(X) = -2X$ and Z dephasing commutes with the Ising evolution, the ring observable $K_j = Z_{j-1} X_j Z_{j+1}$ obeys

$$\Phi_{\gamma_z}(t_*) = e^{-2\gamma_z t_*} \Phi_0(t_*).$$

   A rate defined as transverse attenuation instead would put $\gamma_z/2$ in the generator. For X dephasing, the two endpoint Z factors are damped and the central X is fixed; the dissipator does

not commute with the Ising generator. An exponential for the initial Pauli operator alone therefore does not supply the time-evolved quotient or a uniform noise threshold. The general local Lindblad bounds remain those of §5.4.1.

7. **Discretization and observer replication.** For fixed bounded generators, symmetric Strang splitting has one-step error $O(\Delta t^3)$ and accumulated error $O(T\Delta t^2)$ over fixed time T, with constants depending on the relevant commutators and stability bounds. These are approximation orders, not validated numerical error bars. For the visibility construction, take upper patches u_i and lower overlaps $x_i < u_i, u_{i+1}$ on a K-cycle, $K \geq 3$, with fibers $\mathbb{R}^K$, identity restrictions, $s(u_i) = e_i$ and $s(x_i) = 0$. Its $2K$ covering edges each have residual norm 1, hence $I(s_K) = 2K$. Observer i watches the two edges into u_i and deletes coordinate i, so every local tally is zero. Trace-norm growth for a noncontractive evolution shows why a contractivity premise cannot arbitrarily be dropped from its proof; it does not make complete positivity or Hermiticity necessary for every bound. Transpose, for example, is positive and trace preserving, not completely positive in dimension at least two, and is trace-norm isometric.

`code/tests/test_physical_obstruction.py` checks the declared cycle adapter and finite graph-state examples; `code/tests/test_connected_cumulants.py` checks specified cumulant calculations. `code/tests/test_po000_e2e_extension.py` contains selected algebraic and floating-point controls, including the cluster interference and GHZ examples. Its numerical Duhamel and tangent outputs have the non-certifying scope stated in §§5.10–5.11. Passing these files does not prove unrestricted topological, dynamical or reconstruction claims.

7.10 Related Foundational Frameworks: Protected Histories and Bell Extensions

Discrete protected history carriers, partition refinement quotients, joint future sufficiency, and positive Bell extensions are developed in companion packages:

- **Finite QM** (`F004_papers/006_finite_qm`, §13.8) develops the operational quantum realization, character cancellation on the sign cube, and the separation between linear gluing, support gluing, and positive Bell cone extensions.
- **PO-035** (`002_projection_obstruction/035_zero_container_relational_spacetime...`) establishes defect 4-current conservation and relational spacetime holonomy.

Appendix A: Modular Architecture & Sibling Directory

To maintain focus on quantum lattice dynamics, local correlation bounds, and observer visibility, foundational manifold chart constructions and companion developments are coordinated across the research series:

1. **Decoupled Banach Chart Formulations:** The full hierarchy of Banach-manifold chart-derivative formulas, vertical shear retractions, parameter-chart transitions, and bounded left-inverse failure criteria (formerly Lemmas 3.4–3.14 and Examples 3.4–3.14) is omitted from this 69-page edition. These results are preserved in archived monograph revisions `90e6231546` / `d44f760555` and verified by symbolics in `code/retraction_geometry.py`.
2. **Operational Quantum Foundations:** The operational Jordan–Lie reconstruction, positive cone characterizations, canonical trace witnesses, and central-sector purification obstructions (formerly Appendix E) are imported from companion package `Finite QM` (`F004_papers/006_finite_qm`, §13.8) and formalized in `lean4/Po000JordanLie.lean` and `lean4/Po000JordanTrace.lean`.

3. **Coordinated Sibling Packages:** Sibling packages in the `002_projection_obstruction` series and `coda_atlas` are listed in the Authoritative Sibling Reference Table in §1.2.

Appendix B: Exact Constant Ledger for Theorem 5.1

For an observable supported on $k \in \{1, 2\}$ sites, support counting gives

$$\|[H, A]\|_\infty \le 2kg\|A\|_\infty, \qquad \|[H, [H, A]]\|_\infty \le 4k(k+1)g^2\|A\|_\infty.$$

Thus the one-site and two-site reduced-state second derivatives are bounded by $8g^2$ and $24g^2$, respectively, while each one-site first derivative is bounded by $2g$.

For

$$F_e(t) = \rho_e(t) - \rho_a(t) \otimes \rho_b(t),$$

the product rule and trace-norm tensor multiplicativity give

$$\begin{aligned}\|F_e''(t)\|_1 &\le \|\ddot{\rho}_e(t)\|_1 + \|\ddot{\rho}_a(t)\|_1 + 2\|\dot{\rho}_a(t)\|_1\|\dot{\rho}_b(t)\|_1 + \|\ddot{\rho}_b(t)\|_1 \\ &\le 24g^2 + 8g^2 + 8g^2 + 8g^2 \\ &= 48g^2.\end{aligned}$$

Since $F_e(0) = 0$ and $F_e'(0) = n_e$, the integral Taylor formula and reverse triangle inequality yield

$$D_e(t) = \|F_e(t)\|_1 \ge t\|n_e\|_1 - \frac{t^2}{2}(48g^2) = \nu_e t - 24g^2t^2.$$

Consequently, on $0 < t \le \nu_{\min}/(48g^2)$,

$$D_e(t) \ge \frac{\nu_{\min}}{2}t$$

for every declared active edge with $\nu_e \ge \nu_{\min}$. If at least $\eta|E|$ edges are active and $|E| \ge aN$, then

$$\frac{1}{N}\sum_{e\in E} D_e(t) \ge a\eta(\nu_{\min}t - 24g^2t^2) \ge \frac{a\eta\nu_{\min}}{2}t.$$

There is no causal-cone radius, Lieb–Robinson error, regular-stratum hypothesis, or positive lower time endpoint in this derivation.

Appendix C: Extension Map

The research series coordinates specialized geometric, cohomological, and foundational analyses across distinct packages. Sibling packages, titles, anchors, and scopes are specified authoritatively in §1.2:

- **PO-001** (*From Covering-Edge Descent Complexes to Functorial Presheaves and Coherence Defects*, v2.0.0): Saturated-chain presheaves, degree-2 composition curvature cochains $K_{c,b,a}$, and operator cocycle coboundaries.
- **PO-006 (GS-003)** (*Vector and Quantum Observer Geometries: Gauge Relativity and Trace Metric Bounds*, v0.4.0): Non-scalar observer geometries, Schatten-norm operator bounds, dual witness certificates, and phase metrics.

- **PO-007 (GS-001)** (*M-Compatible Gauge Classification for Banach Presheaf Coboundaries: The ℓ^1 Maximal-Symmetry Endpoint*, v4.7.0): M-compatible gauge classification for Banach presheaves; maximal linear symmetry selecting weighted ℓ^1.
- **PO-008 (GS-002)** (*SO(k)-Invariant Gauge Rigidity for Declared Euclidean Stalks: Local ℓ^2 Selection and Outer ℓ^1 Aggregation*, v2.0.0): $SO(k)$-invariant gauge rigidity on Euclidean stalks; local ℓ^2 selection with outer ℓ^1 aggregation.
- **PO-009 (GS-000)** (*Coordinate-Wise Additivity and Local ℓ^1 Gauge Selection*, v6.1.1): Coordinate-wise additivity and local ℓ^1 gauge selection on finite graph cochains.
- **PO-033** (*Certified Adaptive Quantum Dynamics from Projection Residuals*, v1.0.1): Conditional analytic bounds and uncertified adaptive cluster prototype (heuristic_uncertified / `HEURISTIC_UNCERTIFIED`); Lean scope is conditional finite-sum inequalities over an abstract carrier (`Nat`/`Int`). Validated continuous-time ODE integration and conservative quadrature enclosures on general graphs remain open work.
- **PO-034** (*Higher-Cell Product Retractions, Simplicial Cochains, and Finite Retraction Dynamics*, v0.1.0): Higher-cell product retractions, oblique projectors, simplicial cochain dynamics, and finite-dilation Taylor estimates.
- **PO-035** (*Zero-Container Relational Spacetime, Emergent Lorentz Causality, and Witness Holonomy*, v0.1.0): Defect 4-current conservation ($\partial_\mu J^\mu = 0$), discrete relational spacetime, and witness holonomy.
- **Finite QM** (*Finite Quantum Mechanics*, §§13.5–13.8): Operational Jordan–Lie foundations, character cancellation on the sign cube, positive Bell cone extensions, and central-sector purification obstructions.
- **Archived Monograph Material** (Commits `90e6231546` / `d44f760555`; symbolics in `code/retraction_geometry.py`): Banach graph charts, shear retractions, parameter-chart transitions, bounded-left-inverse failure lemmas, and conditional extension reductions (omitted from the 69-page edition).

These companion extensions close distinct gaps exposed by the correction audit. Earlier series labels are historical only. None silently upgrades another paper's imports, mechanization status, or authority.

Data and Code Availability

The v6.9.16 revision connects existing results in §7.10 and operational quantum foundations without adding a primitive-to-quantum reconstruction. The external repository sources in [26] implement native history quotients, joint refinement and exact Bell extensions in the Rust category, WDP and quantum owners. The external Lean modules `GenesisContinuation.lean` and `GenesisJointOperations.lean` prove abstract future/operation laws and integer-weight CHSH bounds; they do not verify the Rust algorithms or the quantum realization. These external sources are pinned navigation references, not additional bundled or newly mechanized proofs of this manuscript. Finite QM [19, §13.8] states the separate scope of `CarrierTests.lean` and its Rust/Python carrier-test interfaces.

The companion software and formalization archive (`PO-000`, release `6.9.18`, Python code `6.9.6`) accompanies this manuscript and is covered by the deterministic checksum manifest `SHA256SUMS` at the package root. Authority status: none; the software, finite replay receipts, and mechanized modules are mathematical and numerical verification artifacts only and confer no operational authority.

The package consists of the following components:

- **Manuscript and Claim Ledgers:** `projection_obstruction.md`, together with the structured claim ledgers `projection_obstruction_claim_ledger.md`, `projection_obstruction_claim_ledger.yaml`, `docs/claim_dependencies.json`, and `projection_obstruction_change_log.md`.
- **Lean 4 Formalization Suite (19 files in `lean4/`):**
  - *Scoped Lake build modules (Lean 4.34.0 / Mathlib v4.34.0):* `lean4/Po000JordanLie.lean` and `lean4/Po000JordanLieImage.lean` (formalizing the complexified associative C^*-algebra construction and positive cone of the operational Jordan–Lie algebra), and `lean4/Po000JordanTrace.lean` (formalizing the canonical trace witness, idempotent polynomial projections, and positive definiteness of the central trace).
  - *Arithmetic jet constants:* `lean4/po000_short_time_jets.lean` (checking support-factor and factorial-normalization arithmetic, including the constants 24, 64, and 160 in §§5.8 and 6.3).
  - *Foundational hierarchy and observer-visibility modules:* Fifteen `po000_*.lean` modules, including `po000_l001_replica_hierarchy.lean` (`ReplicaHierarchy`, mechanizing the RE3 singleton decomposition and RE3+RE5 unweighted sum of Theorem 4.2), `po000_l005_weighted_hierarchy.lean` (`PO000.WeightedHierarchy`, mechanizing the weighted and orbit singleton cores of Theorem 4.2 items 2–3 and the former counterexample arithmetic for the withdrawn five-way independence claim), `po000_l004_observer_visibility.lean` (`PO000.ObserverVisibility`, mechanizing Propositions 7.4.3–7.4.5, Corollaries 7.4.6–7.4.7, Proposition 7.4.7-B, and §7.4.10), and `po000_l001_projection_transfer.lean` (`ProjectionTransfer`). The scoped Jordan–Lie and trace build and the historical per-file checks have separate dated records in `lean4/README.md` and `lean4/manifest.json`. Their recorded axiom reports lie within {`propext`, `Classical.choice`, `Quot.sound`}; no whole-suite or fresh historical-file elaboration is implied. Parked and package-context-only sources are explicitly labelled there.
- **Python Verification and Exact Replay Suite (`code/` and `code/tests/`):**
  - *TDVP Duhamel and Open-System Bounds (§§5.4.1, 5.10.1–5.10.3):* `code/observable_duhamel.py`, `code/open_system_jets.py`, and `code/tests/test_honeycomb_tdvp_duhamel_integration.py`. The latter compares a sampled Duhamel estimate with floating-point Krylov evolution on an eight-site hexagon-plus-square graph. It supplies neither a honeycomb-lattice example nor a validated continuous-time enclosure; §5.10.A remains a written conditional theorem.
  - *Short-Time Jets, Metric Optimality, Connected Cumulants, and Polynomial Onset (§§5.8, 5.11–5.17, 6.3, 7.5–7.7):* `code/short_time_jets.py`, `code/feasible_tangent_cone.py`, `code/derivation_replays.py`, `code/polynomial_onset.py`, `code/local_persistent_zero.py`, and their corresponding pytest suites in `code/tests/`.
  - *Physical Cycle Adapter, Higher-Degree Graph-State Minimax, and Finite Witnesses (§§7.8–7.9):* `code/physical_obstruction.py`, `code/verify_finite_certificates.py`, `code/verify_certificate.py`, `code/replay.py`, and test suites `code/tests/test_physical_obstruction.py`, `code/tests/test_sharp_onset.py`, and `code/tests/test_po000_e2e_extension.py` (checking selected finite instances related to Theorems 4.3.A, 5.10.B and 7.12, along with the explicitly non-certifying numerical demonstrations described above; no twenty-witness unconstrained certificate is claimed).

To execute the verification suite from the package root:

```
python -m pip install ".[jets]"
python -m pytest -q code/tests/test_metric_phase_derivations.py code/tests/test_connected_cumulants.py
python -m pytest -q code/tests/test_short_time_jets.py code/tests/test_polynomial_onset.py
python -m pytest -q code/tests/test_physical_obstruction.py code/tests/test_sharp_onset.py
```

```
python -m pytest -q code/tests/test_honeycomb_tdvp_duhamel_integration.py
 ↪  code/tests/test_po000_e2e_extension.py
python code/replay.py --output build/verification/current_run.json
python code/verify_certificate.py build/verification/current_run.json
python scripts/regen_sha256sums.py --check
```

Scope and evidence boundaries. Each Lean module or Python test supports only the specific algebraic identity or finite instance it encodes; finite numerical or symbolic replays are not a proof of the general Banach-manifold or infinite-volume theorems. Theorem 3.1 rules out affine-coordinate linearization in a declared affine chart, not arbitrary nonlinear smooth conjugacies. In §5, the raw edge sum $C(t)$ is an overlapping two-site correlation tally, not the global trace norm $\|\rho(t) - \bigotimes_v \rho_v(t)\|_1$, and Theorems 5.1, 5.5, and 5.7 use direct local commutator estimates rather than Lieb–Robinson bounds (whereas Theorem 5.10.A incorporates Lieb–Robinson light-cone weights for spatially localized observables). Quantum Hilbert spaces and dynamical generators enter §§5–7 as declared host assumptions rather than derivations from obstruction data, and repairable mass R is not defined in this paper. Companion extensions (such as PO-034 and PO-035) address higher-cell adapters and spacetime holonomy separately.

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26. *Native joint operations and a Genesis-to-Bell experiment*, The Symbol Game repository, revision `4bdf074a1b18cdeb710d991336e35f8f62979542`, 2026. Joint experiment; stable history carrier. External source descriptions; their proof and execution scopes are stated separately.

Supplemental formalization: uncertainty and correction (23 October 2026)

The standalone supplemental build `python tools/replay_projection_bounds.py` compiles `lean4/ProjectionBounds.lean` with Lean 4.35.0 and replays every named theorem in

`lean4/ProjectionCensus.lean`. The 20-theorem result and exact source hashes are recorded in `lean4/projection_axiom_census.json`; the text axiom output is `lean4/projection_axiom_report.txt`. This supplements the existing census without replacing its historical scope. Dependencies, source pins and licenses are shipped in `lean4/projection_vendor/`; no sibling paper or network fetch is needed. Only `propext`, `Quot.sound` and `Classical.choice` are allowed.

Bound	Lean result	Explicit premise retained
Robust rate and onset	`lowerRate_le`, `uncertain_onset`, `half_slope_window`	A two-sided rate enclosure and the actual Taylor lower estimate on its horizon
Correction deadline	`correction_window_iff`, `zero_rate_window`, `safe_horizon`, `exhausted_budget`	Positive rate for division; a separate zero-rate branch; independently valid budgets and horizon
Actual corrected update	`corrected_step`, `accumulated_budget`	A nonexpansive evolution, residual enclosure and correction displacement bound
Finite transported jumps	`transported_jump_bound`	The variation-of-constants identity, an integral bound, and contractive transports
Product and observed residual uncertainty	`factor_replacement_bound`, `product_residual_uncertainty`, `product_defect_uncertainty`, `observed_residual_bound`, `observed_defect_interval`	A factor-replacement path with each step bounded by the state distance; a declared Lipschitz observation

The scalar results are over the real numbers. The jump and residual theorems are over arbitrary normed additive groups, so they apply to a trace-norm carrier once its physical obligations are proved. In particular, the constant $k + 1$ is derived by telescoping k factor replacements and adding the original state discrepancy. The finite sum of jump norms is derived from the supplied transported-jump identity. Lean does not construct the continuous propagator, prove the density-matrix tensor/partial-trace estimates, or certify numerical integration in this supplement. Those statements remain written analytic premises. The new bounds do not turn a supplied error radius into an observed or statistically justified radius.

Three compiled examples check the exact allowed budget, a strictly excessive budget, and a rate completely explained by uncertainty. The companion Python examples test exact arithmetic on supplied budgets; the fresh Lean replay checks the general implications above.